Mapping Recaptures

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Contents

Mapping Genotype Recaptures in Central GA

This markdown will visualize the spatial distribution of bears with *spatial recaptures* from hair samples collected in central Georgia (2023 and 2024). These samples were genotyped via capture-seq (Rapid Genomics) or GT-seq (GTSeek). Capture-seq data were filtered to avoid tightly linked loci (no SNPs within 500kb with $r^2 > 0.8$) and focus on informative markers (MAF > 0.2) and combined with GT-seq data (which have a broader range of MAF), resulting in a total of 1325 SNP loci (see `4_combineGTSeek` for details on combining the data from the two marker panels). Genotypes were subjected to a clone analysis in `COLONY` (Wang 2016) and KING kinship coefficients (Manichaikul et al. 2010) were calculated in `PLINK2` (Chang et al. 2015). Inferences from the two approaches were combined following methods outlined in `10_recapFinal`.

Here, we'll plot the recaptures for bears as a way to assess a couple of things:

1. Confidence in recapture data - are samples spaced too far apart? Are we confident in the combined inferences from `COLONY` and `PLINK`?
2. Evidence for moving activity centers - do we have sufficient data to fit a model where activity centers are allowed to move between years? Does this appear to be happening in the dataset?

Load data

We need to load the recapture objects and the locations of the hair snares to visualize the distribution of recaptures. . .

```
#Load the sampleMetadata file
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)

#Load capture history objects
```

```r
load("../scr/CGA_CH.Rda")

#Load hair snare location data
load("../sampleinfo/snareLocations.Rda")

#Load the state space
library(raster)
```

```
## Loading required package: sp
```

```r
S360 <- raster("../input/state-space360.tif")

#Source functions to calculate some basic summary stats
source("../input/sumstats_4d.R")
```

Females

In the plots that follow, all detections for individuals with spatial recaptures are plotted on the same map. Primary sampling occasions are shown in different colors, while secondary sampling occasions are plotted with different symbols. In cases where the same individual is detected at multiple hair snares *in the same week*, points are connected with a dotted line (colored to indicate the year of the movement). Each plot also gives the total number of detections for a bear, and the number of hair snares J, weeks K, and years T where the individual was detected.

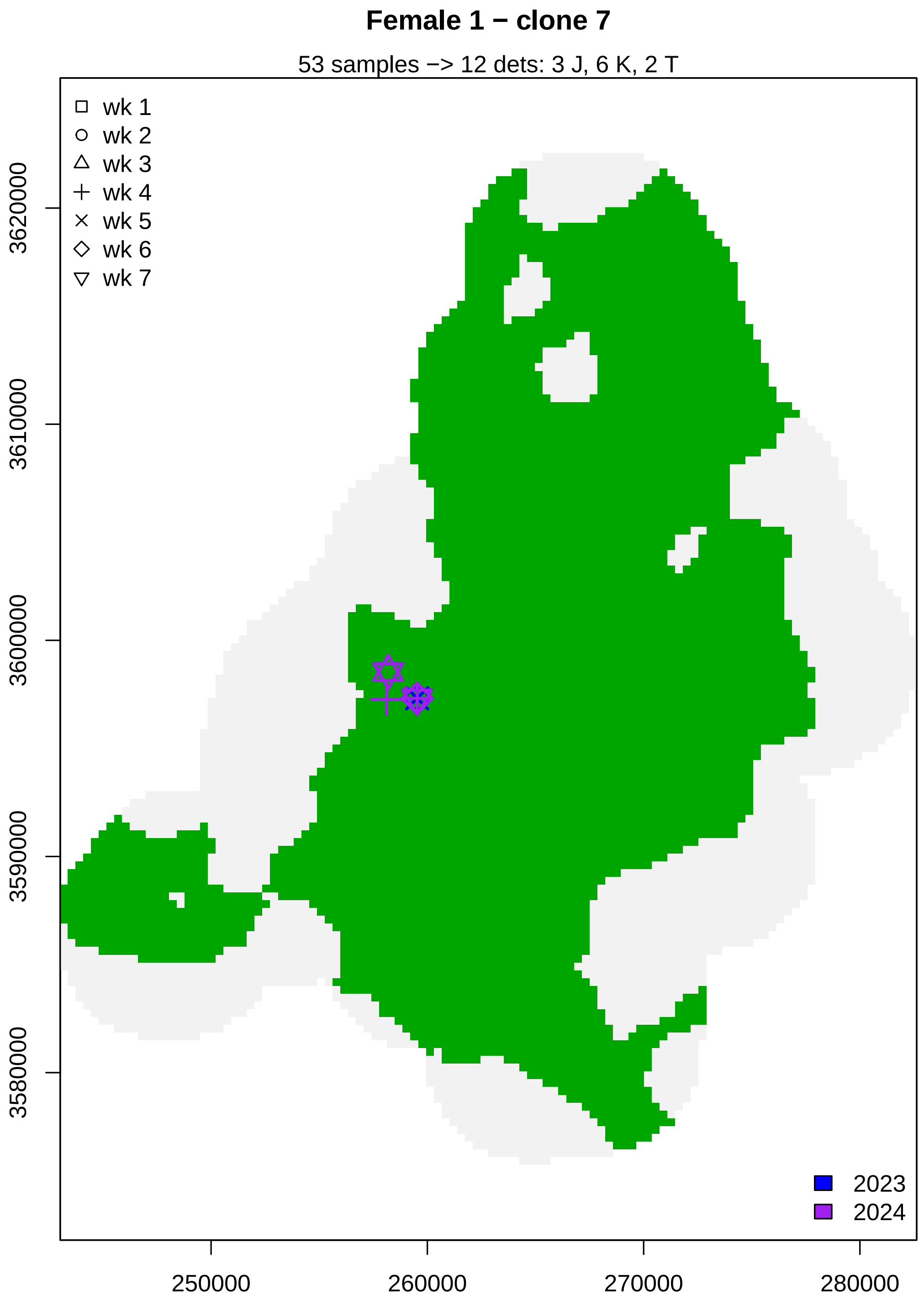
First, for females…

```r
#Females
caphist <- Fch$y
Fclones <- Fch$clone

sumstats_4d(caphist)
```

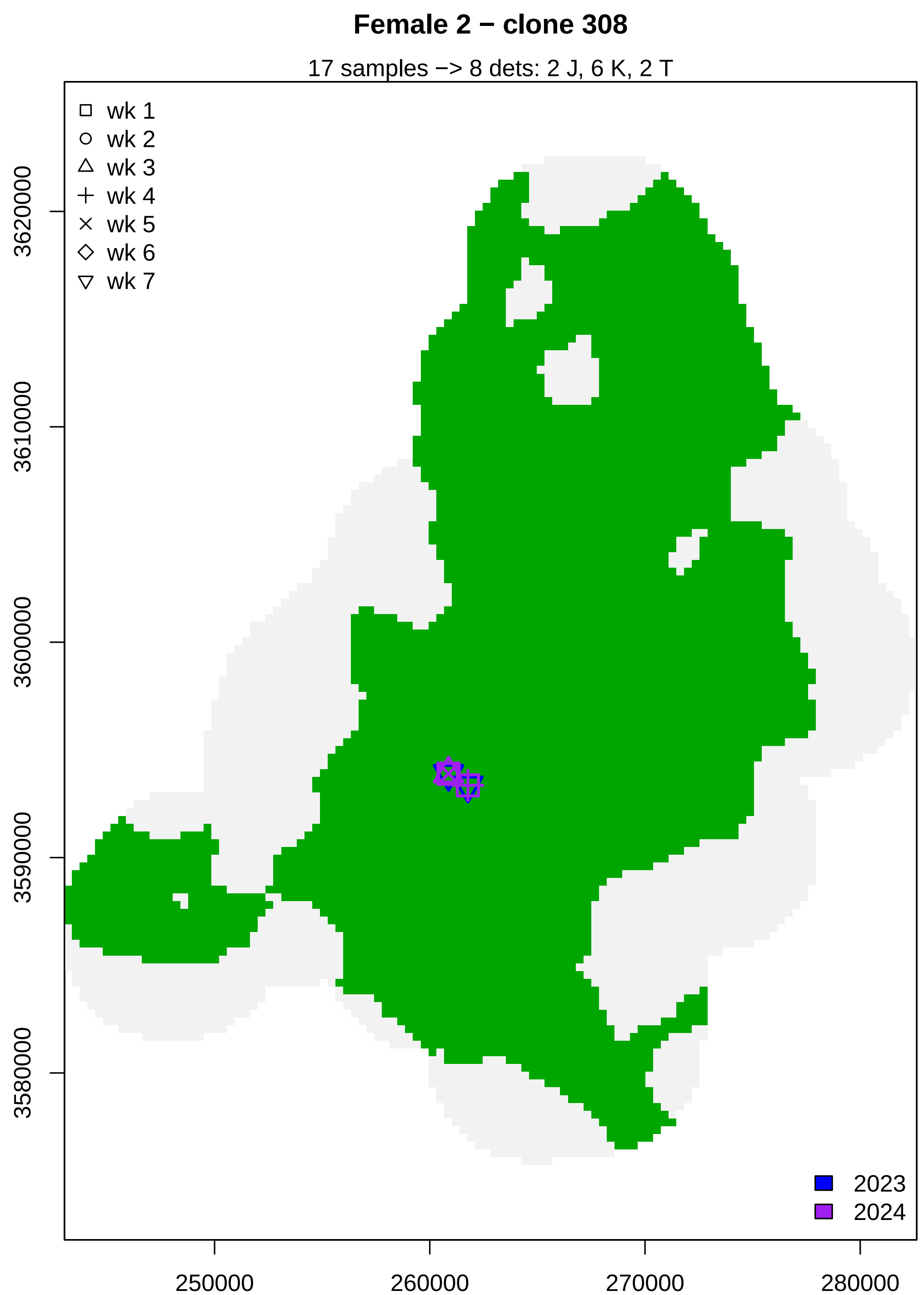
```
##       ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##     81.000000     252.000000       3.111111      31.000000       1.641975
##     ss.avgyrs
##      1.308642
```

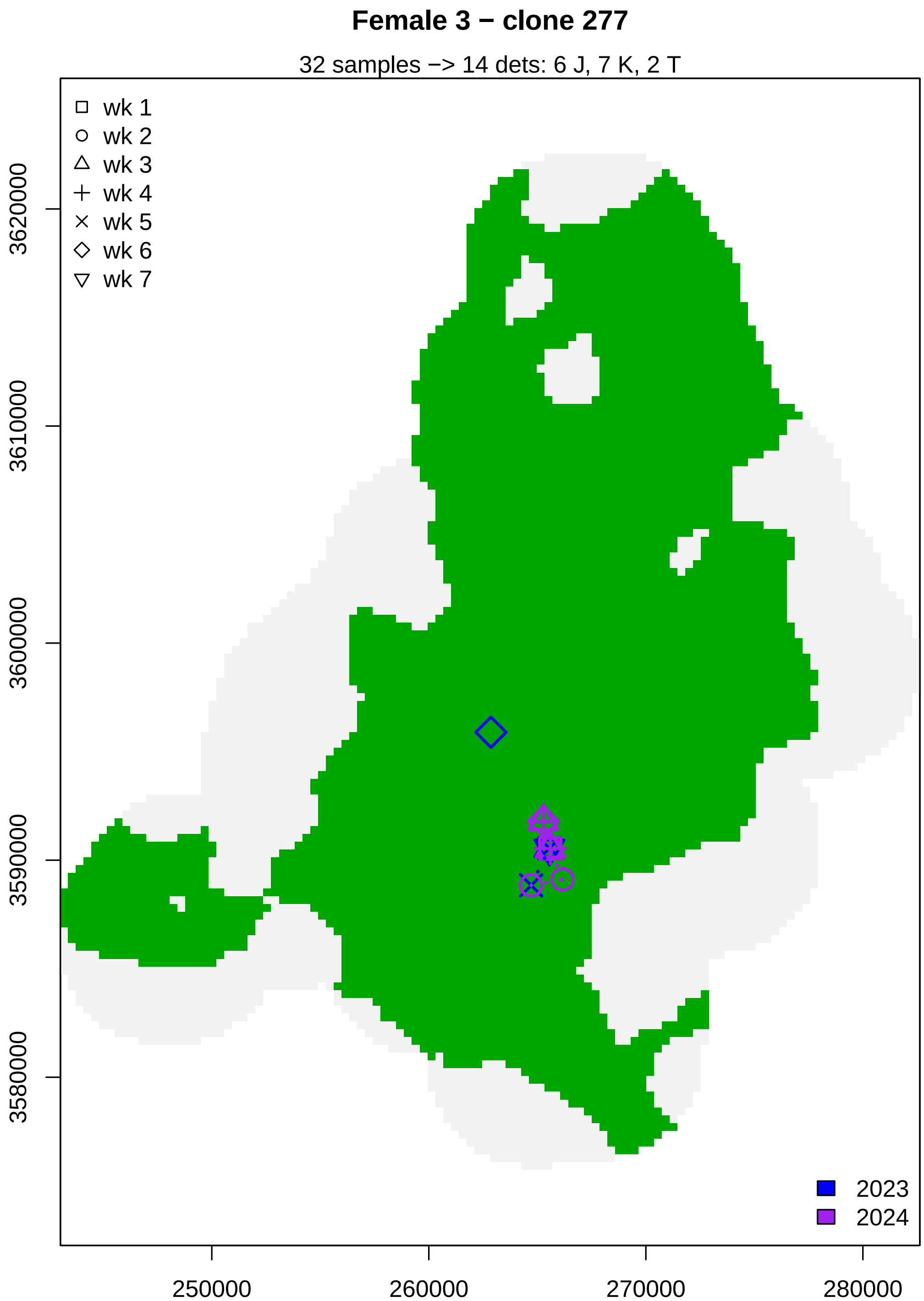
```r
nind <- dim(caphist)[1]

yr.col <- c("blue","purple")
wk.symbol <- 0:6

for(i in 1:nind) {
  i.ch <- caphist[i,,,]

  nTraps <- sum(rowSums(i.ch) > 0)
  if(nTraps > 1) {
    plot.me <- as.data.frame(which(i.ch == 1, arr.ind = TRUE))
    plot(S360, legend = FALSE, main = paste("Female",i,"- clone", Fclones[i]))
    mtext(paste(length(which(smd$finalClone == Fclones[i])), "samples ->",
                nrow(plot.me), "dets:",
```

```
              nTraps, "J,",
              length(unique(plot.me$dim2)), "K,",
              length(unique(plot.me$dim3)), "T"))
    points(x=locs[plot.me$dim1,1], locs[plot.me$dim1,2],
           pch = wk.symbol[plot.me$dim2], col = yr.col[plot.me$dim3],
           cex = 2, lwd = 2)

    for(r in 2:nrow(plot.me)) {
      if(plot.me$dim3[r]==plot.me$dim3[r-1] &
         plot.me$dim2[r]==plot.me$dim2[r-1]) {
        lines(x=c(locs[plot.me$dim1[r-1],1], locs[plot.me$dim1[r],1]),
              y=c(locs[plot.me$dim1[r-1],2], locs[plot.me$dim1[r],2]),
              lty="dotted", col = yr.col[plot.me$dim3[r]],
              lwd = 2.5)
      }
    }

    legend("topleft", bty = "n", pch = 0:6, legend = paste("wk",c(1:7)))
    legend("bottomright", bty = "n", fill = yr.col, legend = c("2023","2024"))

  }
}
```

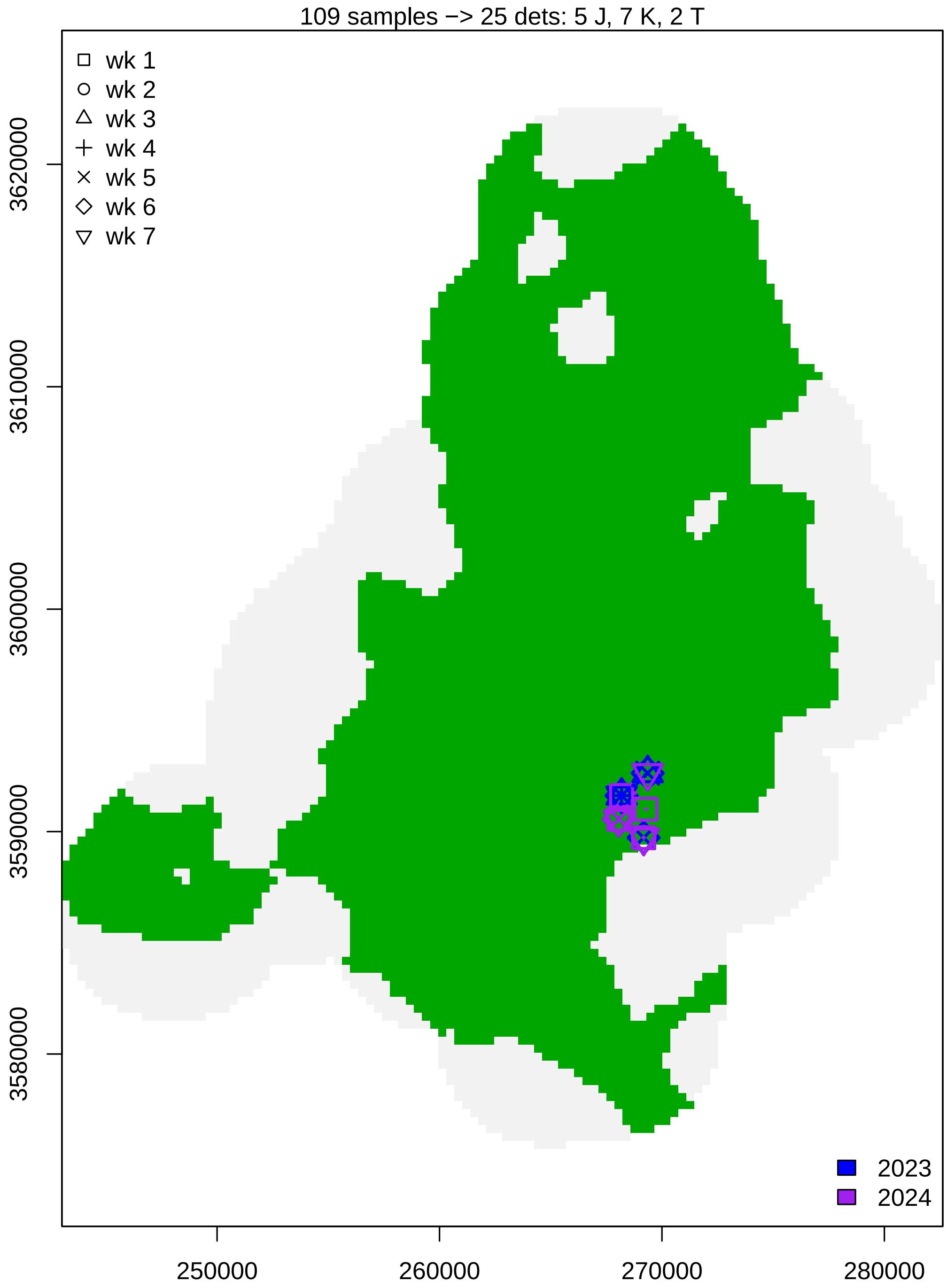


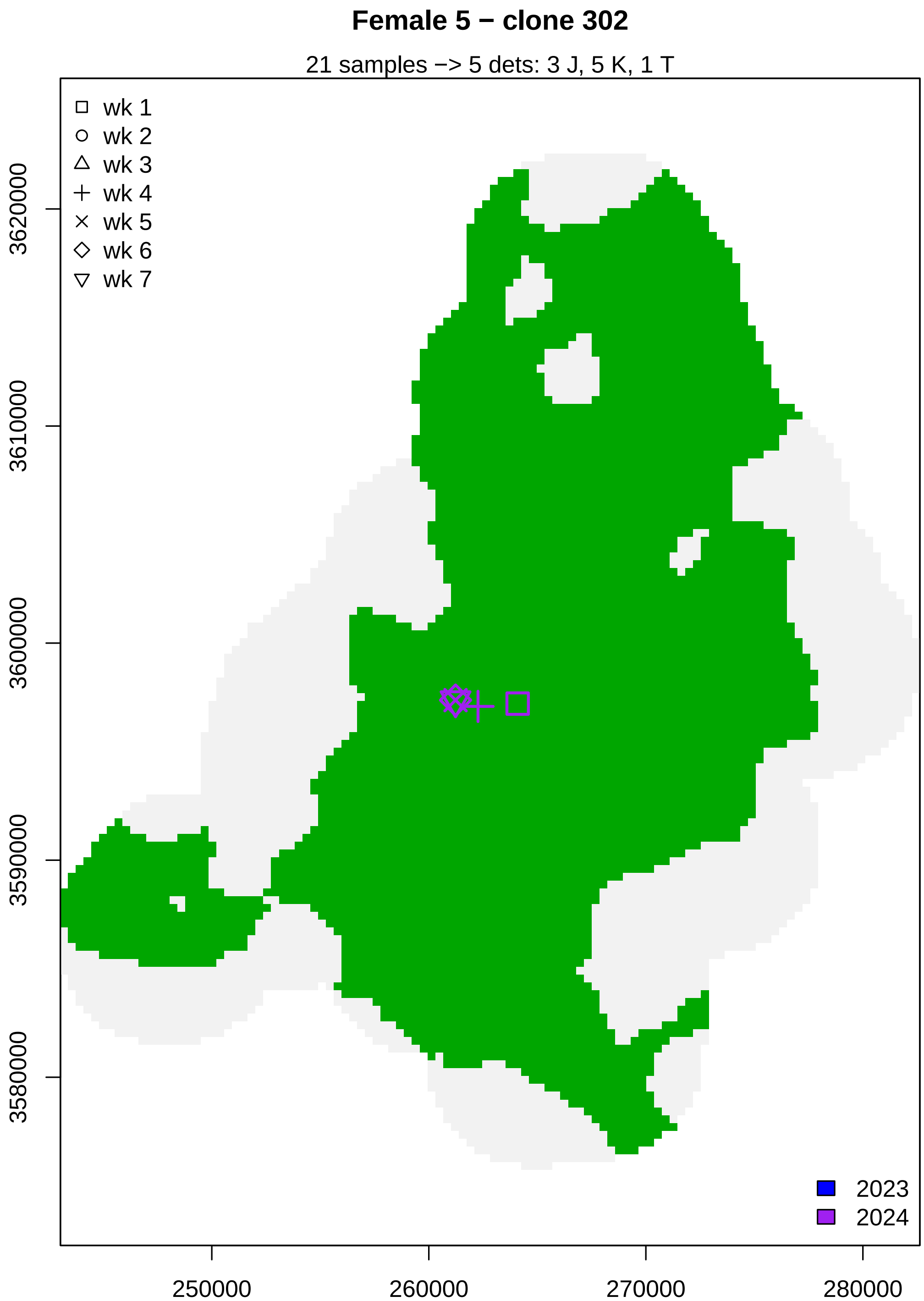
Female 1 − clone 7
53 samples −> 12 dets: 3 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

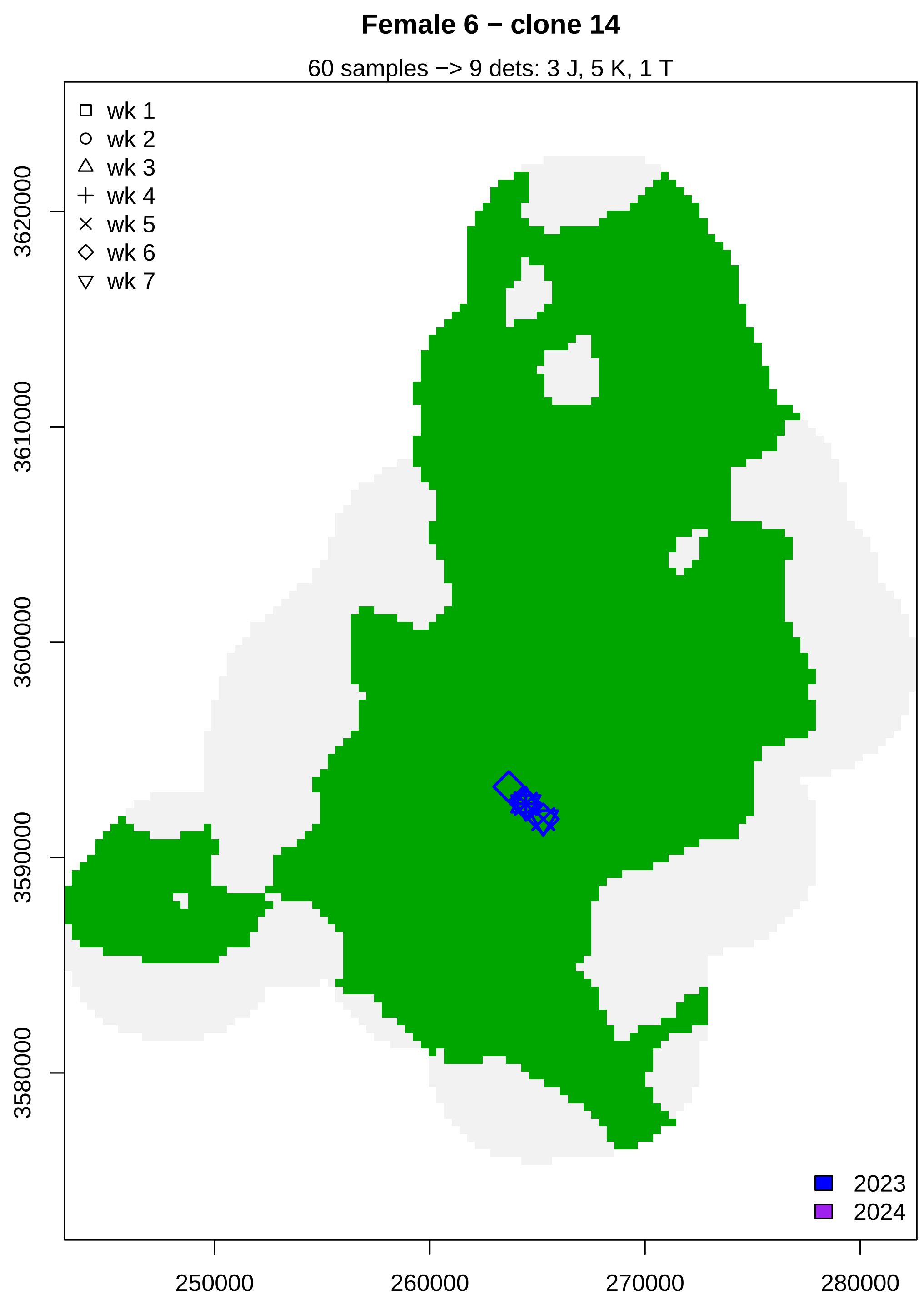

Female 2 − clone 308
17 samples −> 8 dets: 2 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000


Female 3 − clone 277
32 samples −> 14 dets: 6 J, 7 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

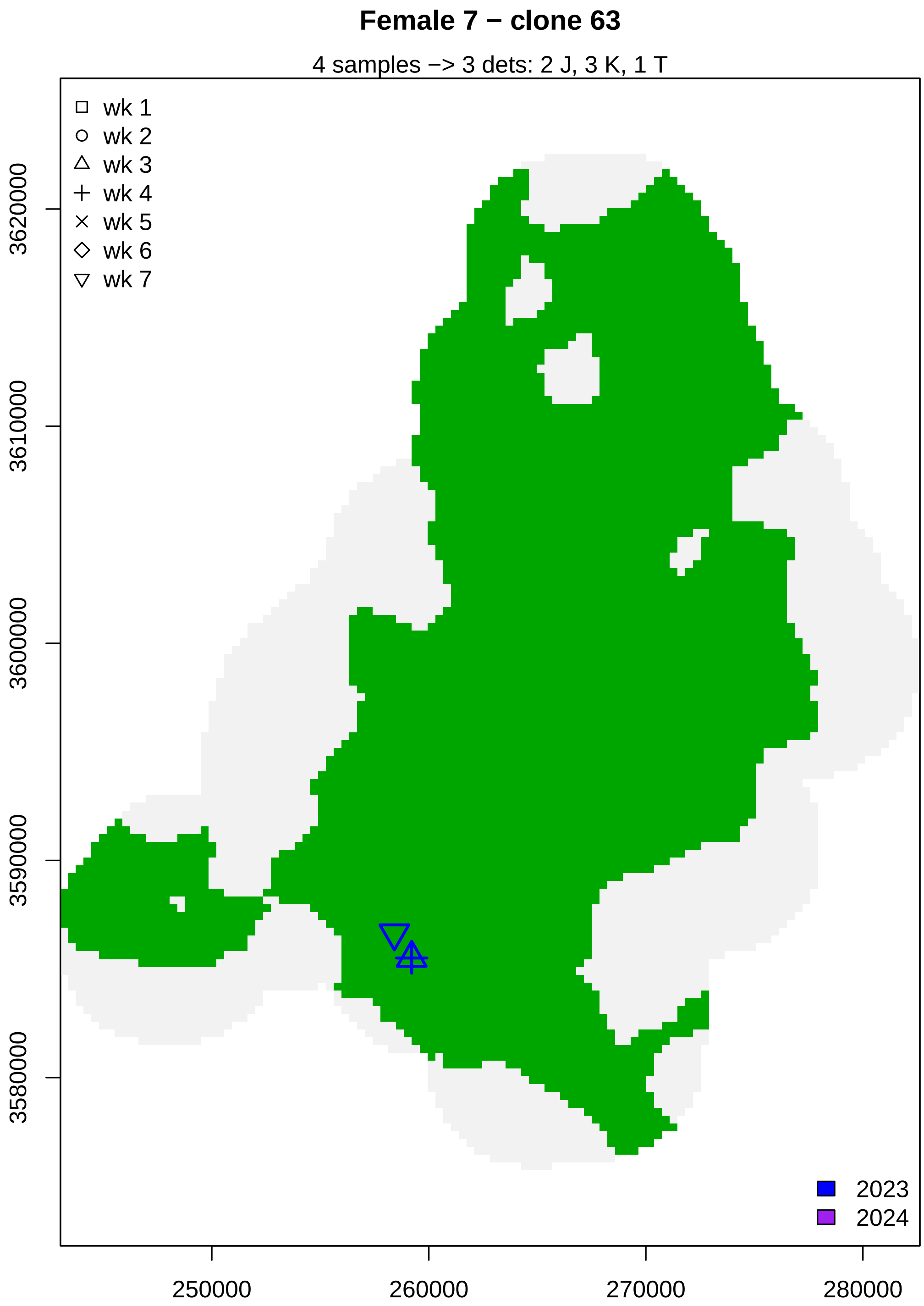
Female 4 − clone 4

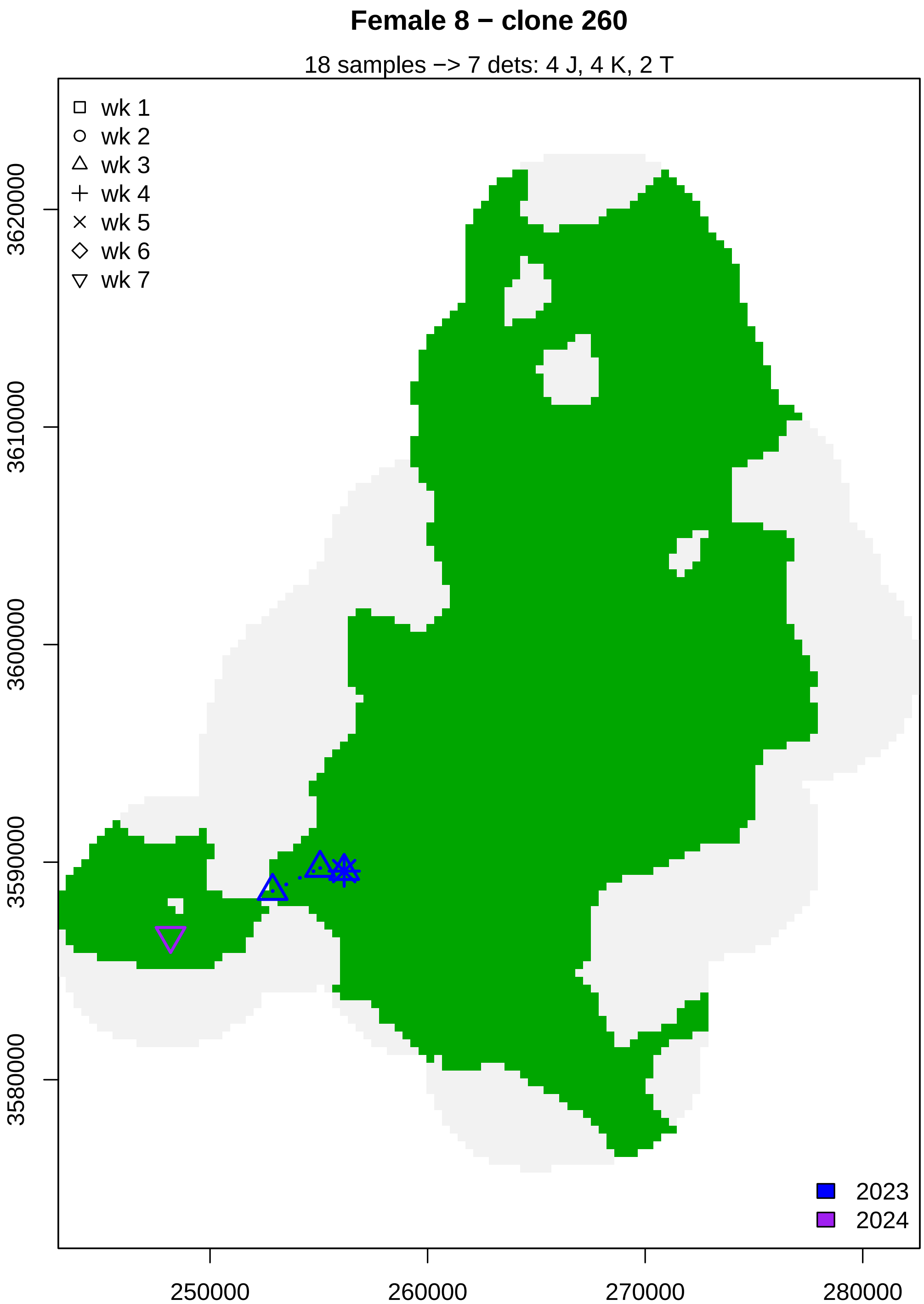


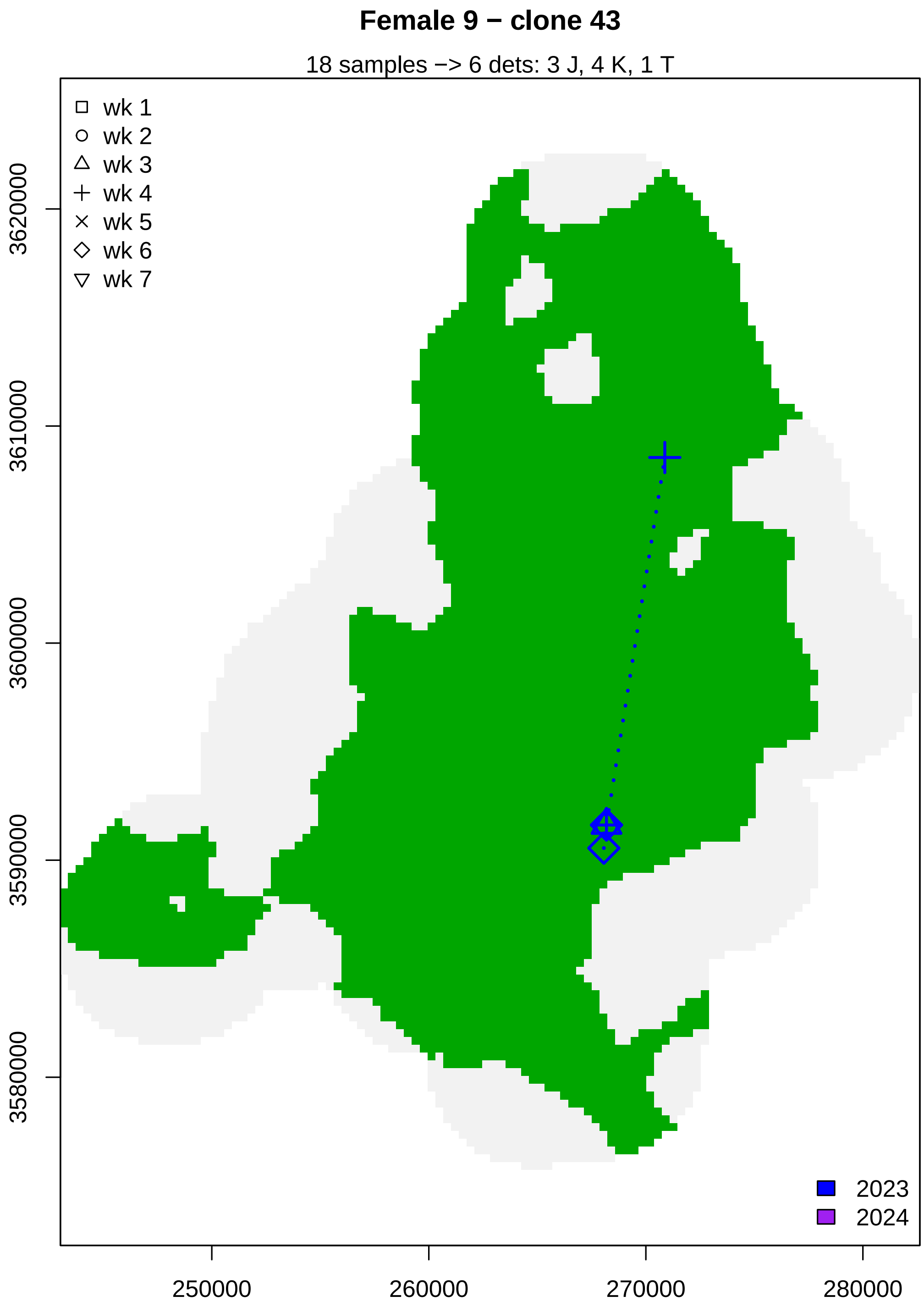

Female 5 − clone 302
21 samples −> 5 dets: 3 J, 5 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

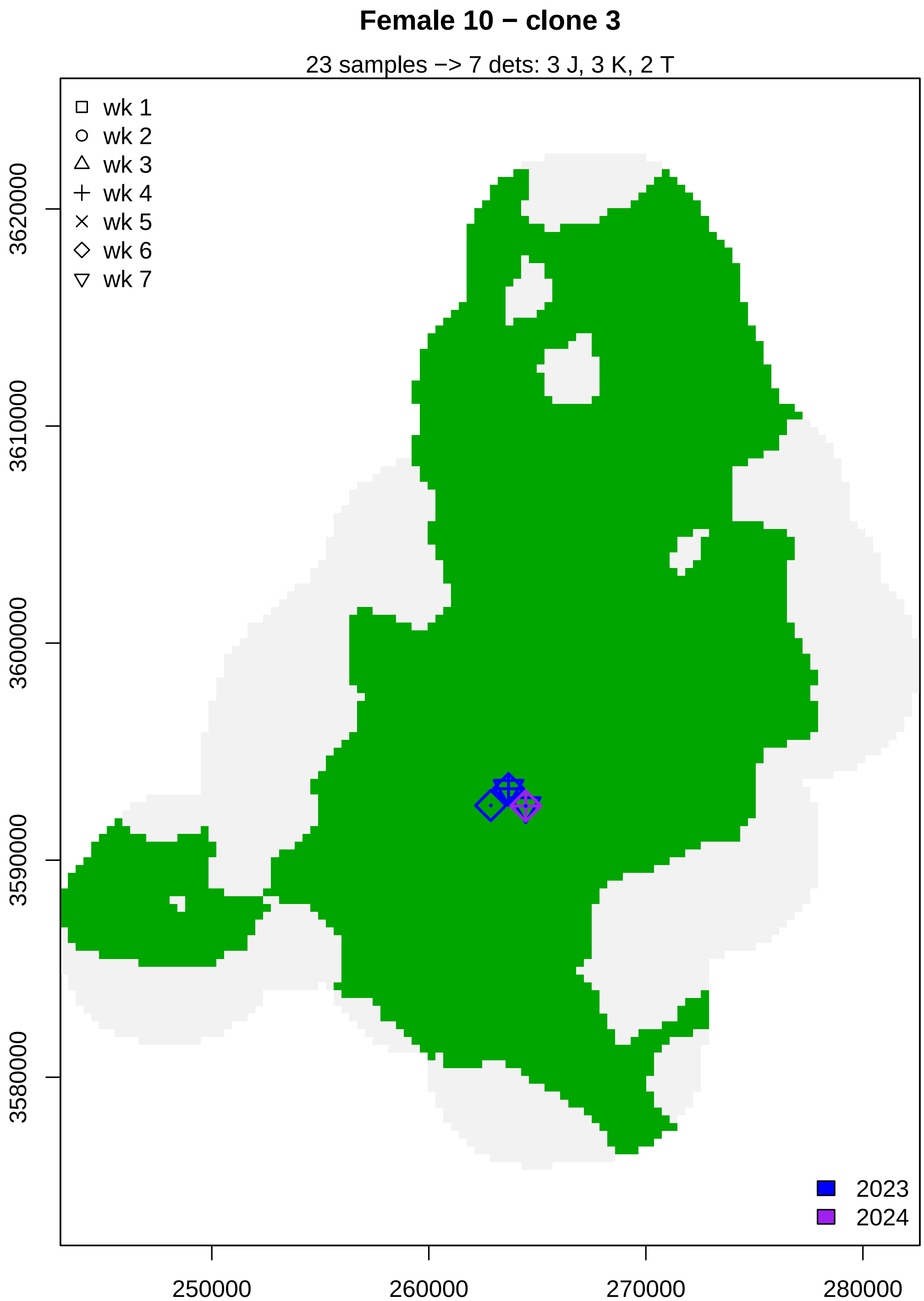


Female 6 − clone 14
60 samples −> 9 dets: 3 J, 5 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

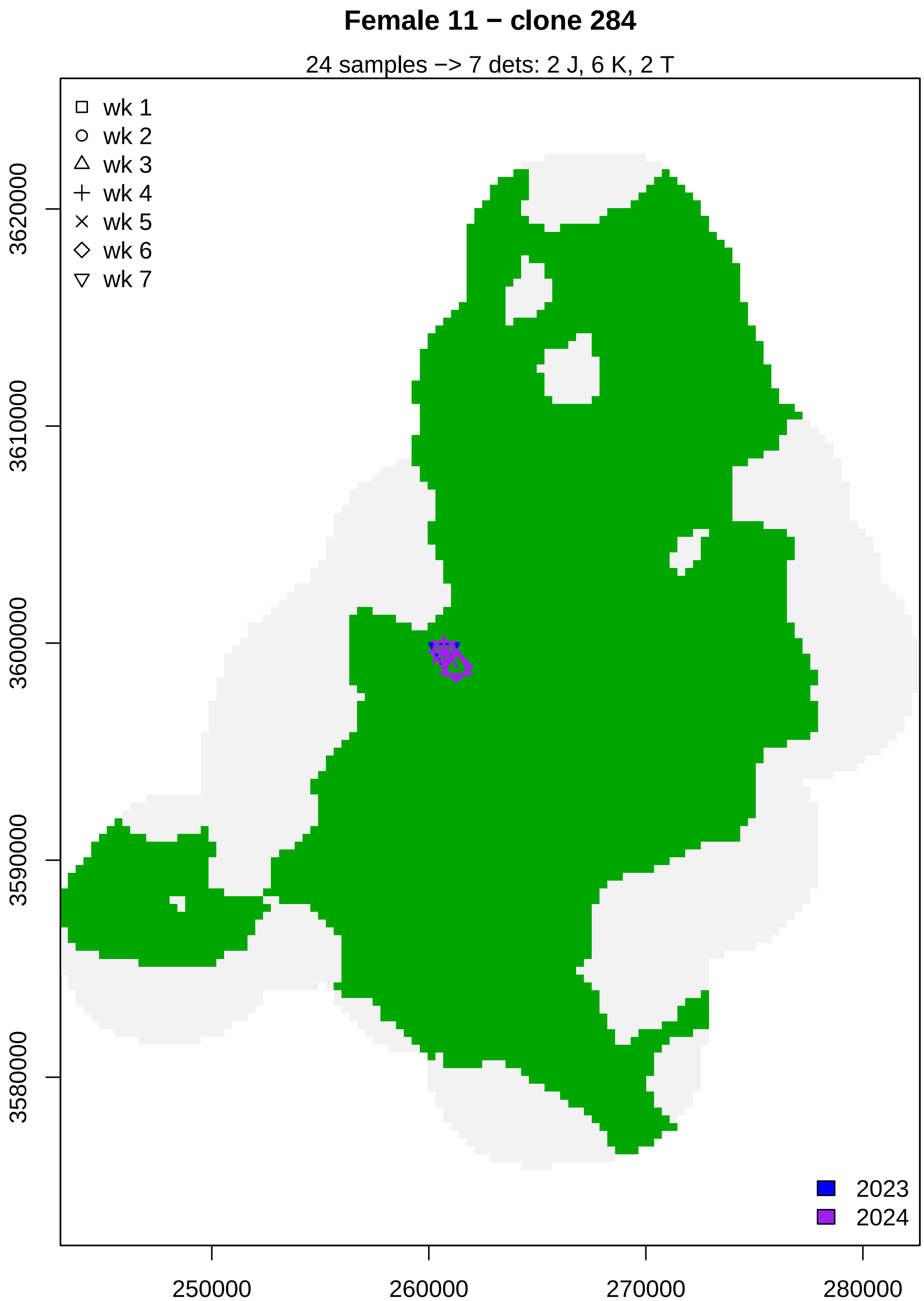

Female 7 − clone 63
4 samples −> 3 dets: 2 J, 3 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000


Female 8 − clone 260
18 samples −> 7 dets: 4 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

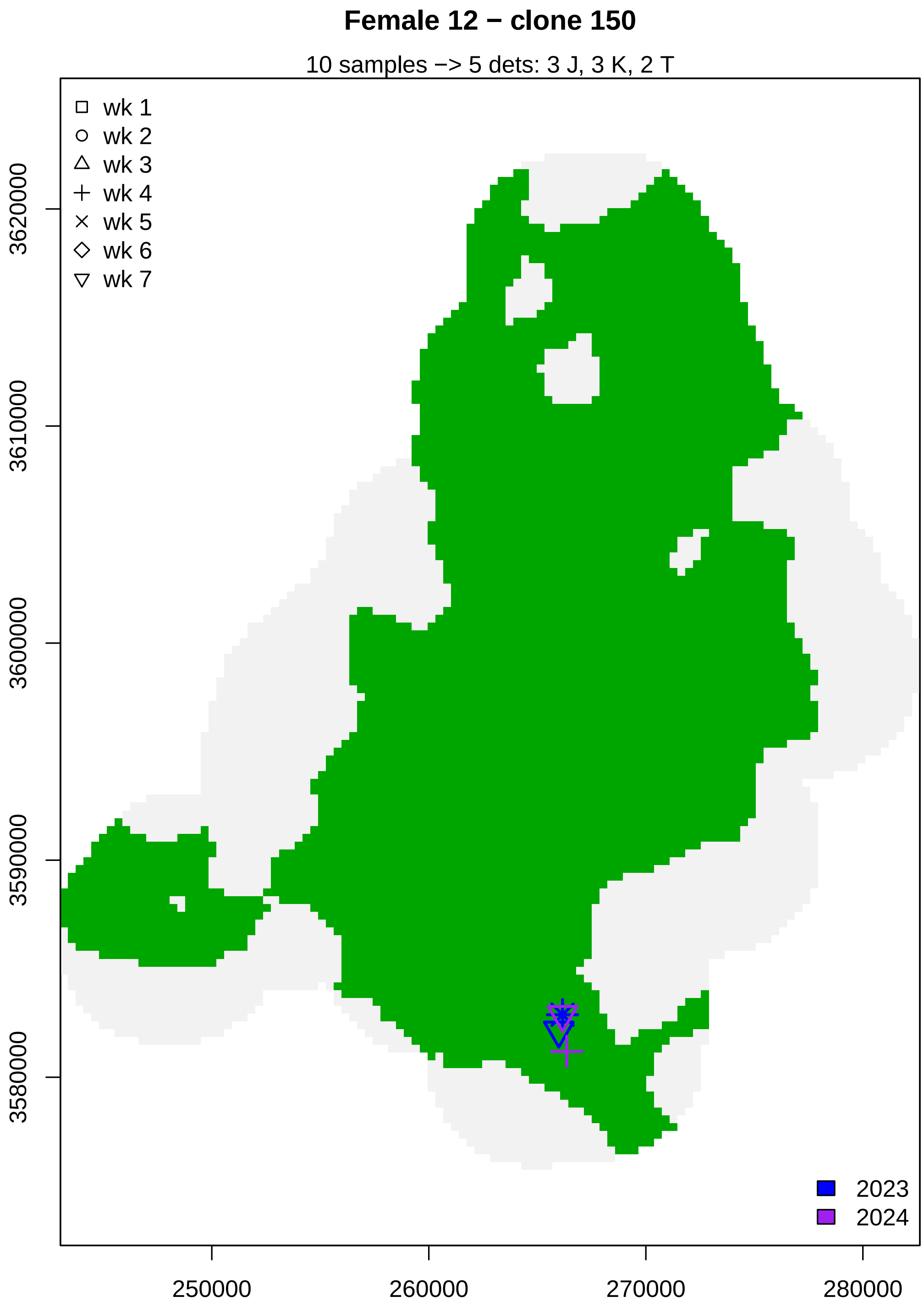

Female 9 − clone 43
18 samples −> 6 dets: 3 J, 4 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

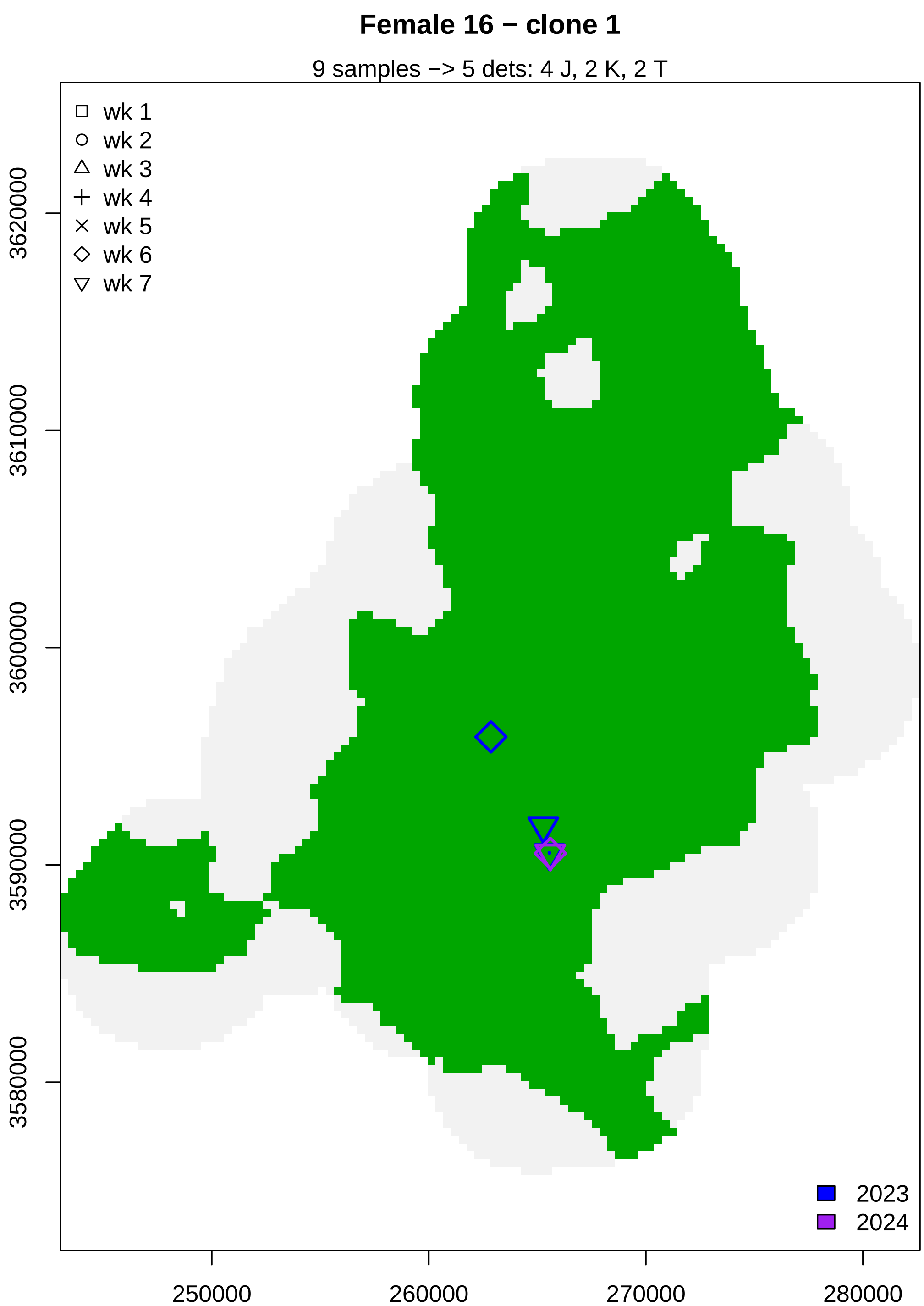


Female 10 − clone 3
23 samples −> 7 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

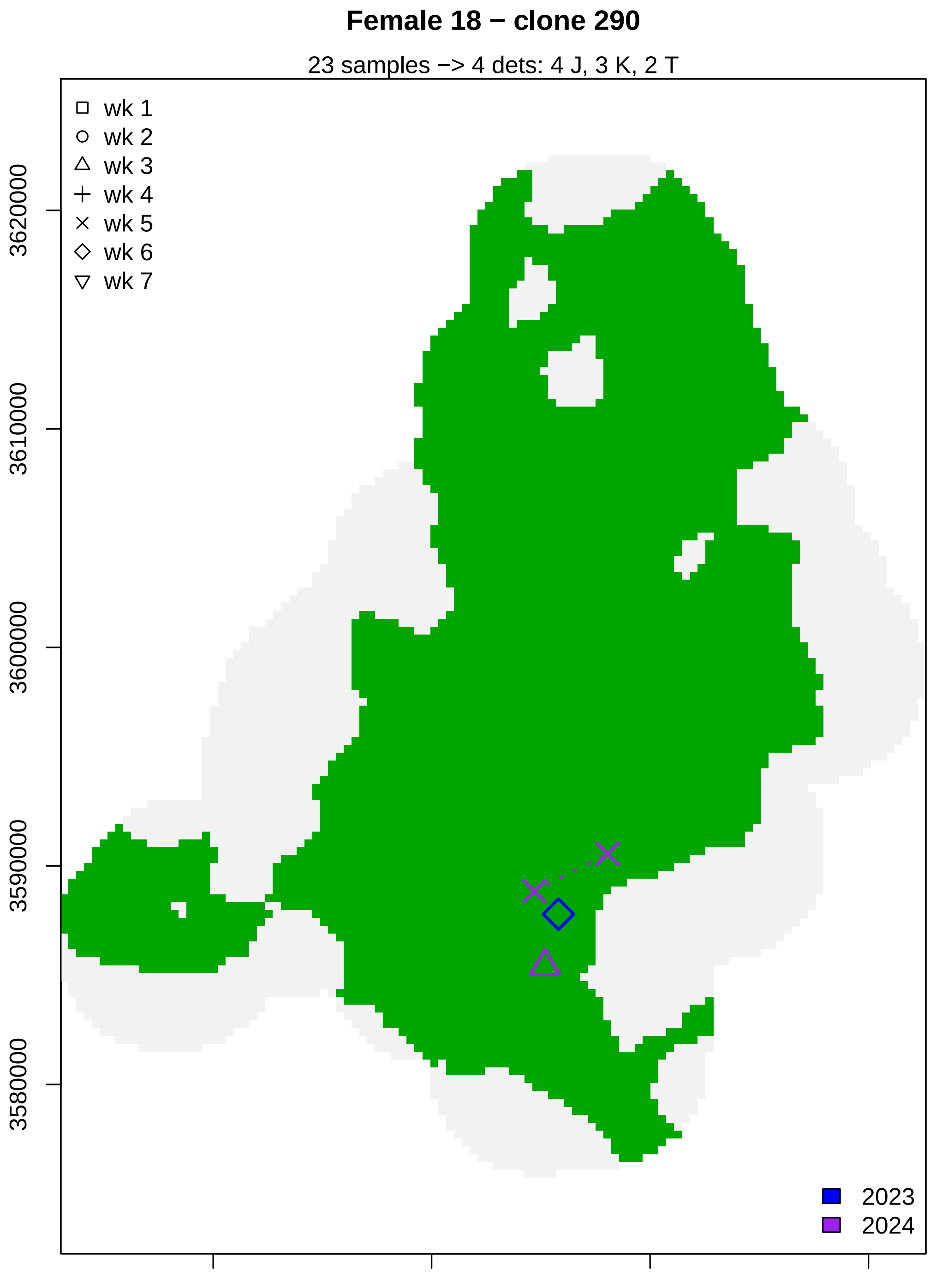


Female 11 − clone 284
24 samples −> 7 dets: 2 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

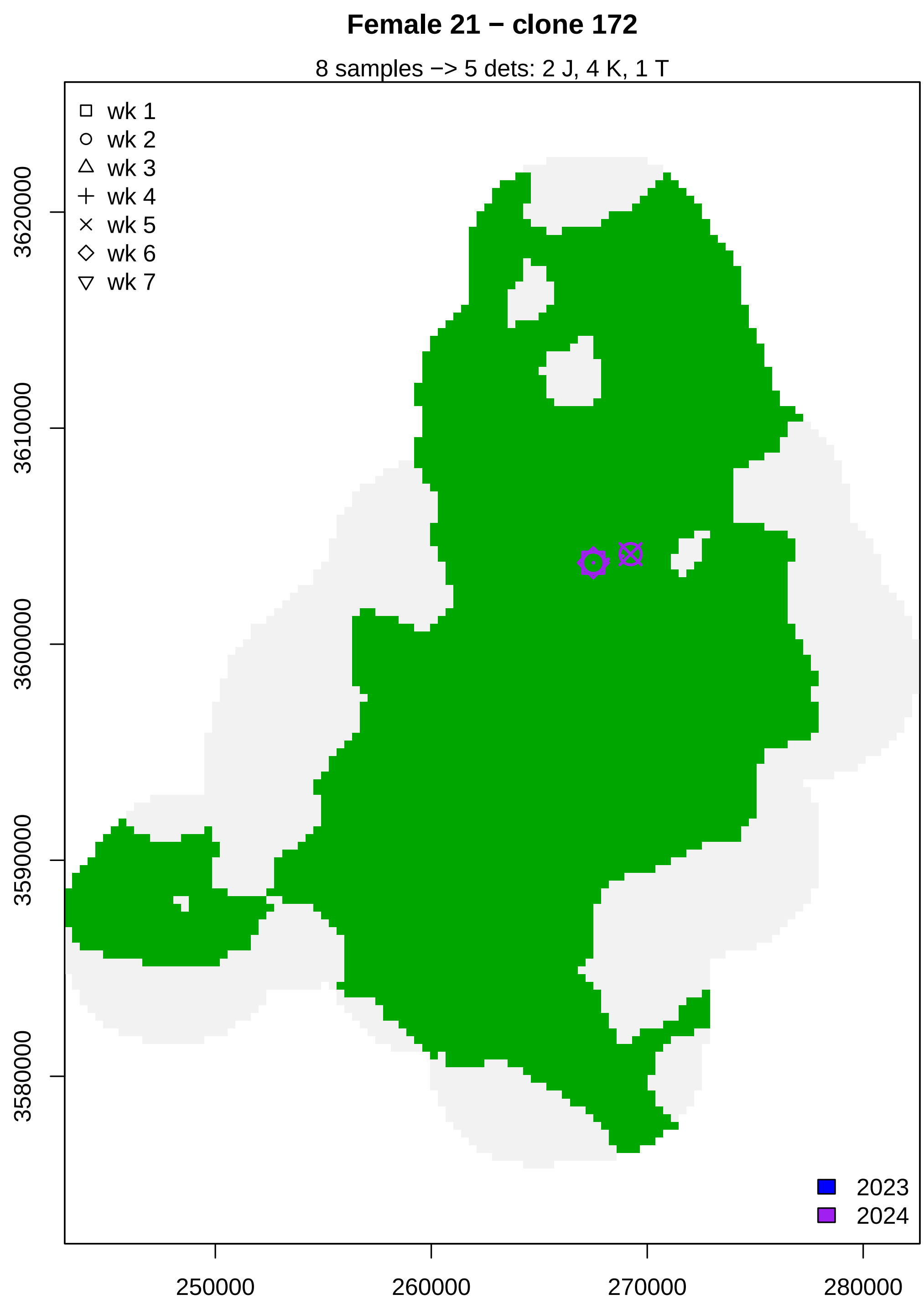

Female 12 − clone 150
10 samples −> 5 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



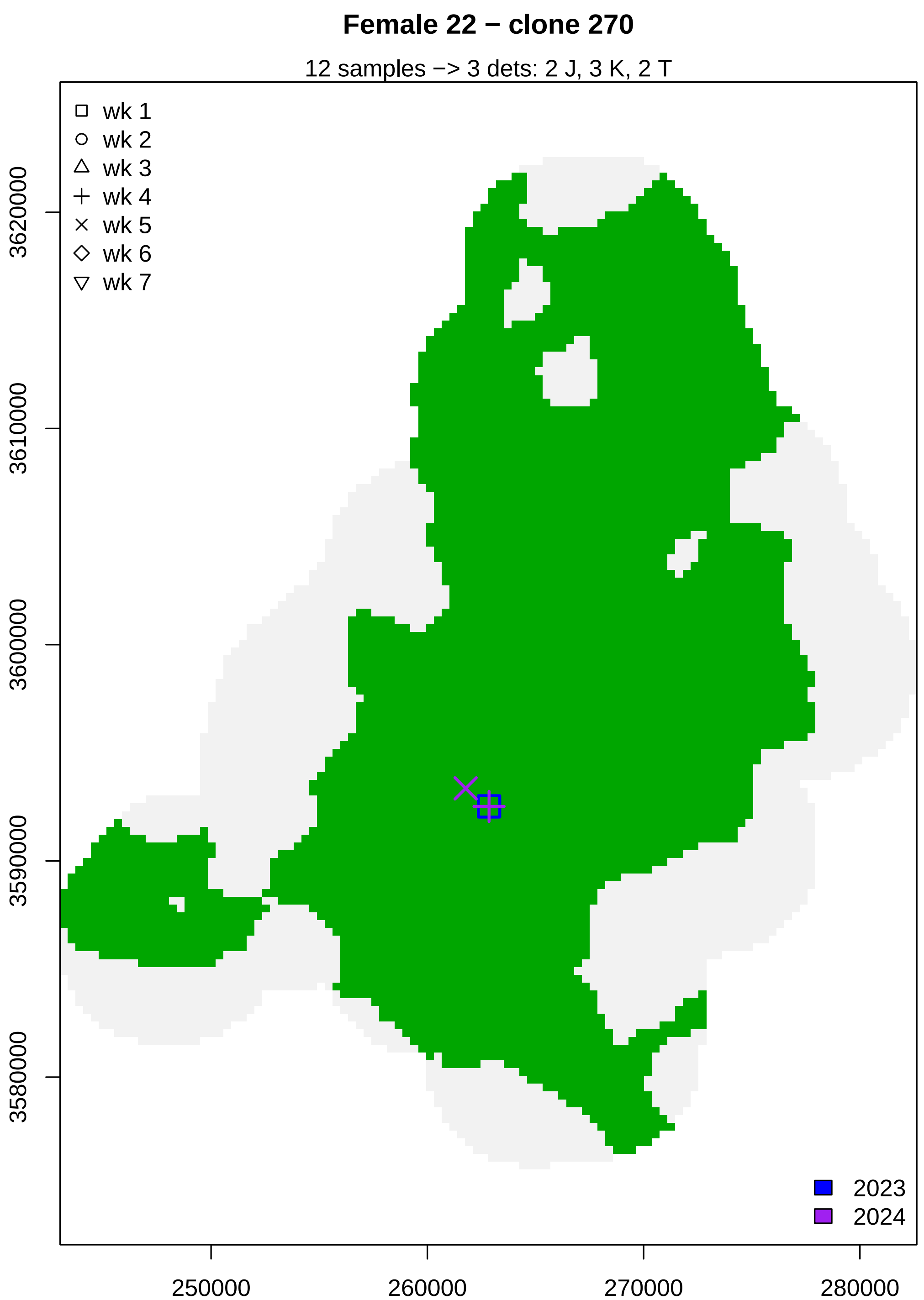
Female 16 − clone 1
9 samples −> 5 dets: 4 J, 2 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



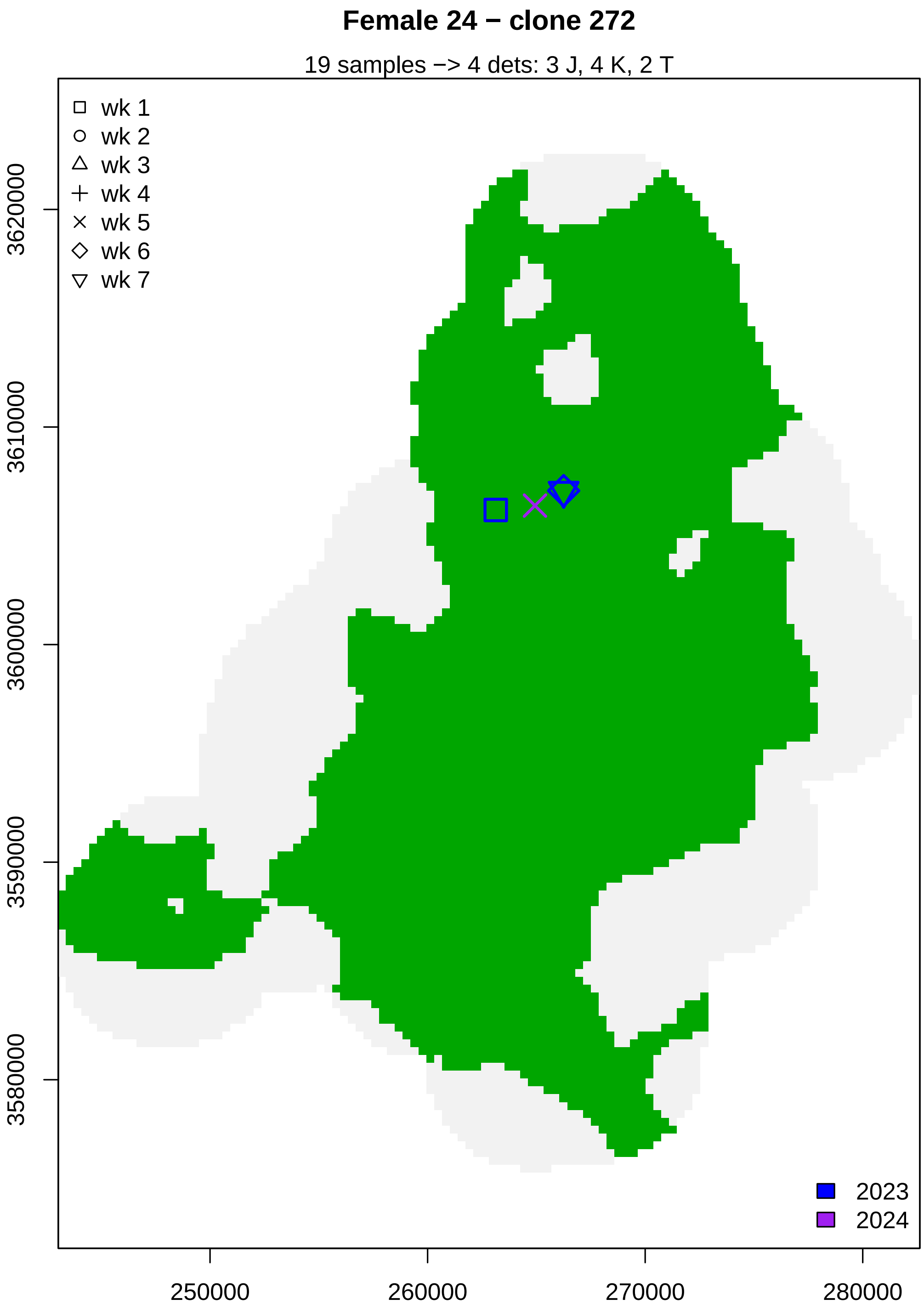
Female 18 − clone 290
23 samples −> 4 dets: 4 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

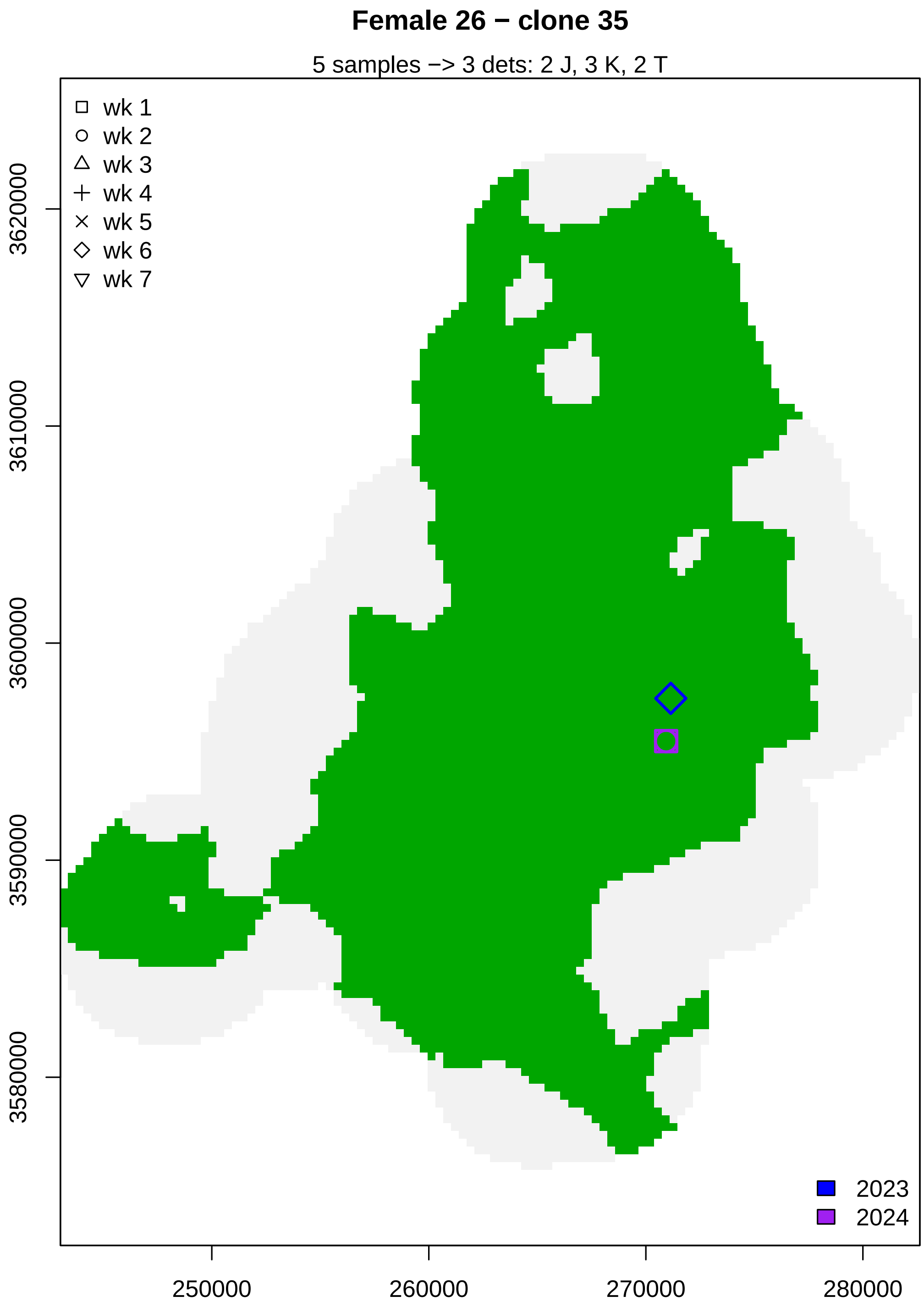


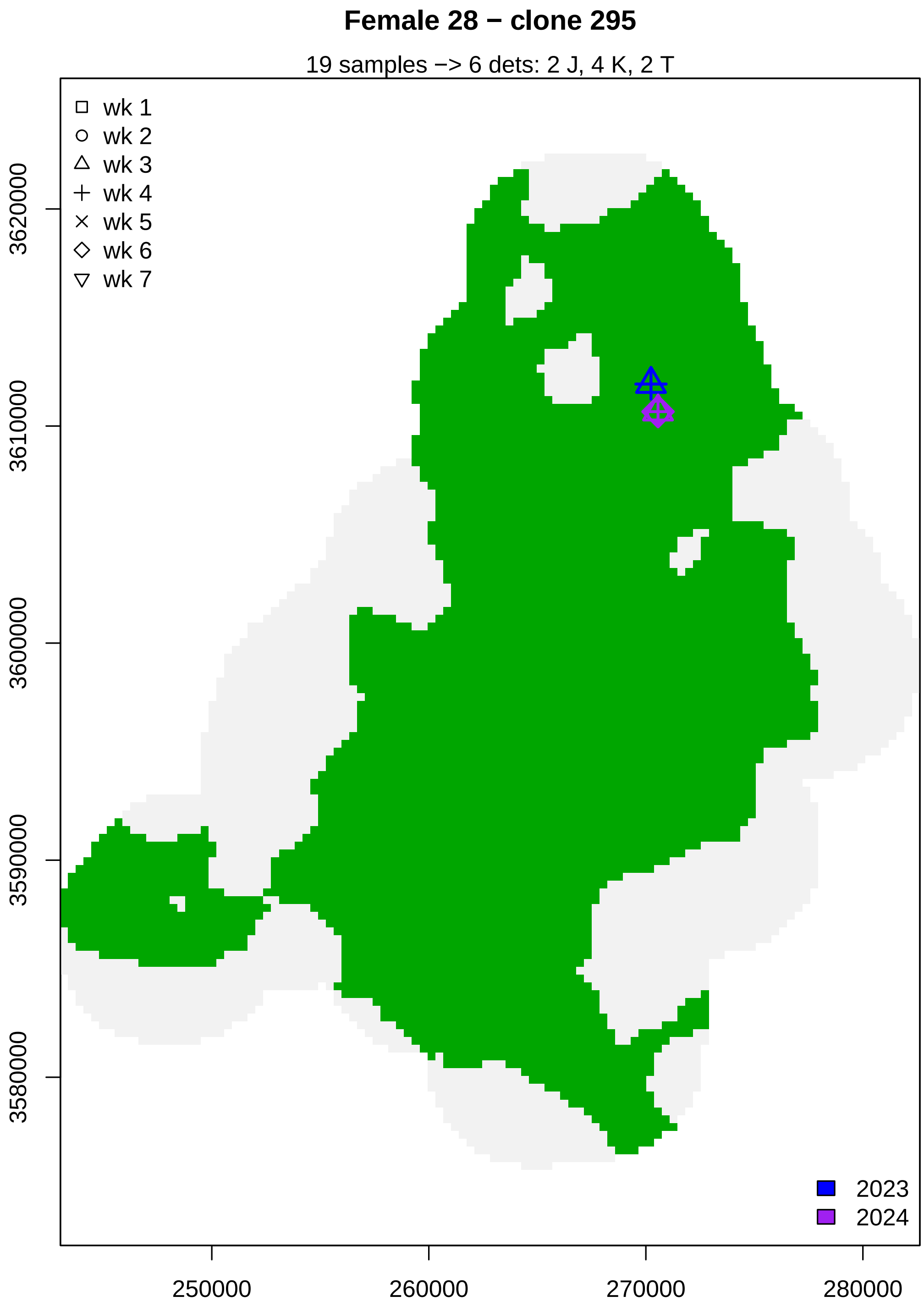
Female 21 − clone 172
8 samples −> 5 dets: 2 J, 4 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



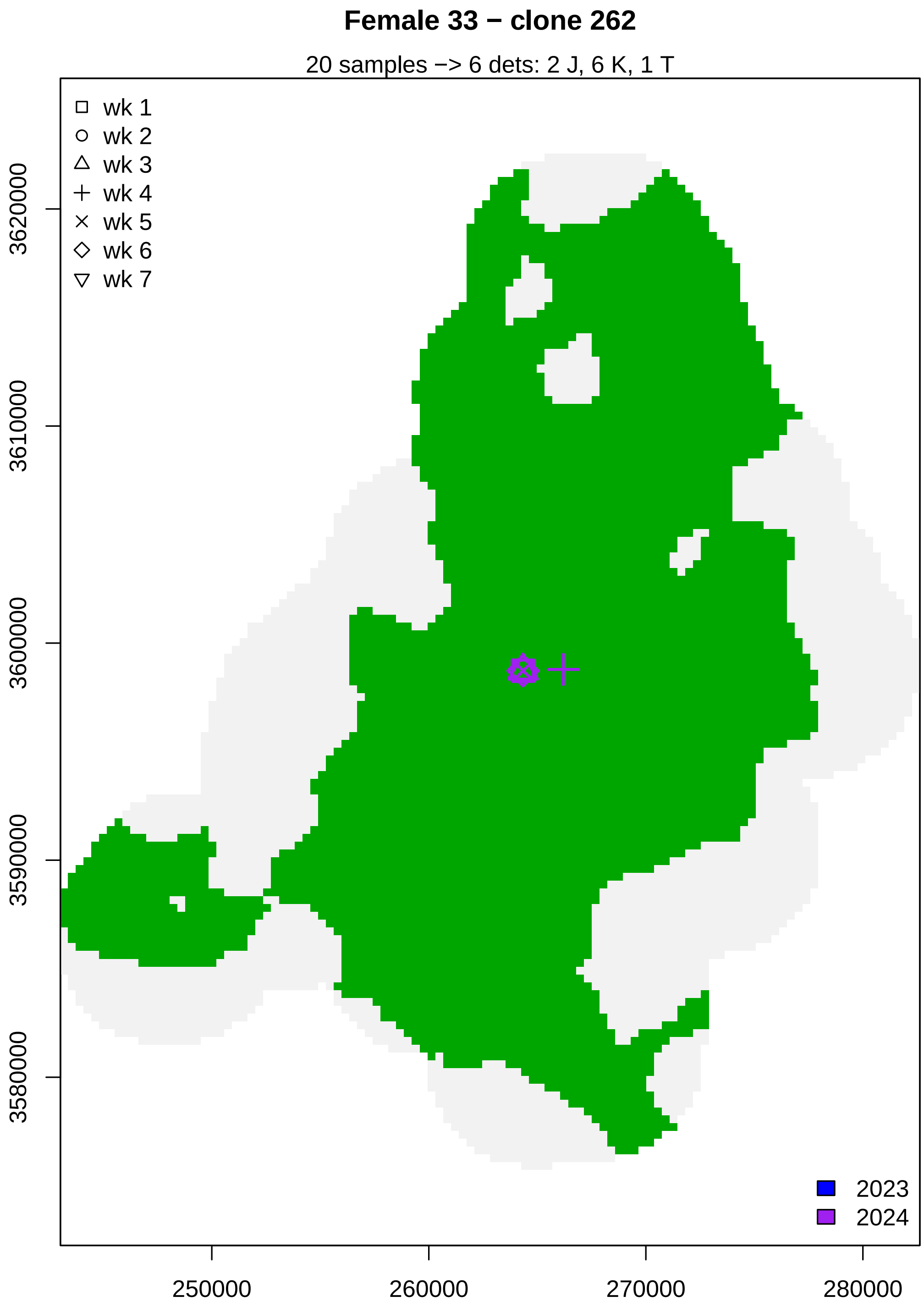
Female 22 − clone 270
12 samples −> 3 dets: 2 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

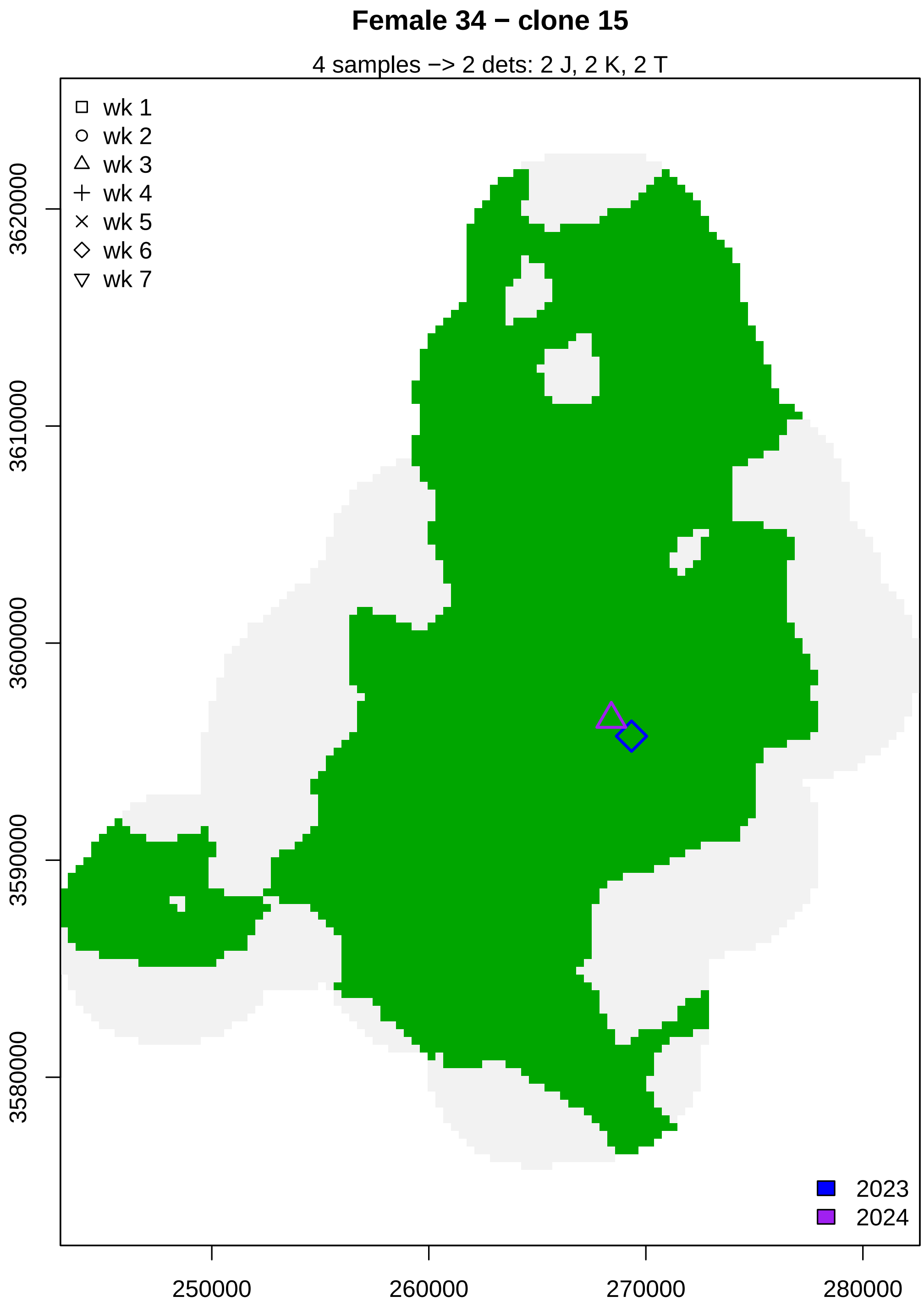

Female 24 − clone 272
19 samples −> 4 dets: 3 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024


Female 26 − clone 35
5 samples −> 3 dets: 2 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

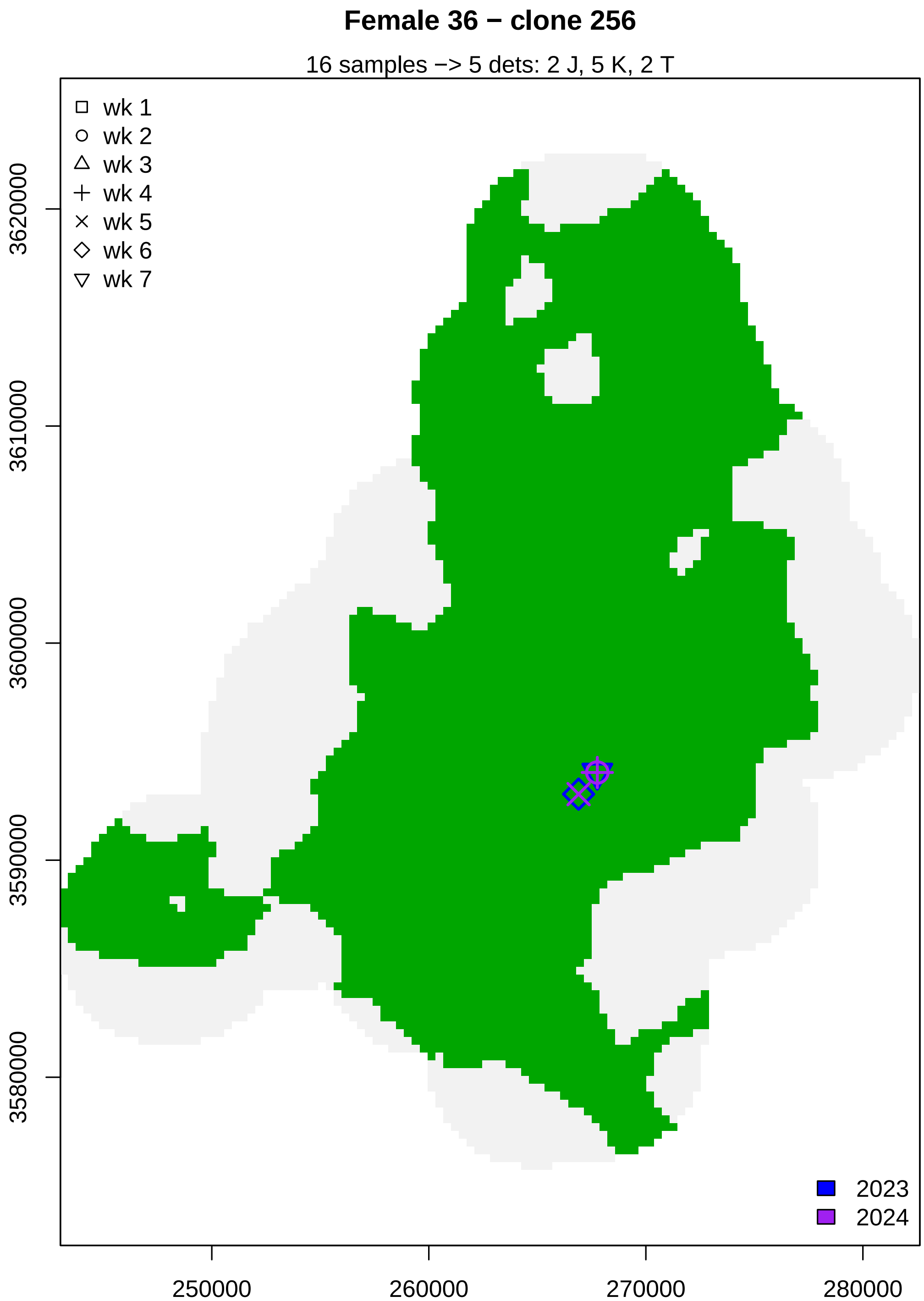


Female 28 − clone 295
19 samples −> 6 dets: 2 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

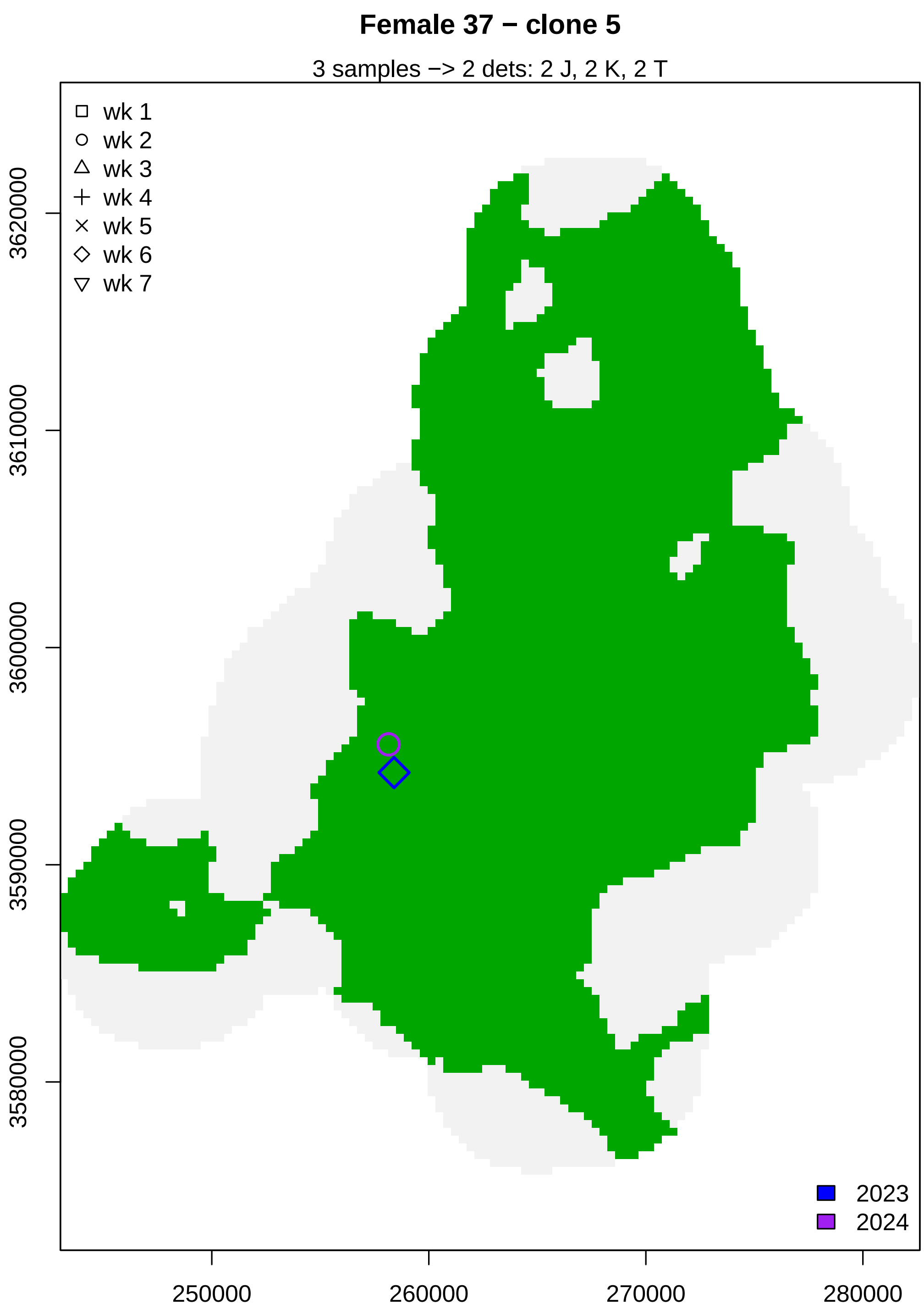

Female 33 − clone 262
20 samples −> 6 dets: 2 J, 6 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

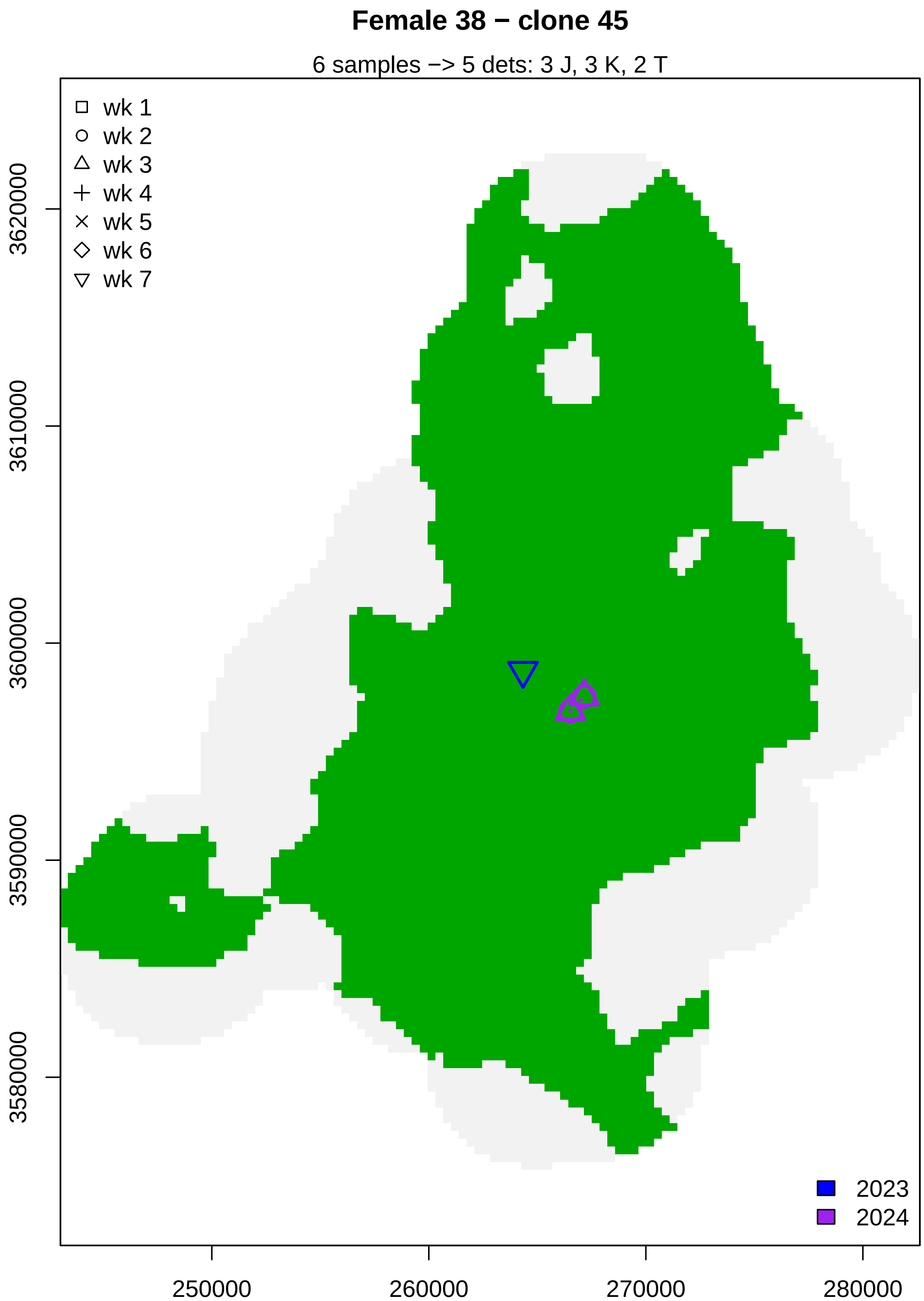


Female 34 − clone 15
4 samples −> 2 dets: 2 J, 2 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

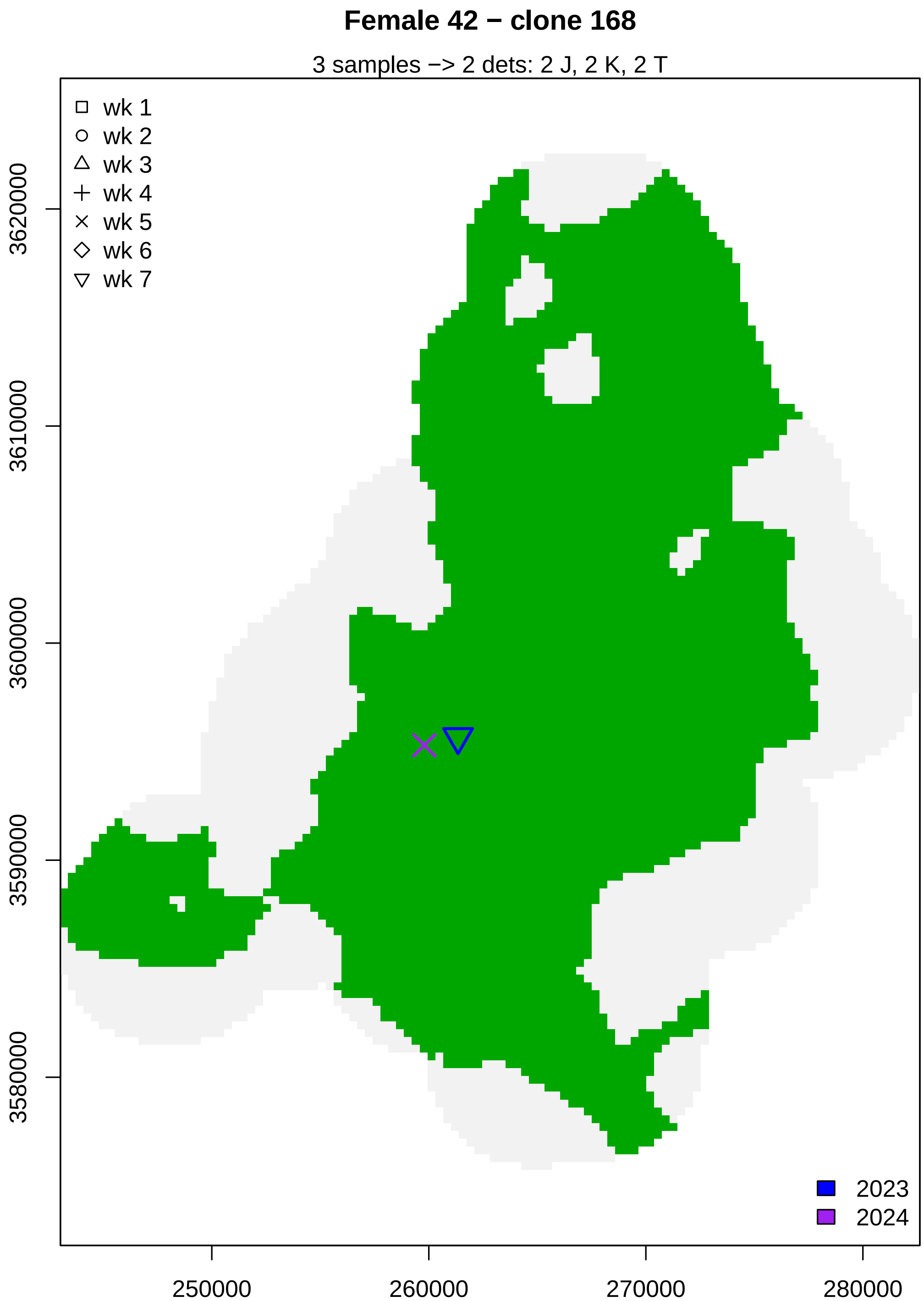


Female 36 − clone 256
16 samples −> 5 dets: 2 J, 5 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024


Female 37 − clone 5
3 samples −> 2 dets: 2 J, 2 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024



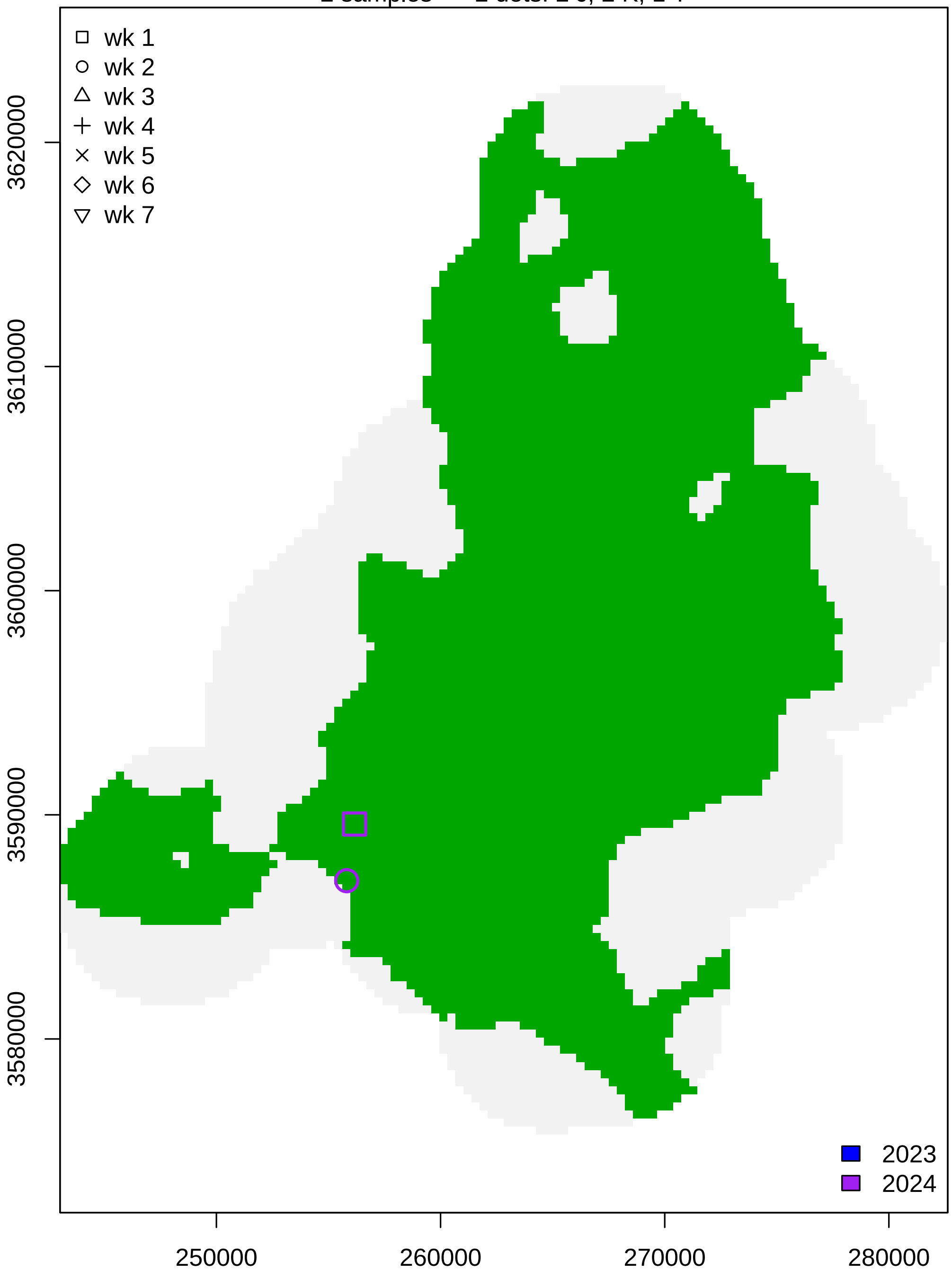
Female 38 − clone 45
6 samples −> 5 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

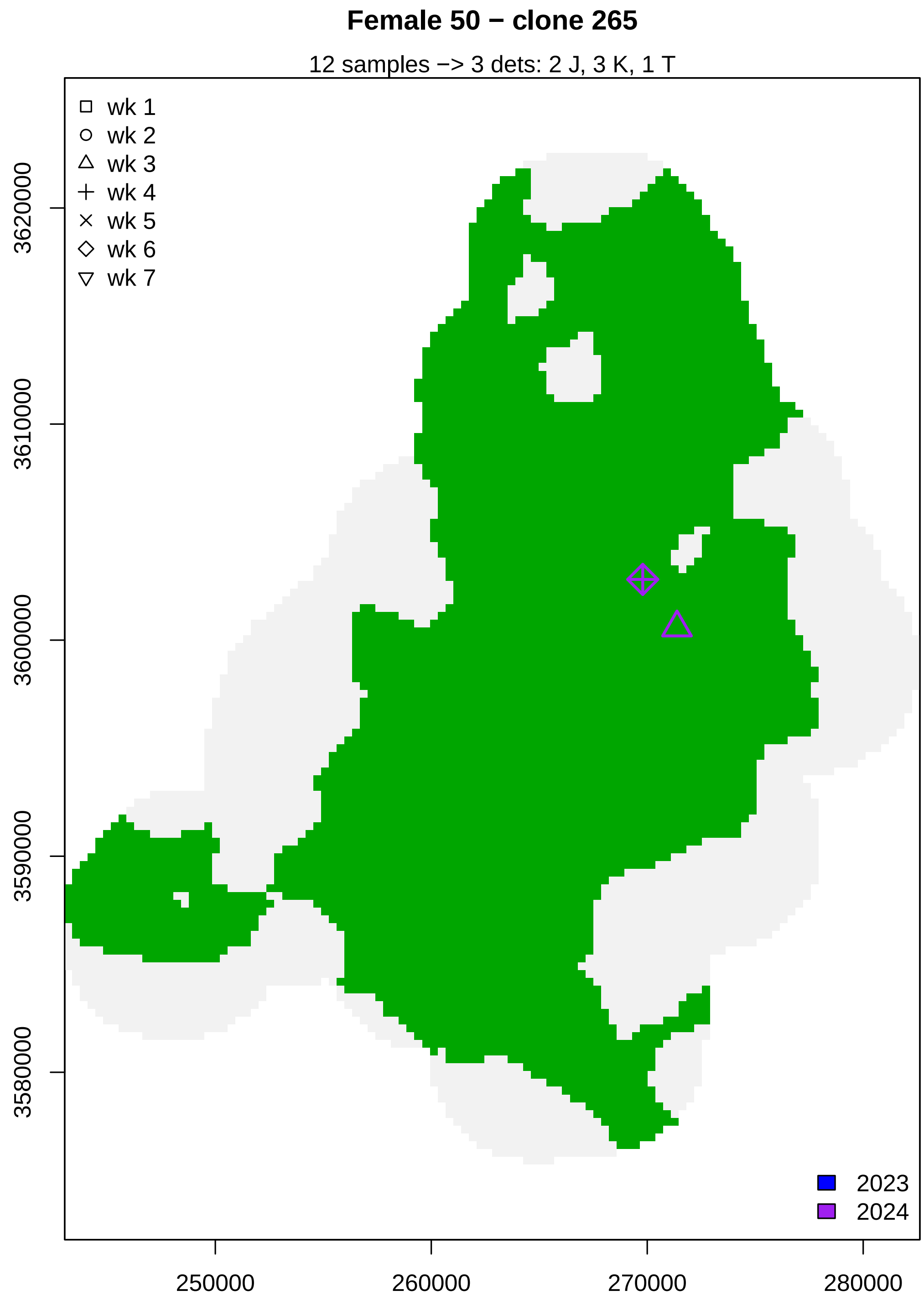


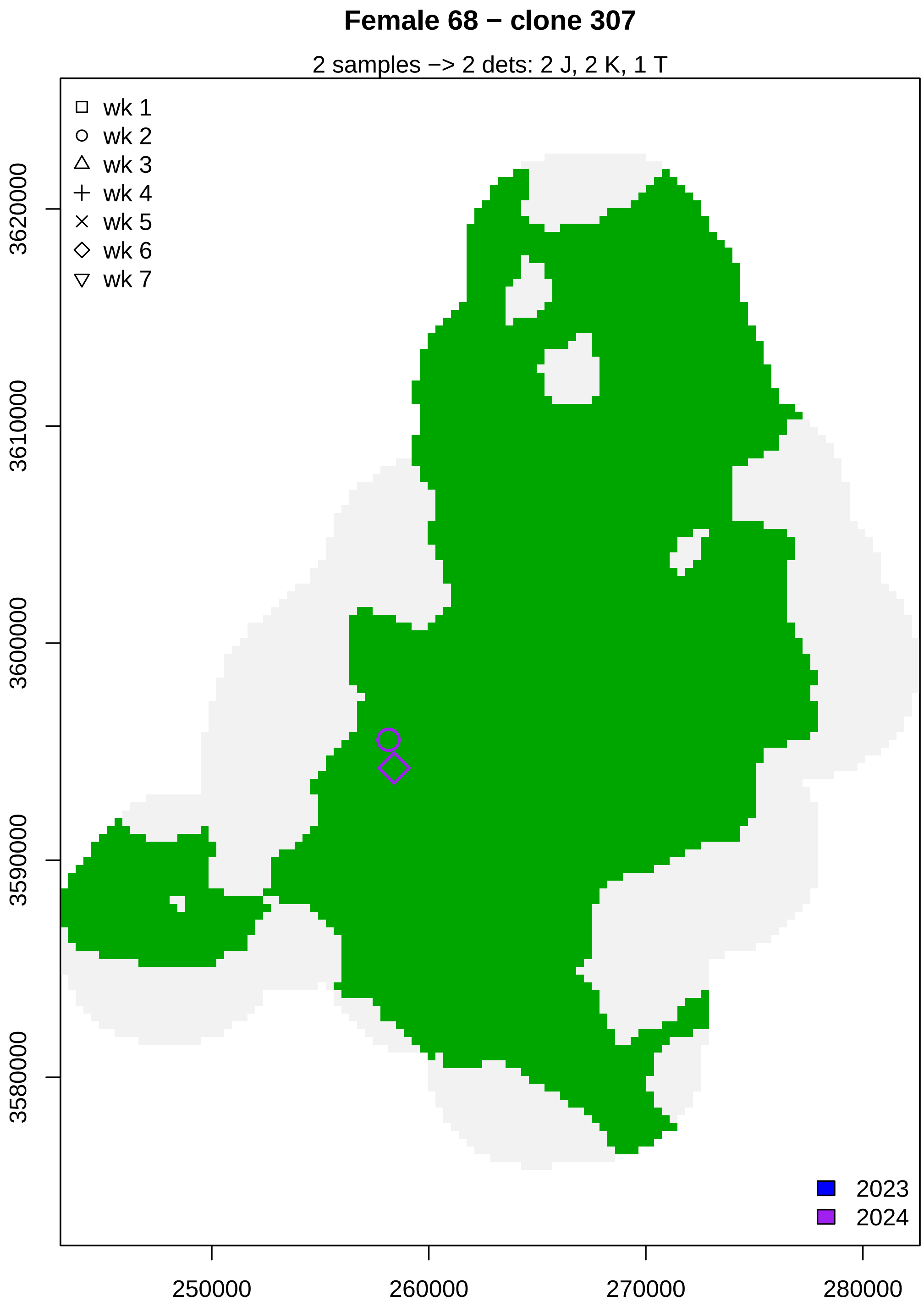
Female 42 − clone 168
3 samples −> 2 dets: 2 J, 2 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

Female 45 − clone 213

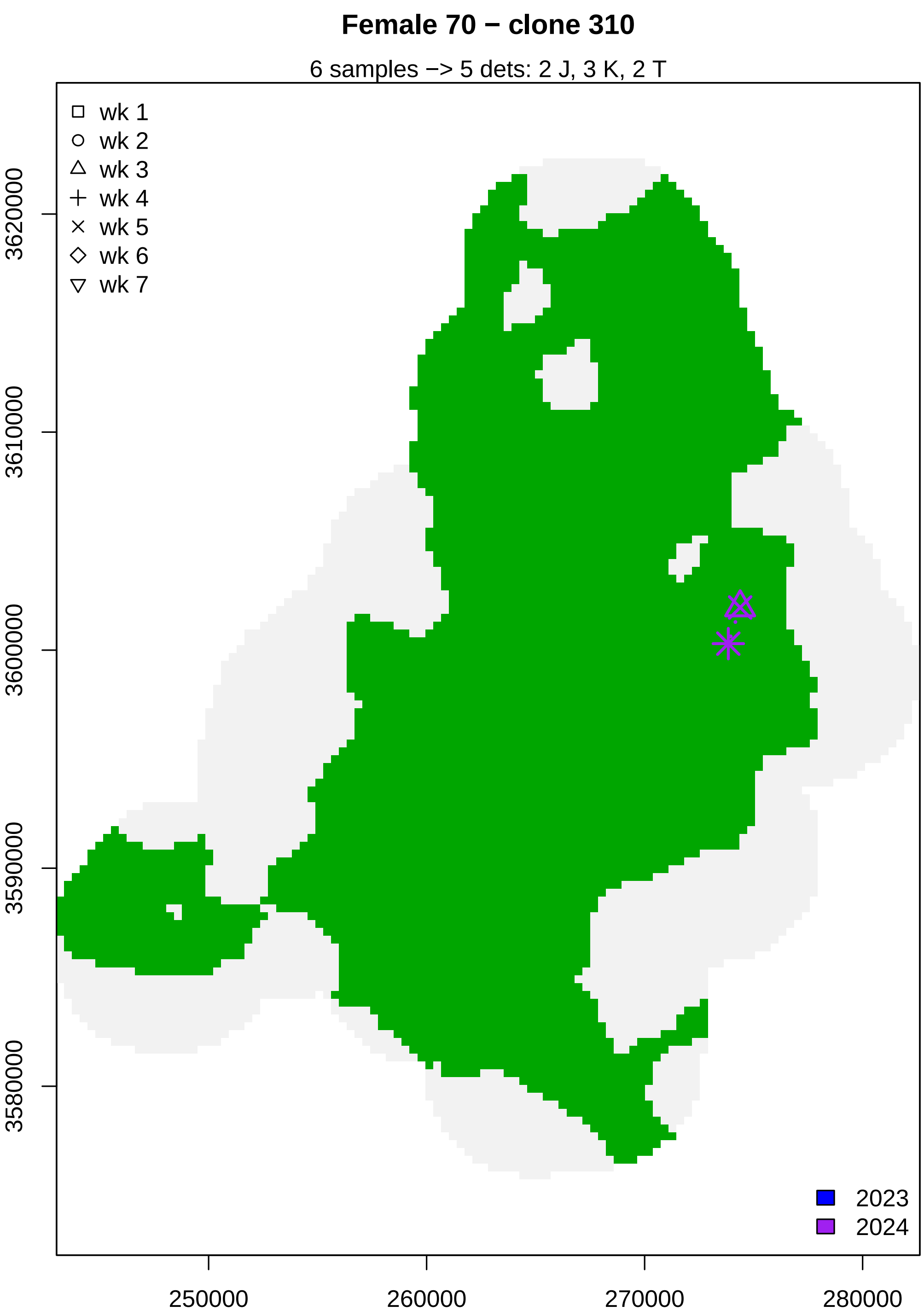
2 samples −> 2 dets: 2 J, 2 K, 1 T



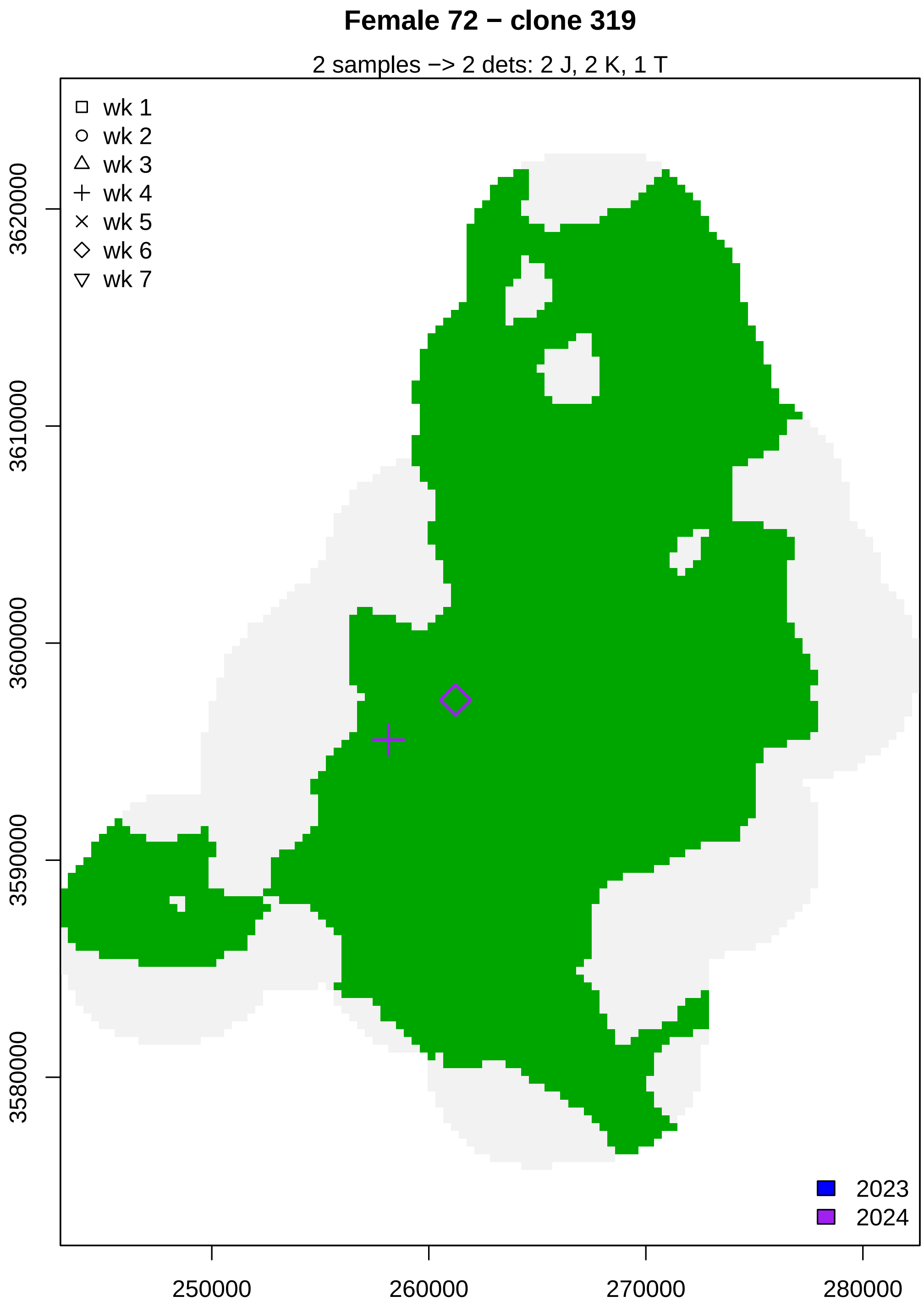

Female 50 − clone 265
12 samples −> 3 dets: 2 J, 3 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024



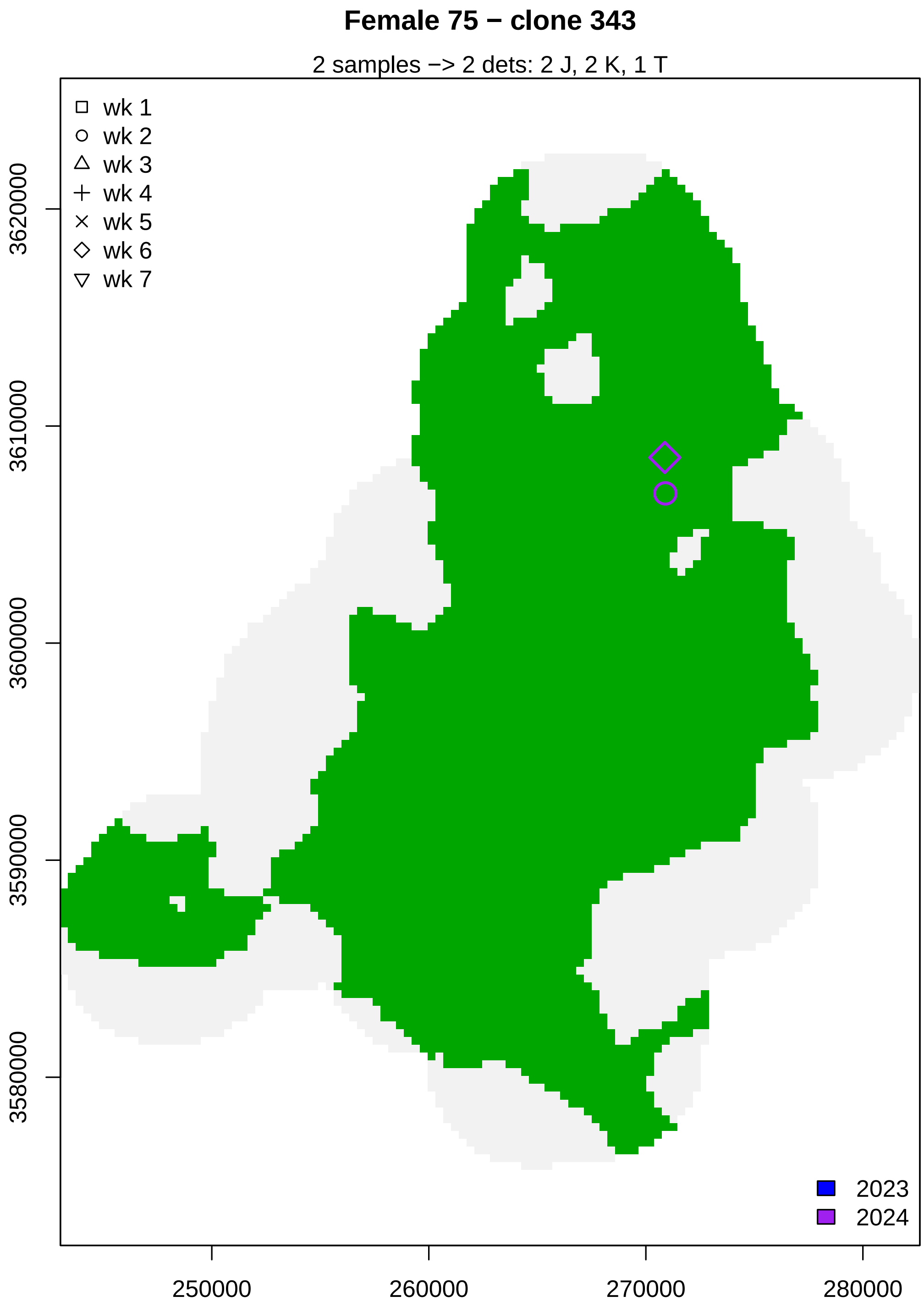
Female 68 − clone 307
2 samples −> 2 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



Female 70 − clone 310
6 samples −> 5 dets: 2 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



Female 72 − clone 319
2 samples −> 2 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



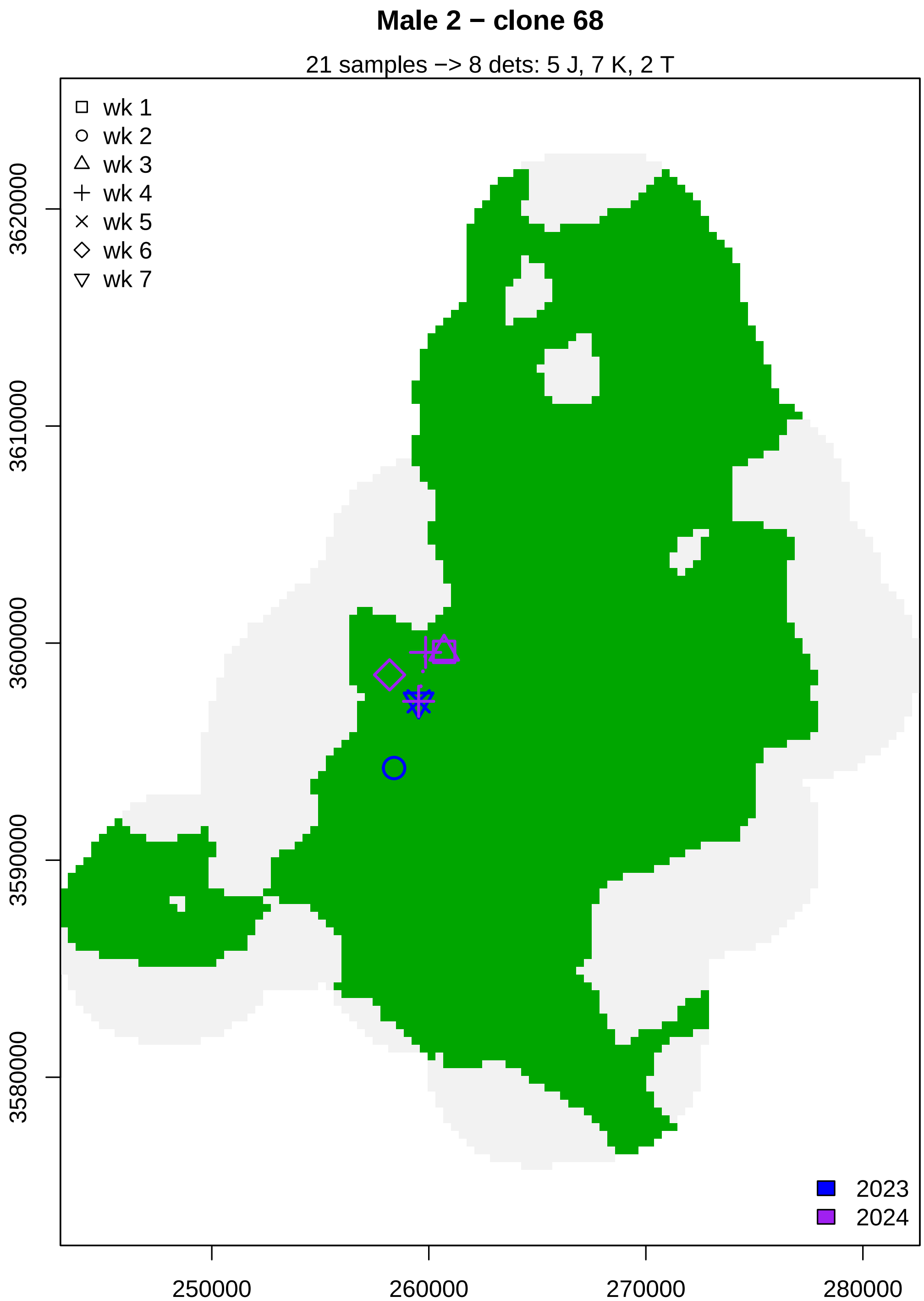
Males

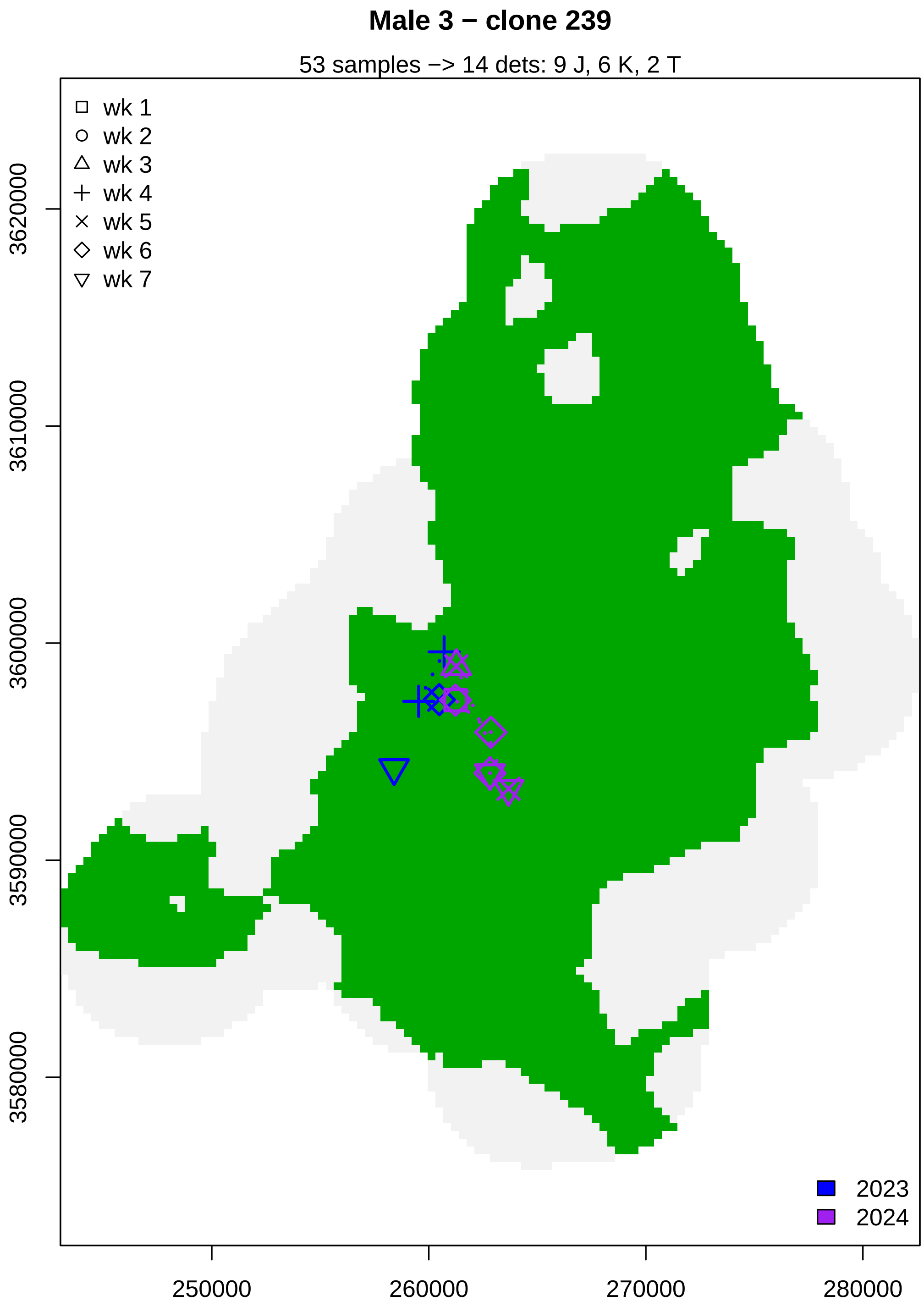
Then for males. . .

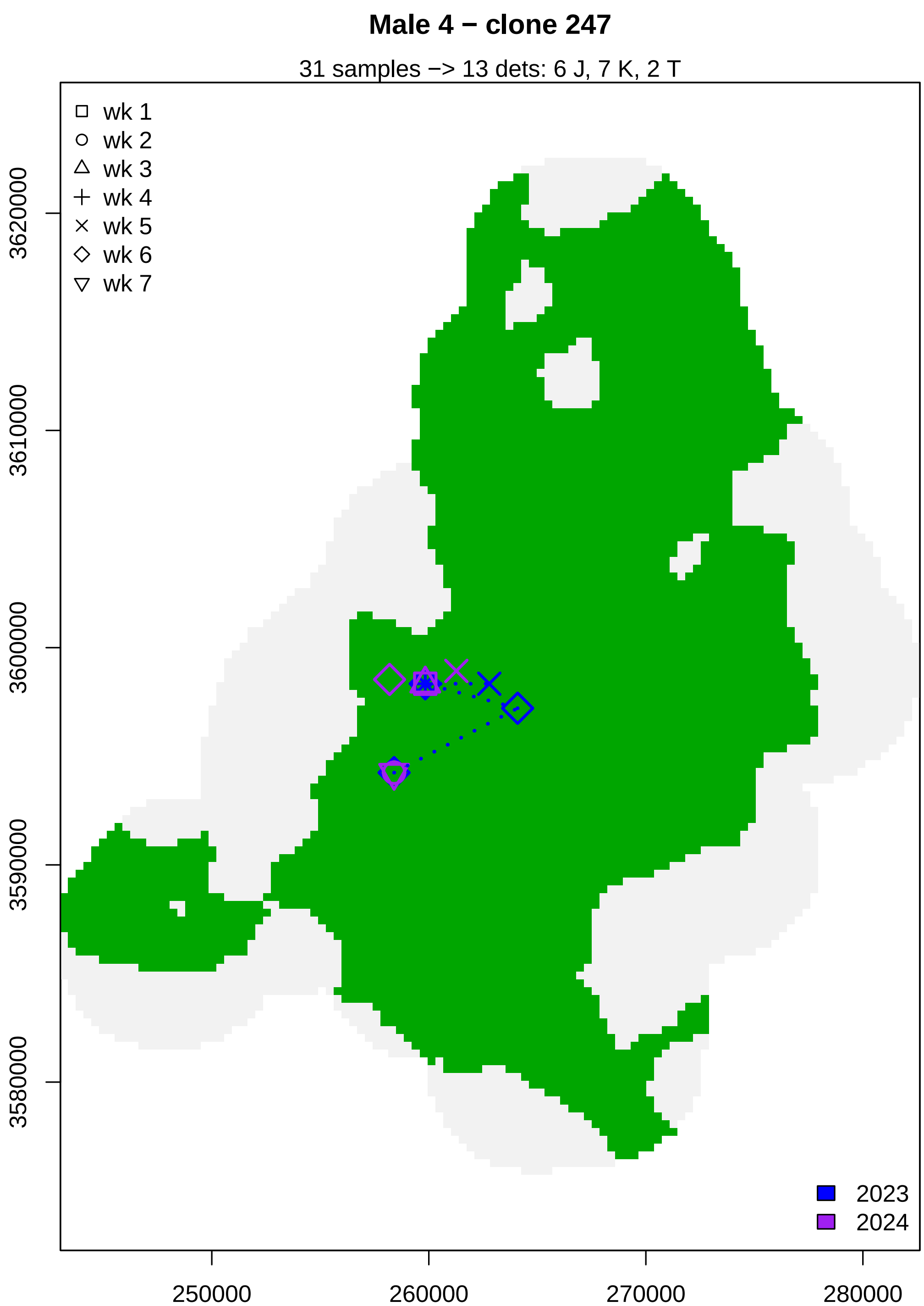
```r
#Males
caphist <- Mch$y
Mclones <- Mch$clone

sumstats_4d(caphist)
```

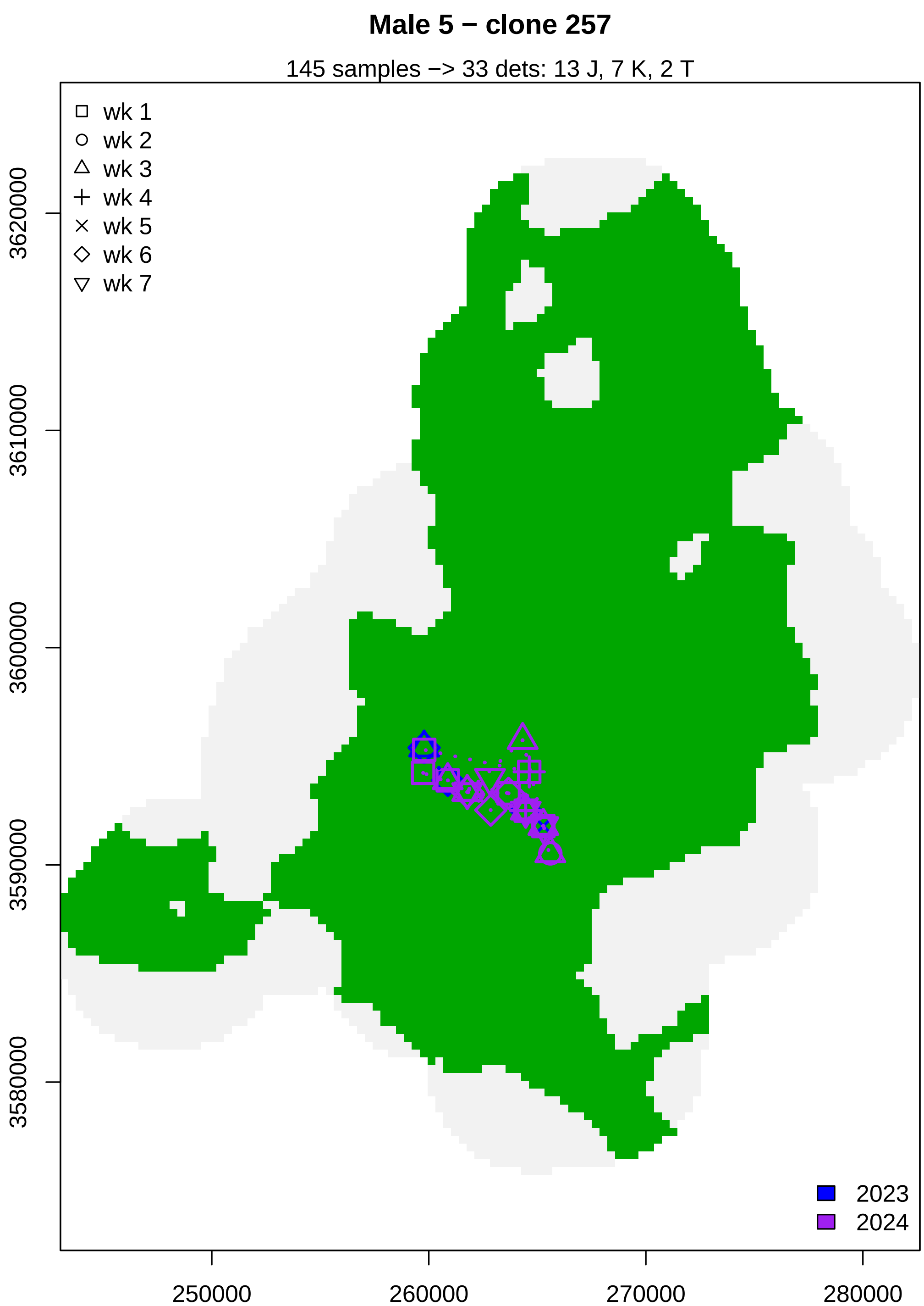
```
##        ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##      66.000000     241.000000       3.651515      26.000000       2.439394
##      ss.avgyrs
##       1.227273
```

```r
nind <- dim(caphist)[1]

yr.col <- c("blue","purple")
wk.symbol <- 0:6

for(i in 1:nind) {
  i.ch <- caphist[i,,,]

  nTraps <- sum(rowSums(i.ch) > 0)
  if(nTraps > 1) {
    plot.me <- as.data.frame(which(i.ch == 1, arr.ind = TRUE))
    plot(S360, legend = FALSE, main = paste("Male",i,"- clone", Mclones[i]))
    mtext(paste(length(which(smd$finalClone == Mclones[i])), "samples ->",
                nrow(plot.me), "dets:",
                nTraps, "J,",
                length(unique(plot.me$dim2)), "K,",
                length(unique(plot.me$dim3)), "T"))
    points(x=locs[plot.me$dim1,1], locs[plot.me$dim1,2],
           pch = wk.symbol[plot.me$dim2], col = yr.col[plot.me$dim3],
           cex = 2, lwd = 2)

    for(r in 2:nrow(plot.me)) {
      if(plot.me$dim3[r]==plot.me$dim3[r-1] &
         plot.me$dim2[r]==plot.me$dim2[r-1]) {
        lines(x=c(locs[plot.me$dim1[r-1],1], locs[plot.me$dim1[r],1]),
              y=c(locs[plot.me$dim1[r-1],2], locs[plot.me$dim1[r],2]),
              lty="dotted", col = yr.col[plot.me$dim3[r]],
              lwd = 2.5)
      }
    }

    legend("topleft", bty = "n", pch = 0:6, legend = paste("wk",c(1:7)))
    legend("bottomright", bty = "n", fill = yr.col, legend = c("2023","2024"))

  }
}
```


Male 2 − clone 68
21 samples −> 8 dets: 5 J, 7 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
2023
2024
250000
260000
270000
280000


Male 3 − clone 239
53 samples −> 14 dets: 9 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

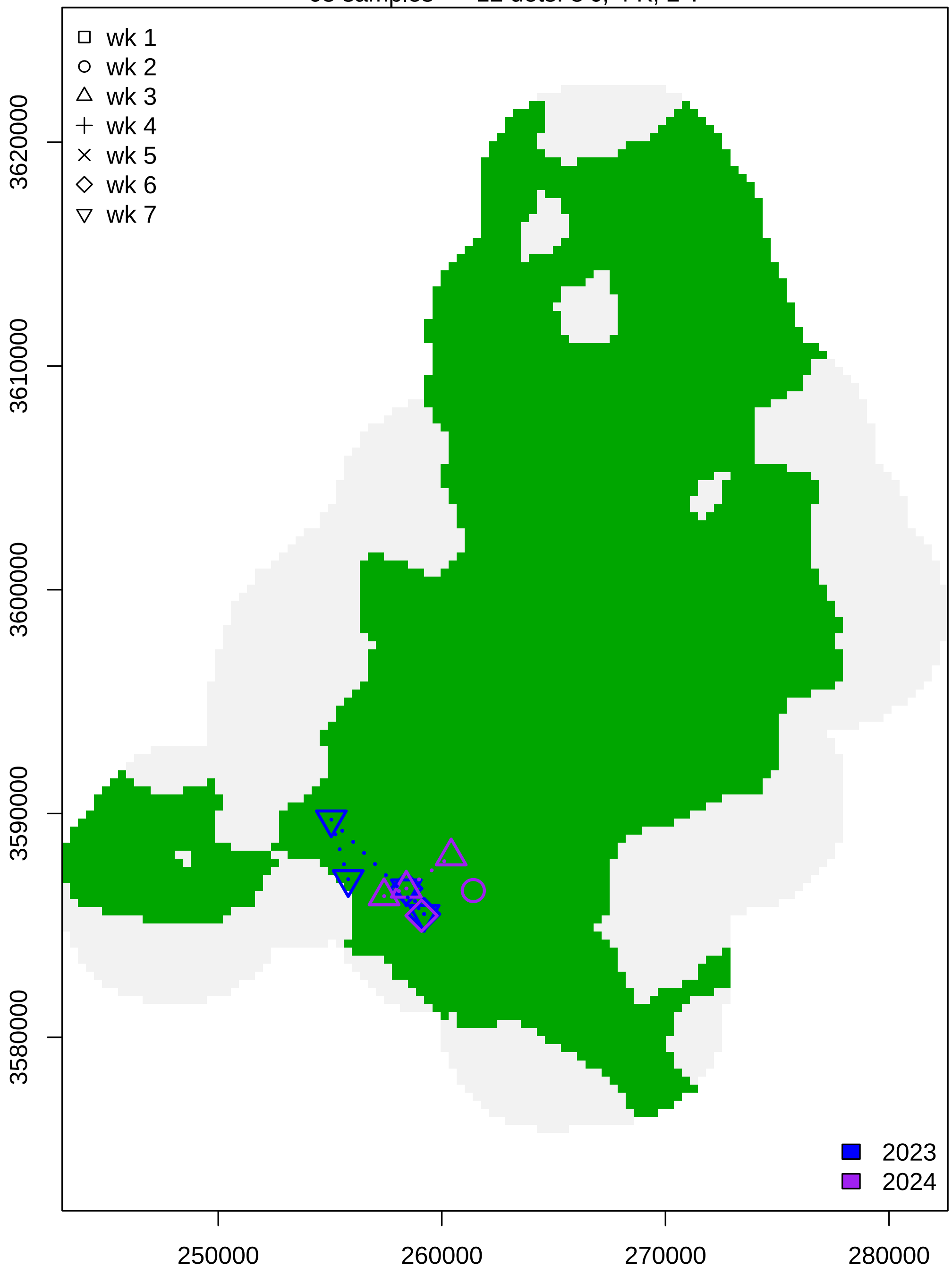

Male 4 − clone 247
31 samples −> 13 dets: 6 J, 7 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

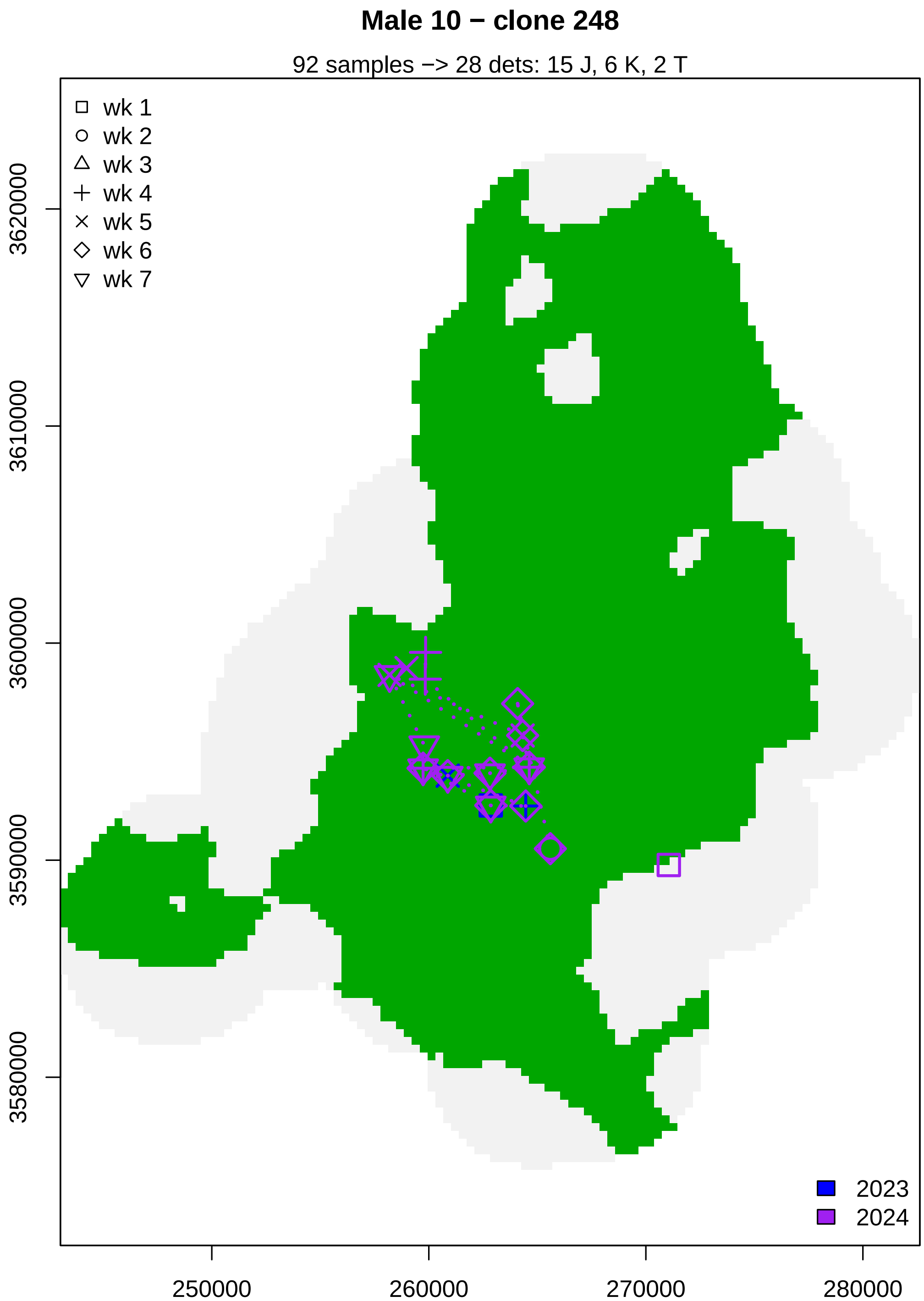


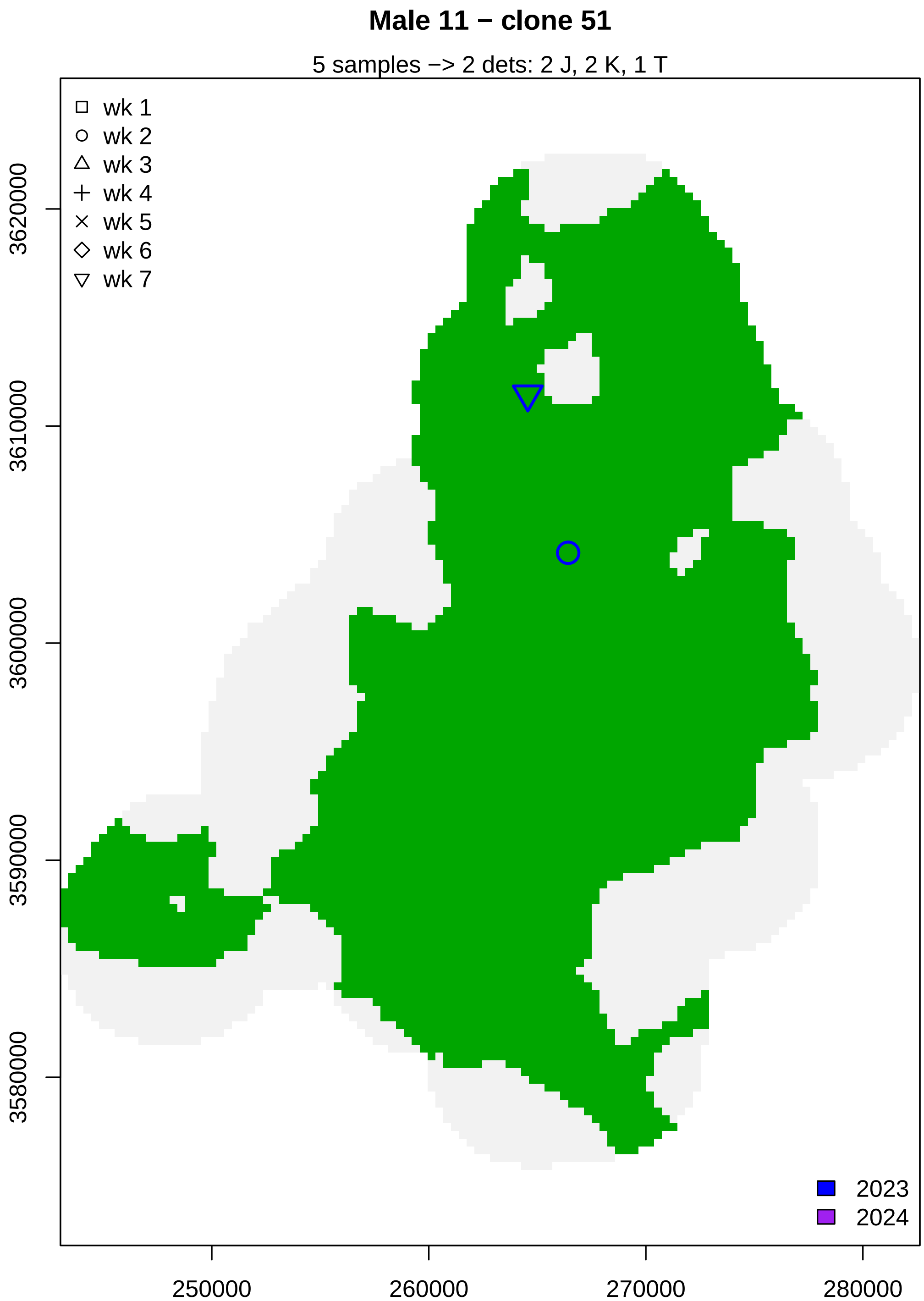
Male 5 − clone 257
145 samples −> 33 dets: 13 J, 7 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

Male 8 − clone 33

68 samples −> 12 dets: 8 J, 4 K, 2 T



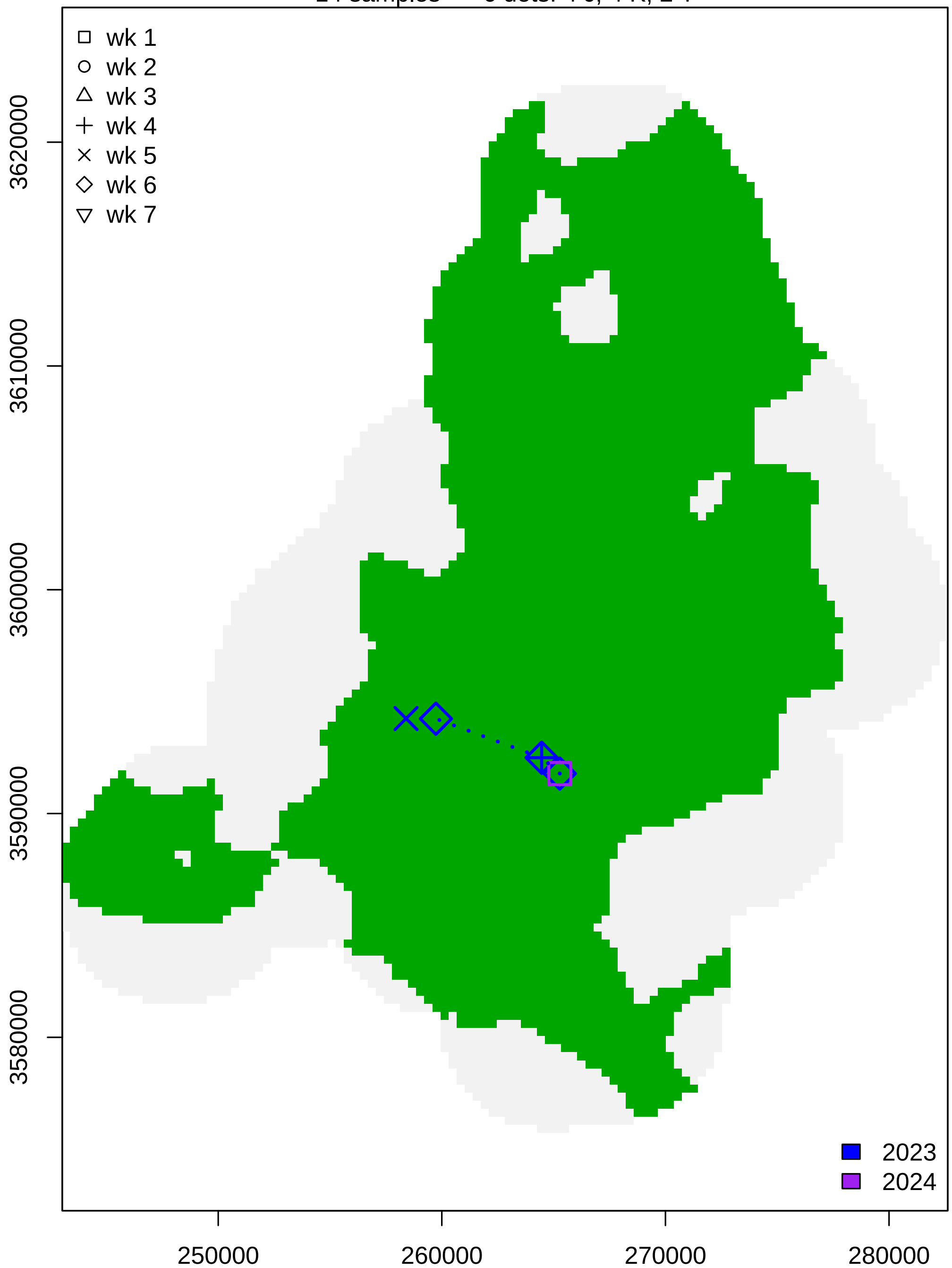

Male 10 − clone 248
92 samples −> 28 dets: 15 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



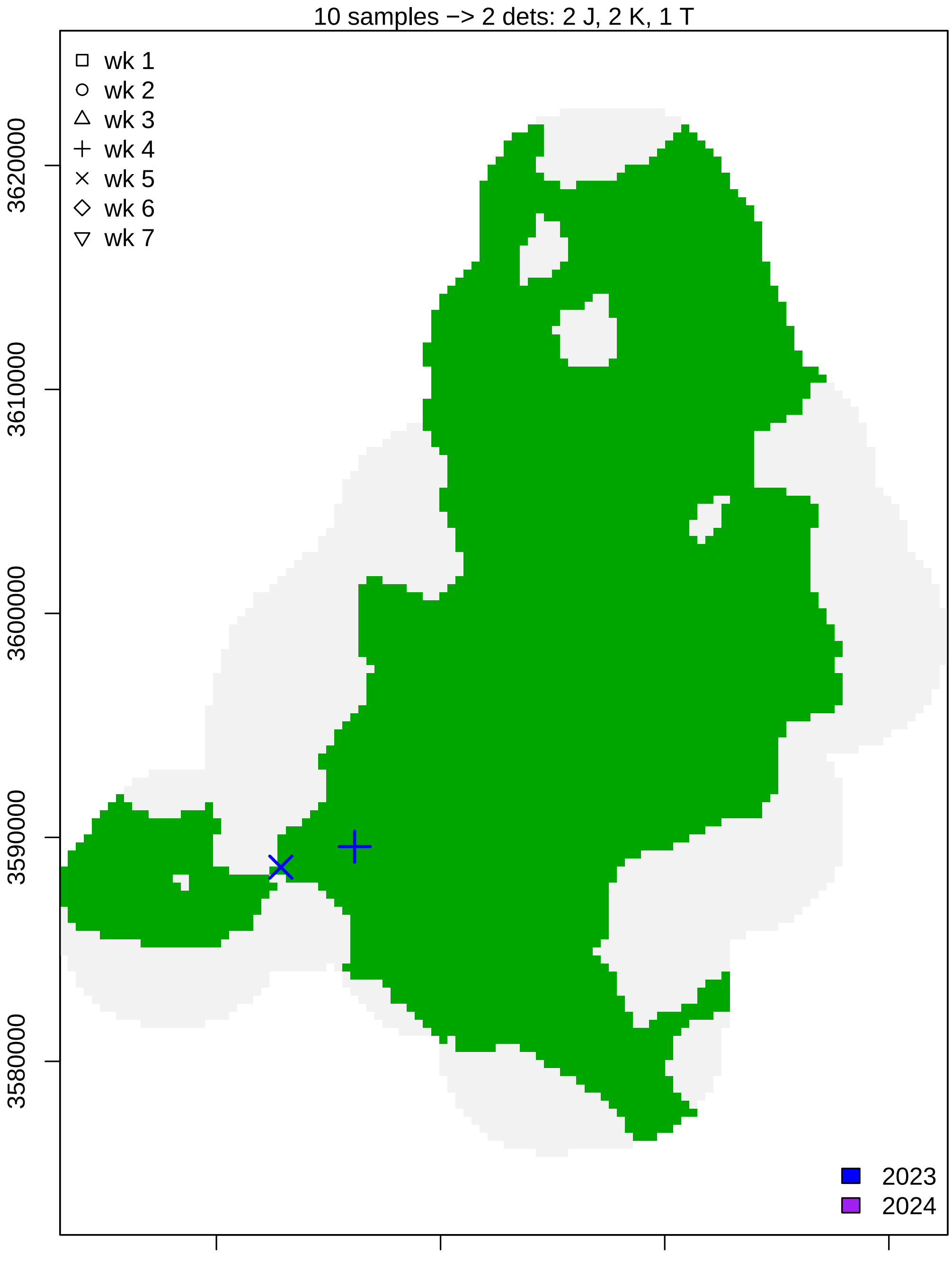
Male 11 − clone 51
5 samples −> 2 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

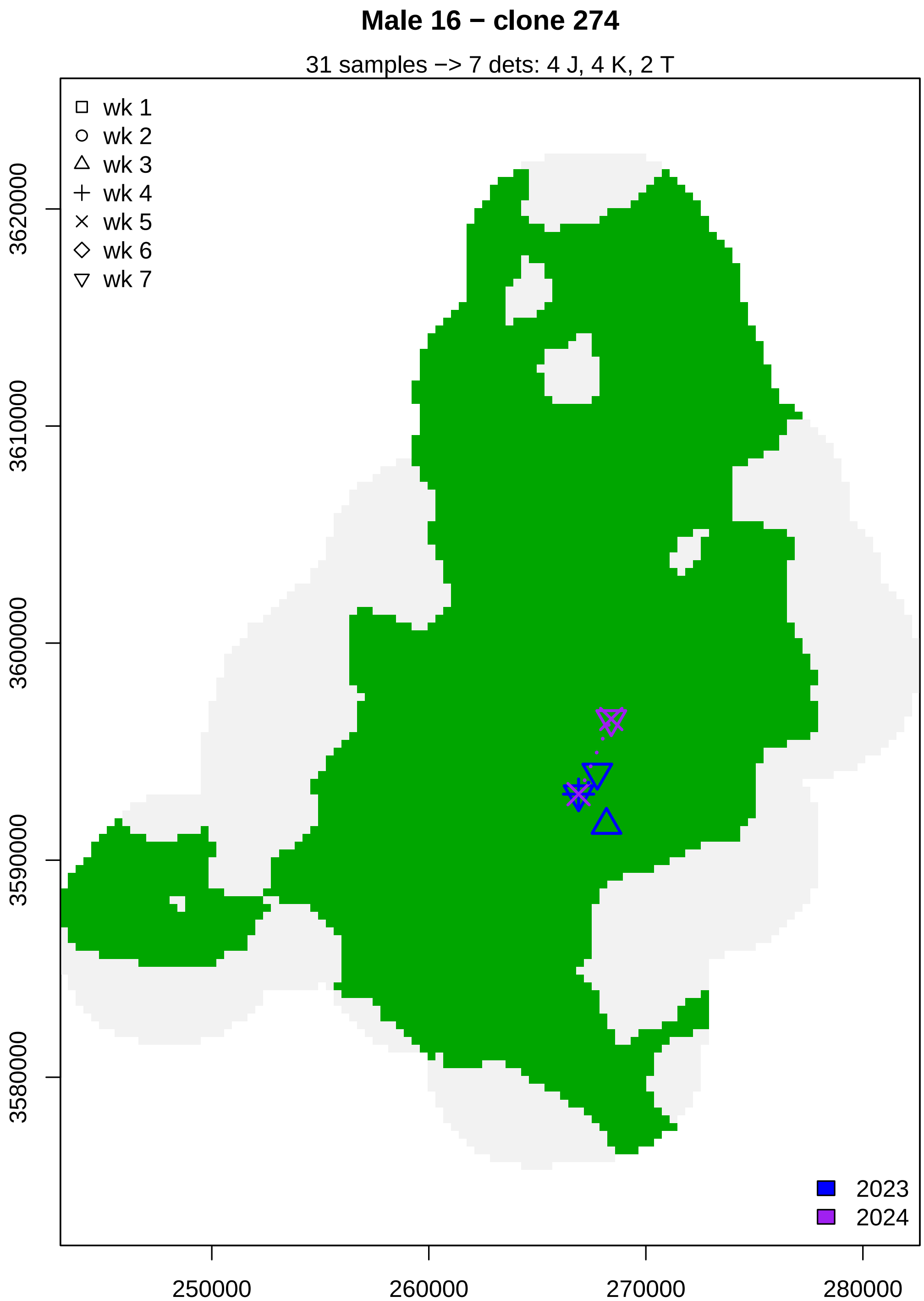
Male 13 − clone 285

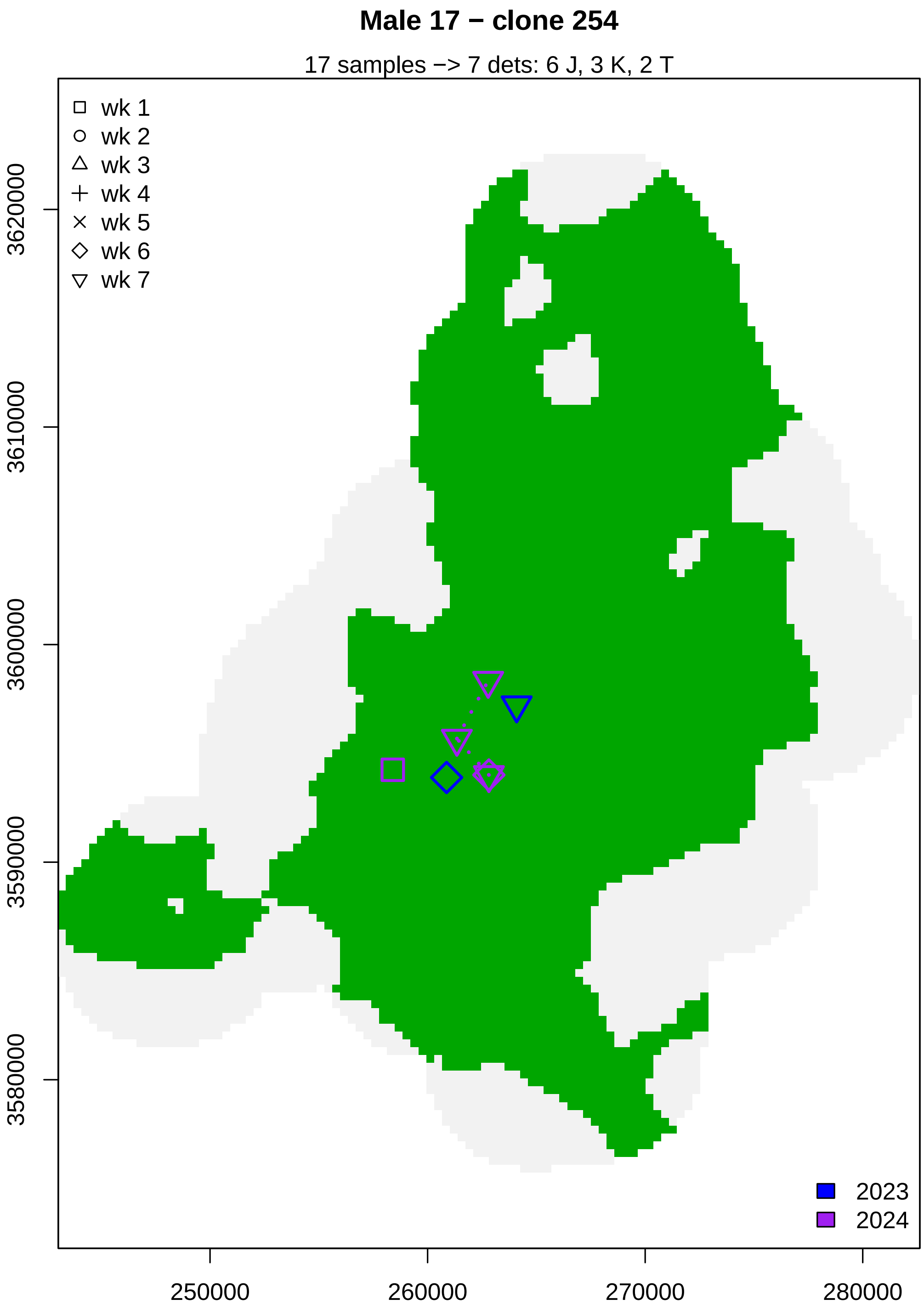
14 samples −> 6 dets: 4 J, 4 K, 2 T



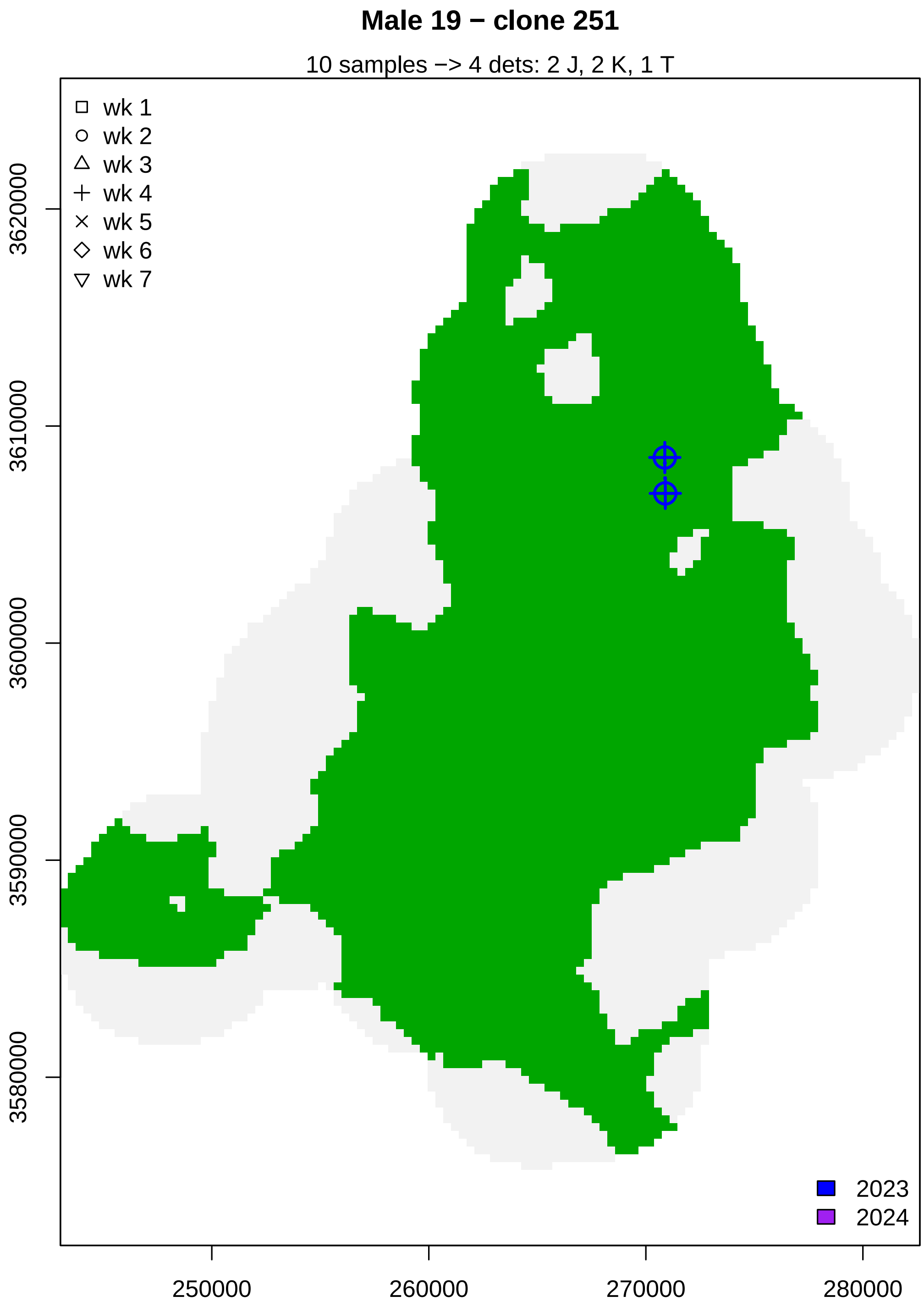
Male 15 − clone 243

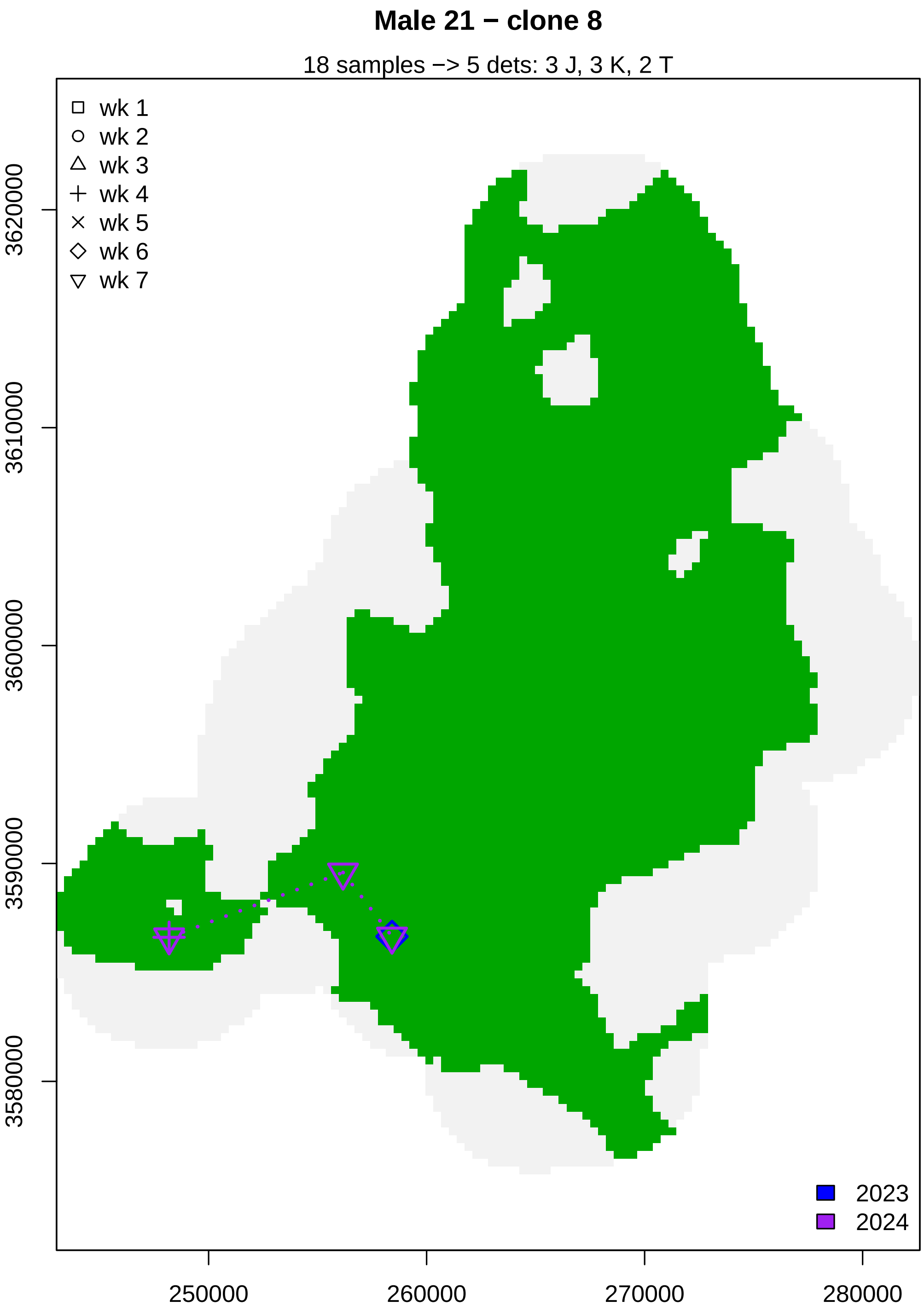



Male 16 − clone 274
31 samples −> 7 dets: 4 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

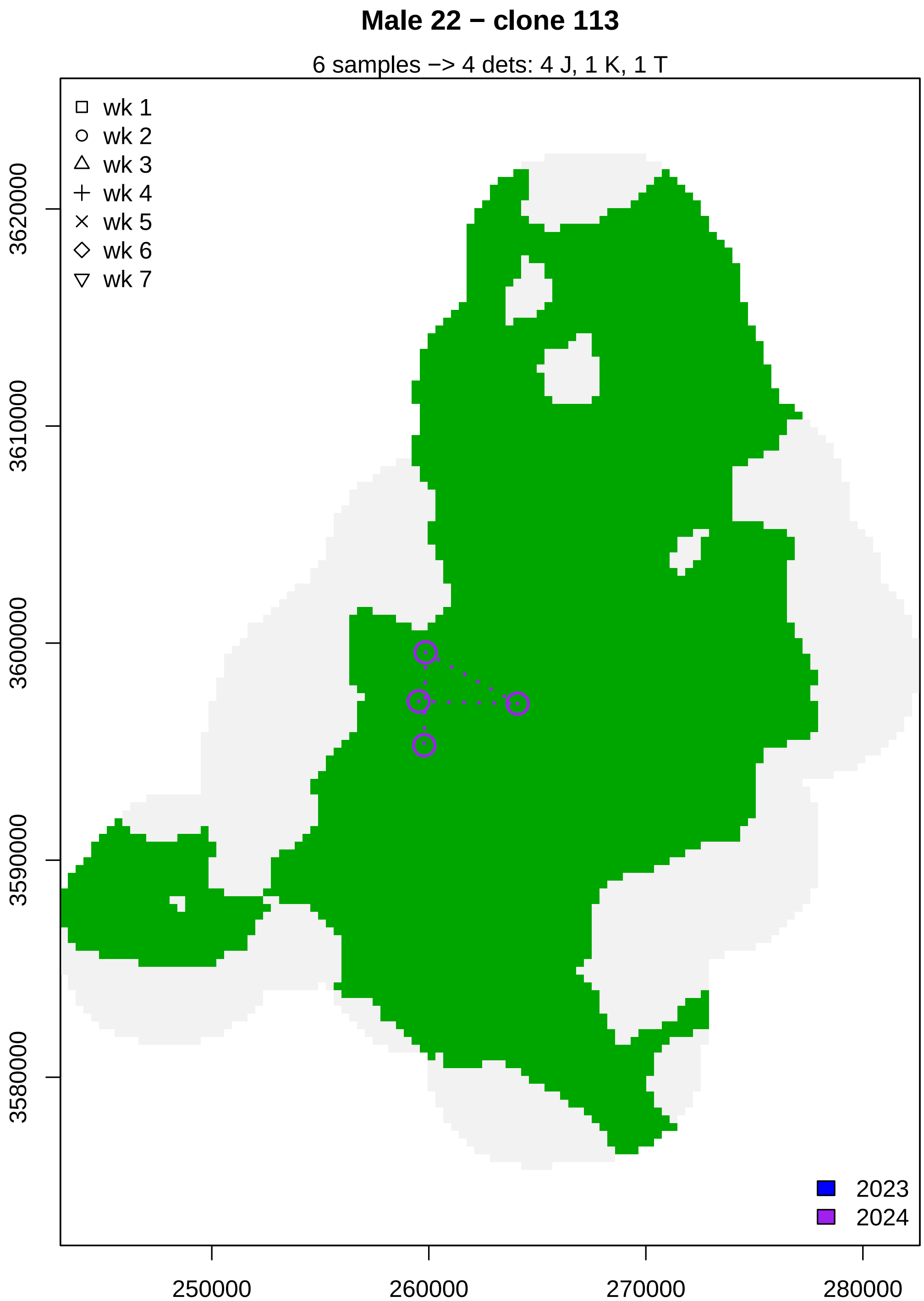


Male 17 − clone 254
17 samples −> 7 dets: 6 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

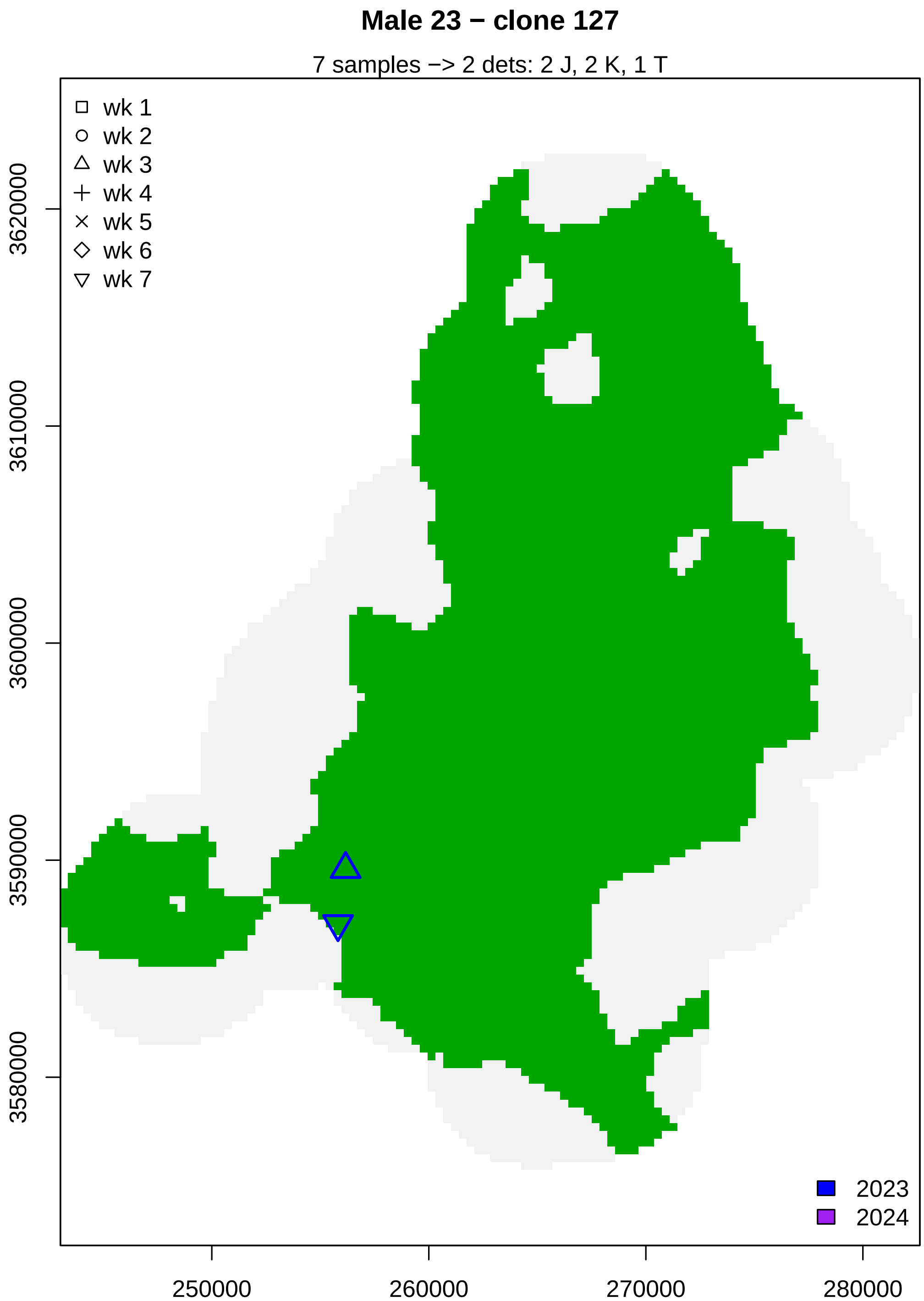

Male 19 − clone 251
10 samples −> 4 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
2023
2024
250000
260000
270000
280000



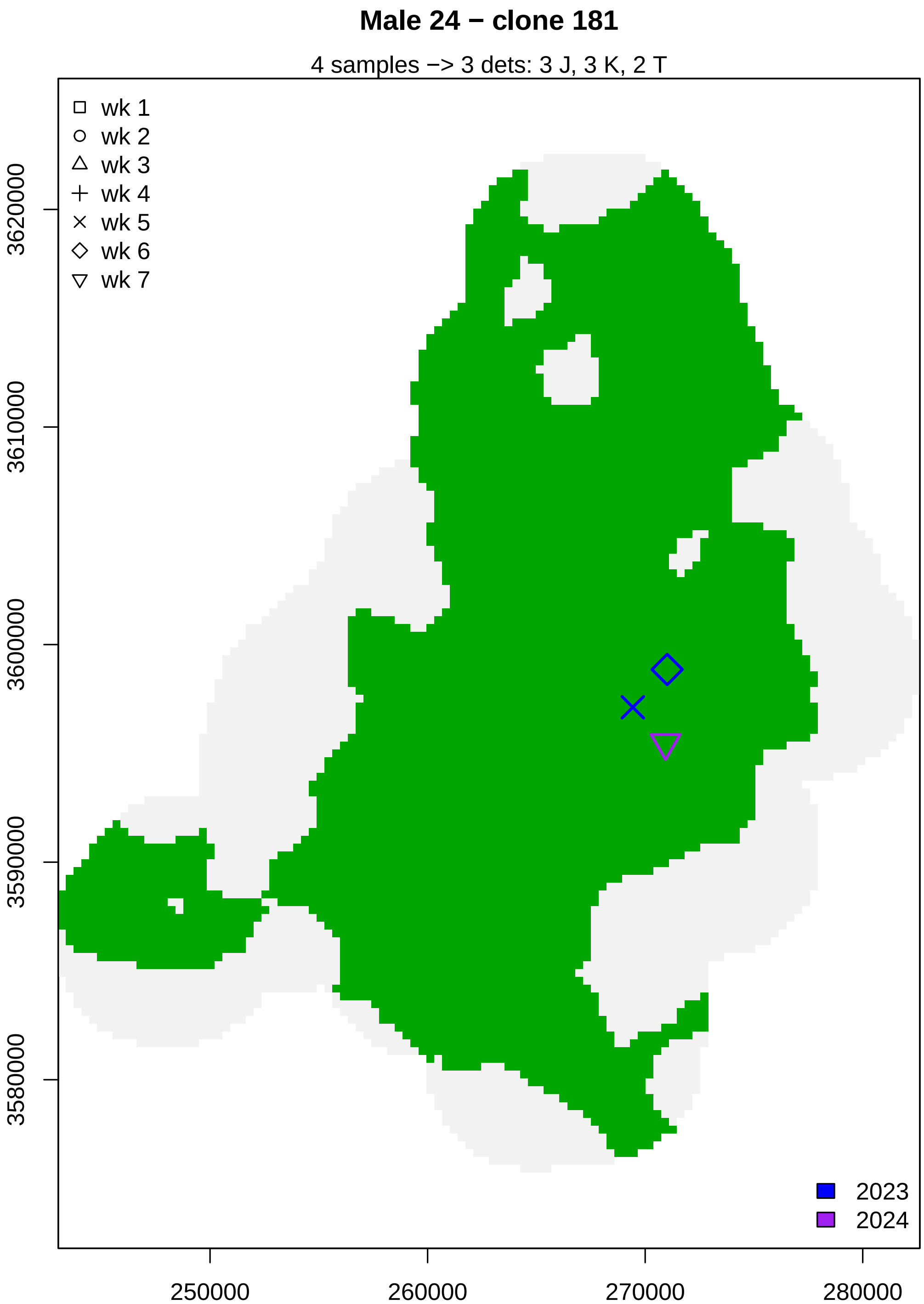
Male 21 − clone 8
18 samples −> 5 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

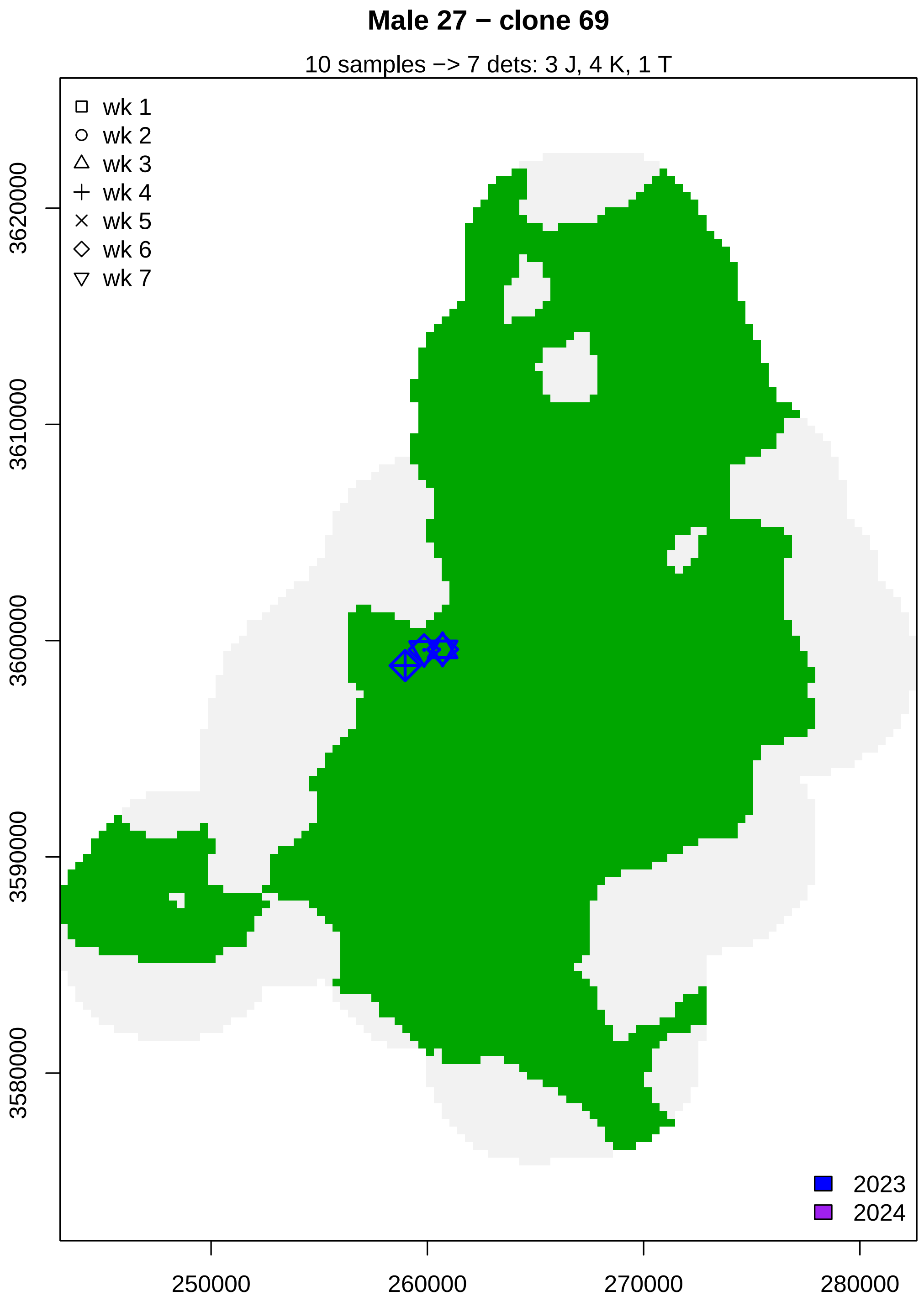


Male 22 − clone 113
6 samples −> 4 dets: 4 J, 1 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
2023
2024
250000
260000
270000
280000

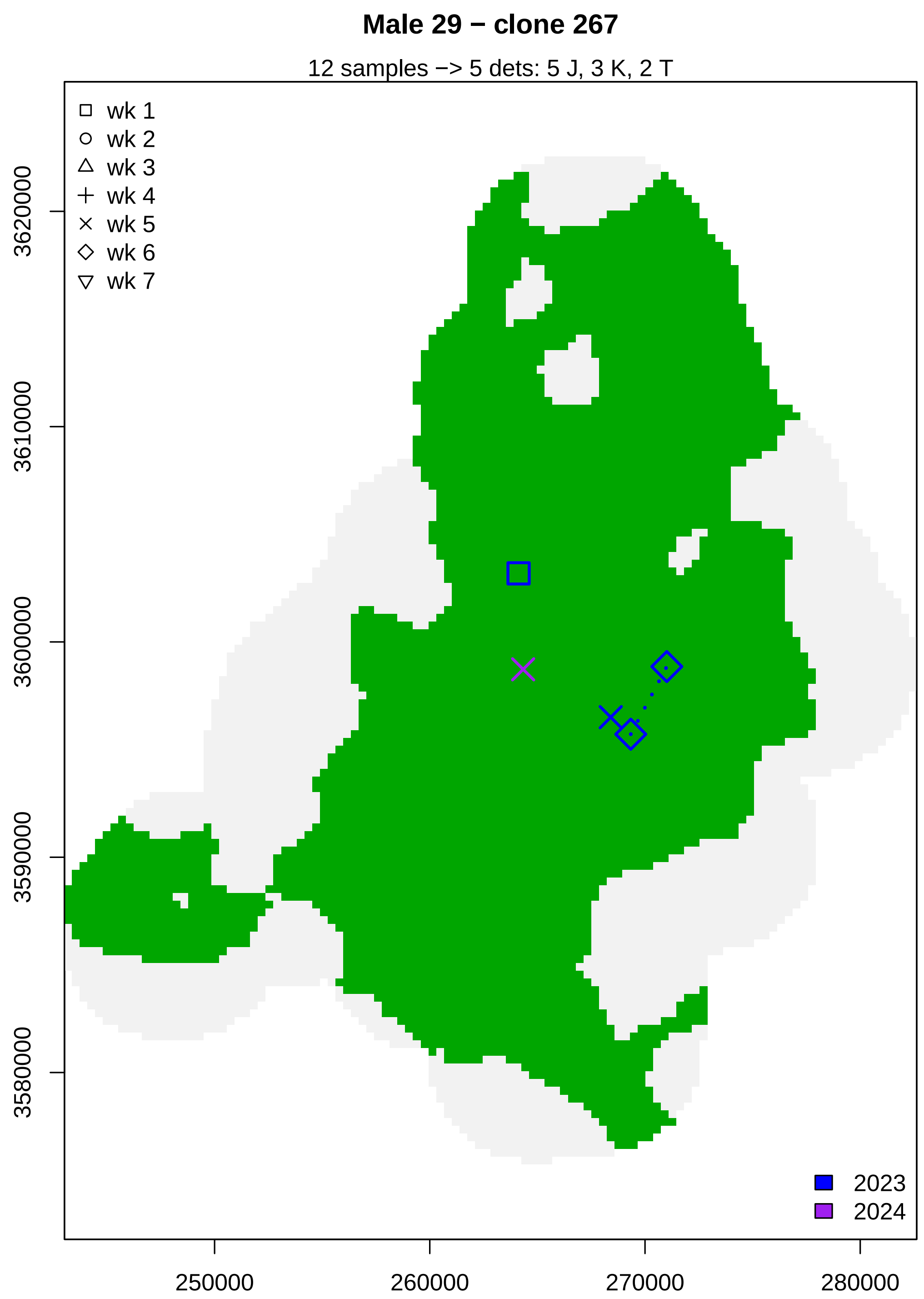


Male 23 − clone 127
7 samples −> 2 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
2023
2024
250000
260000
270000
280000

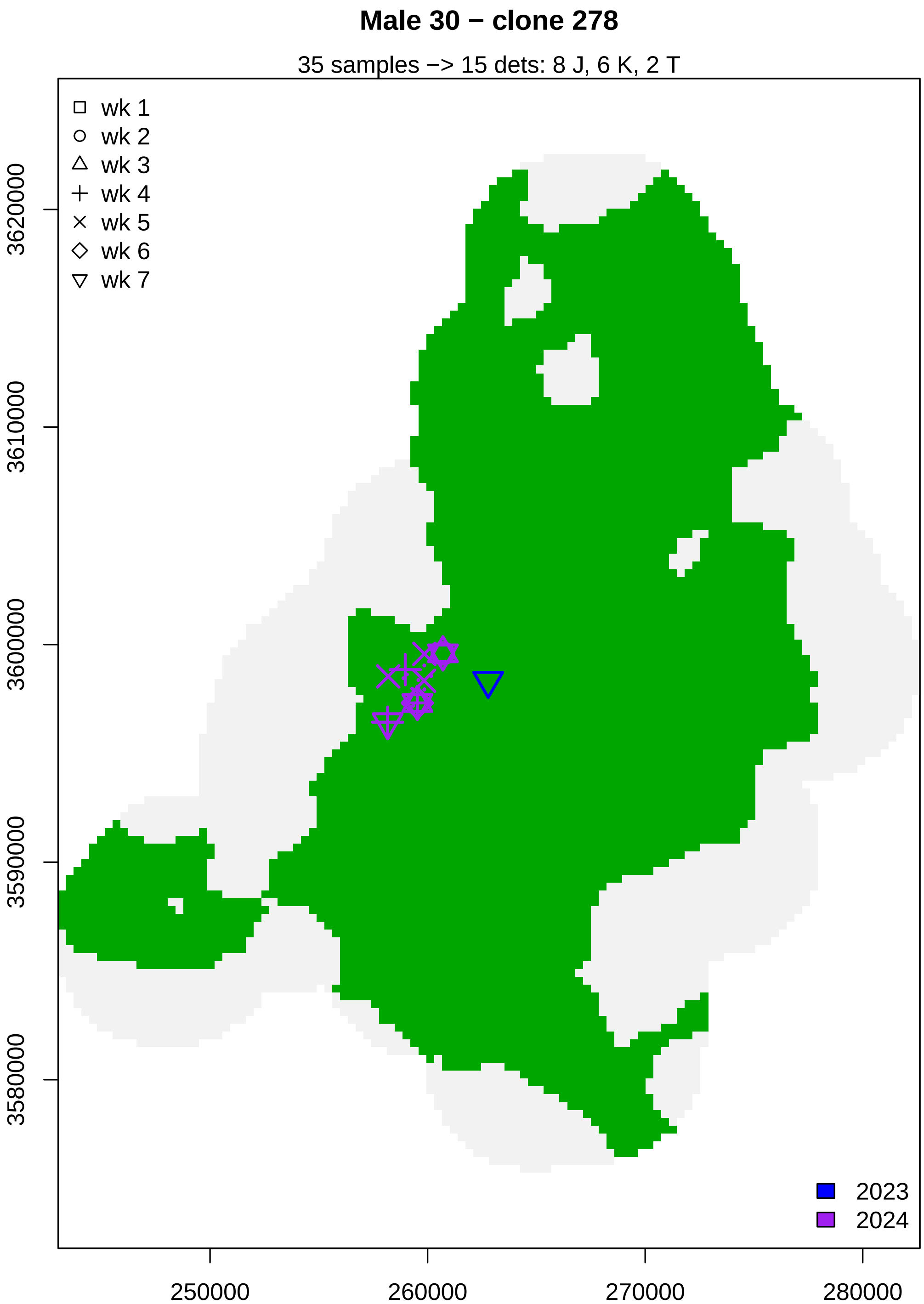

Male 24 − clone 181
4 samples −> 3 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



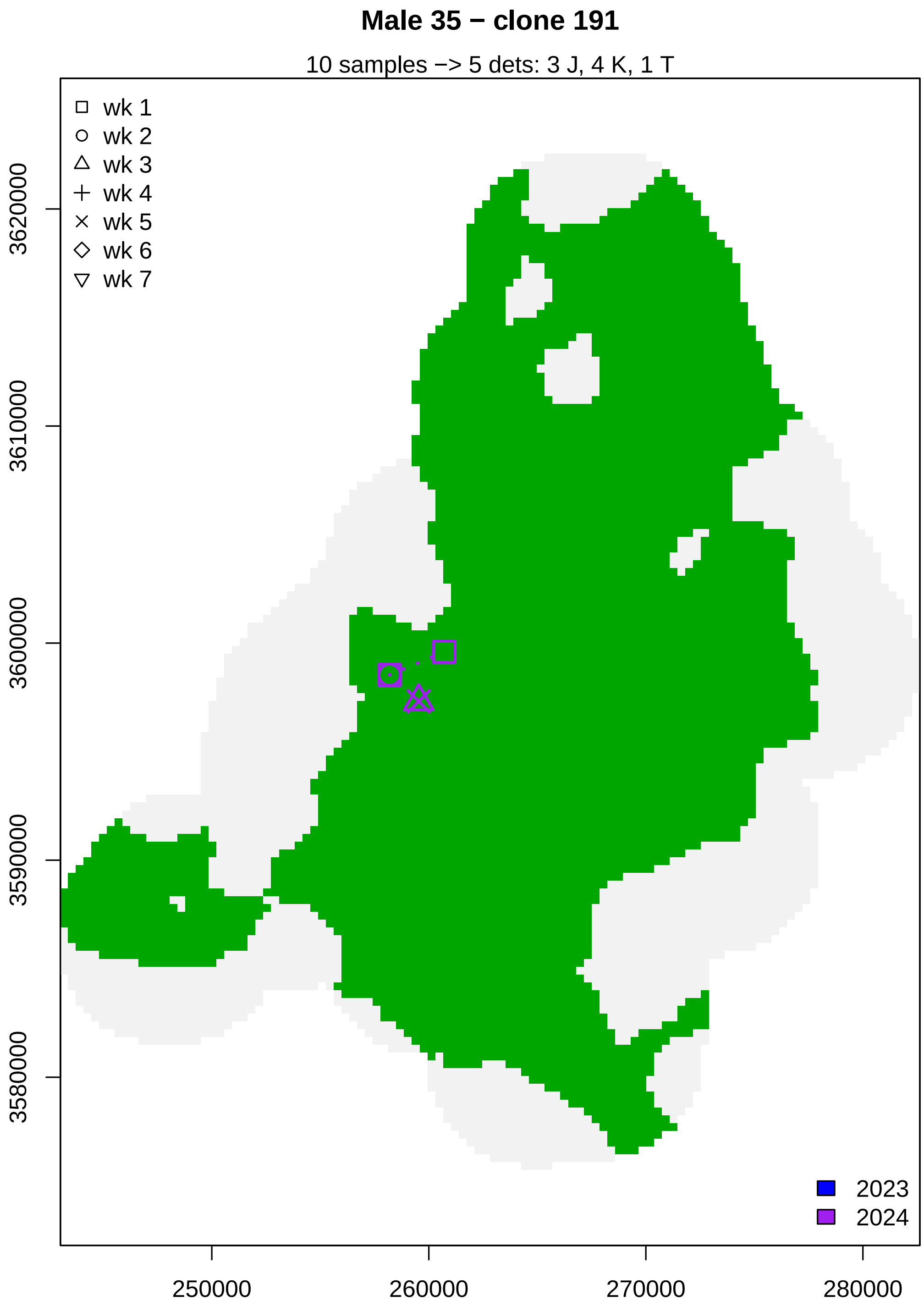
Male 27 − clone 69
10 samples −> 7 dets: 3 J, 4 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024



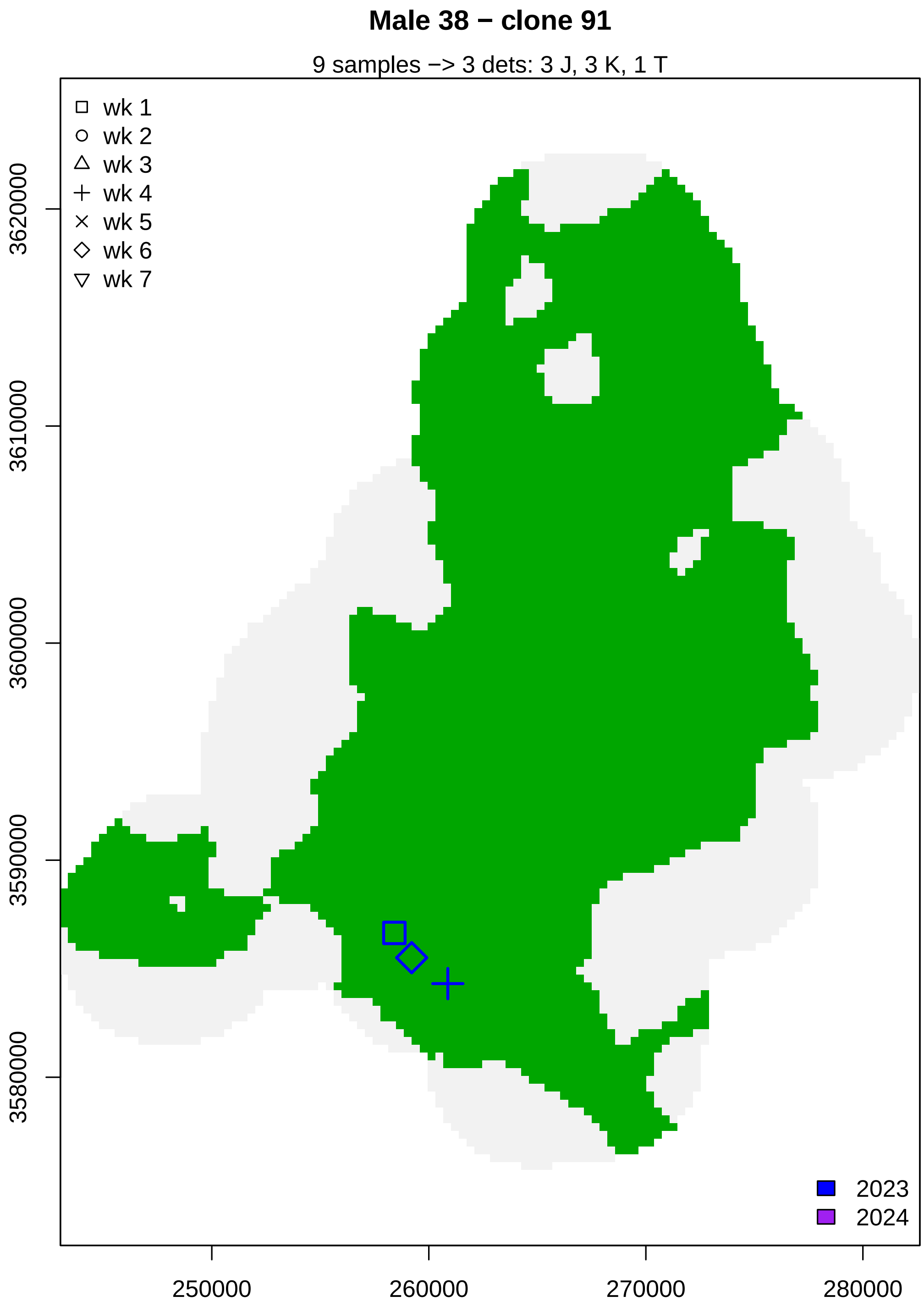
Male 29 − clone 267
12 samples −> 5 dets: 5 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024



Male 30 − clone 278
35 samples −> 15 dets: 8 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



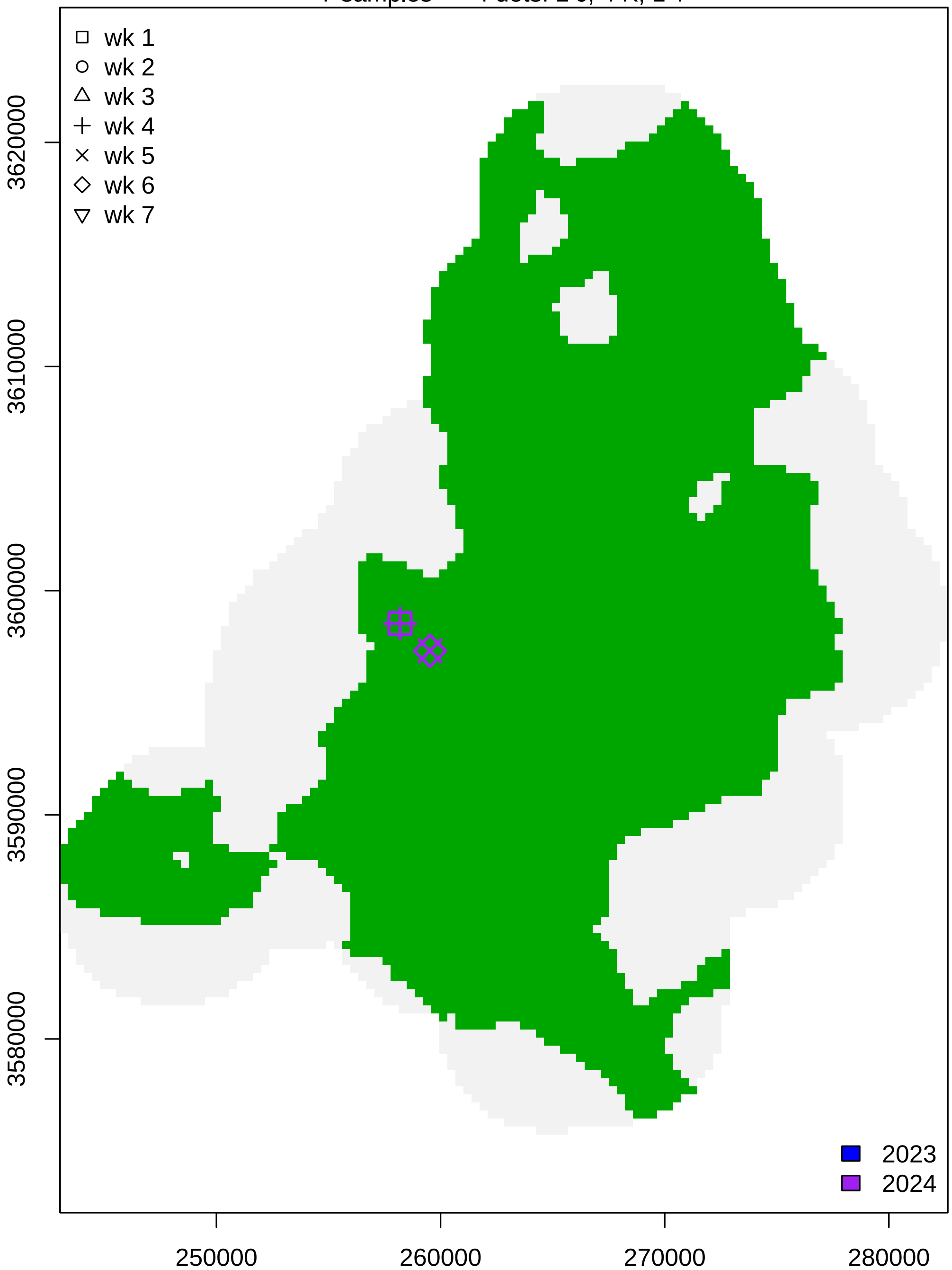
Male 35 − clone 191
10 samples −> 5 dets: 3 J, 4 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

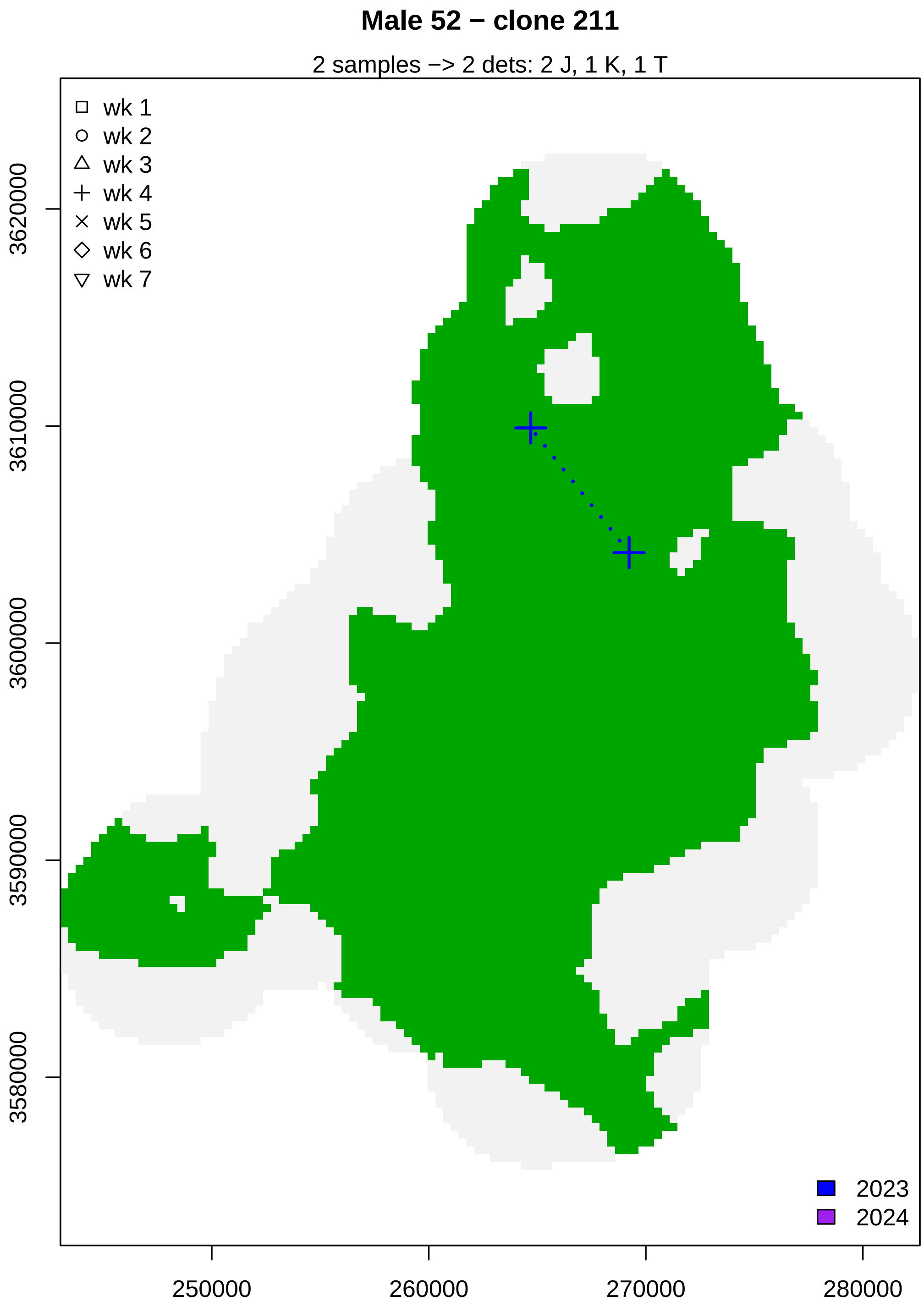


Male 38 − clone 91
9 samples −> 3 dets: 3 J, 3 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

Male 45 − clone 229

7 samples −> 4 dets: 2 J, 4 K, 1 T

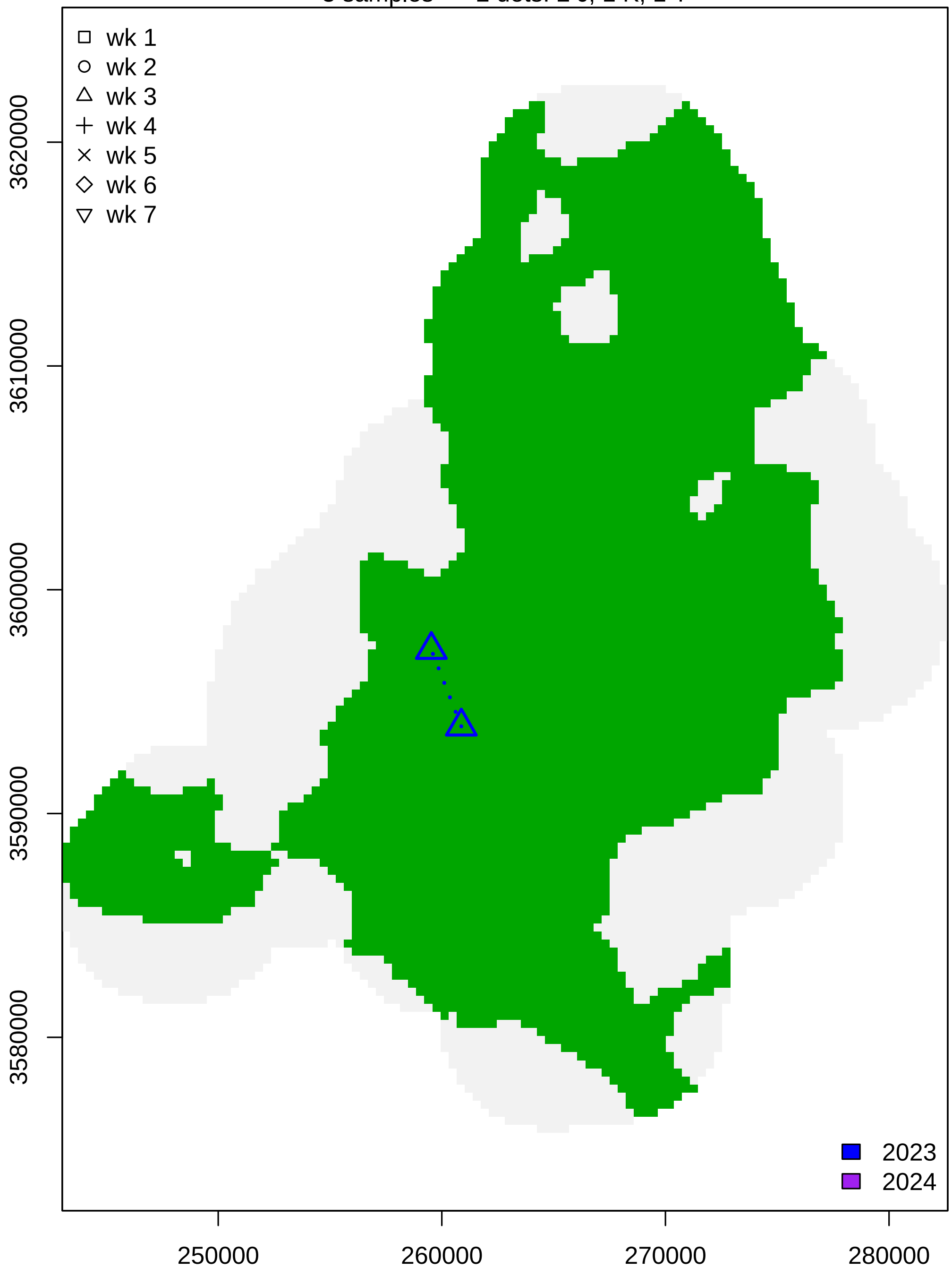


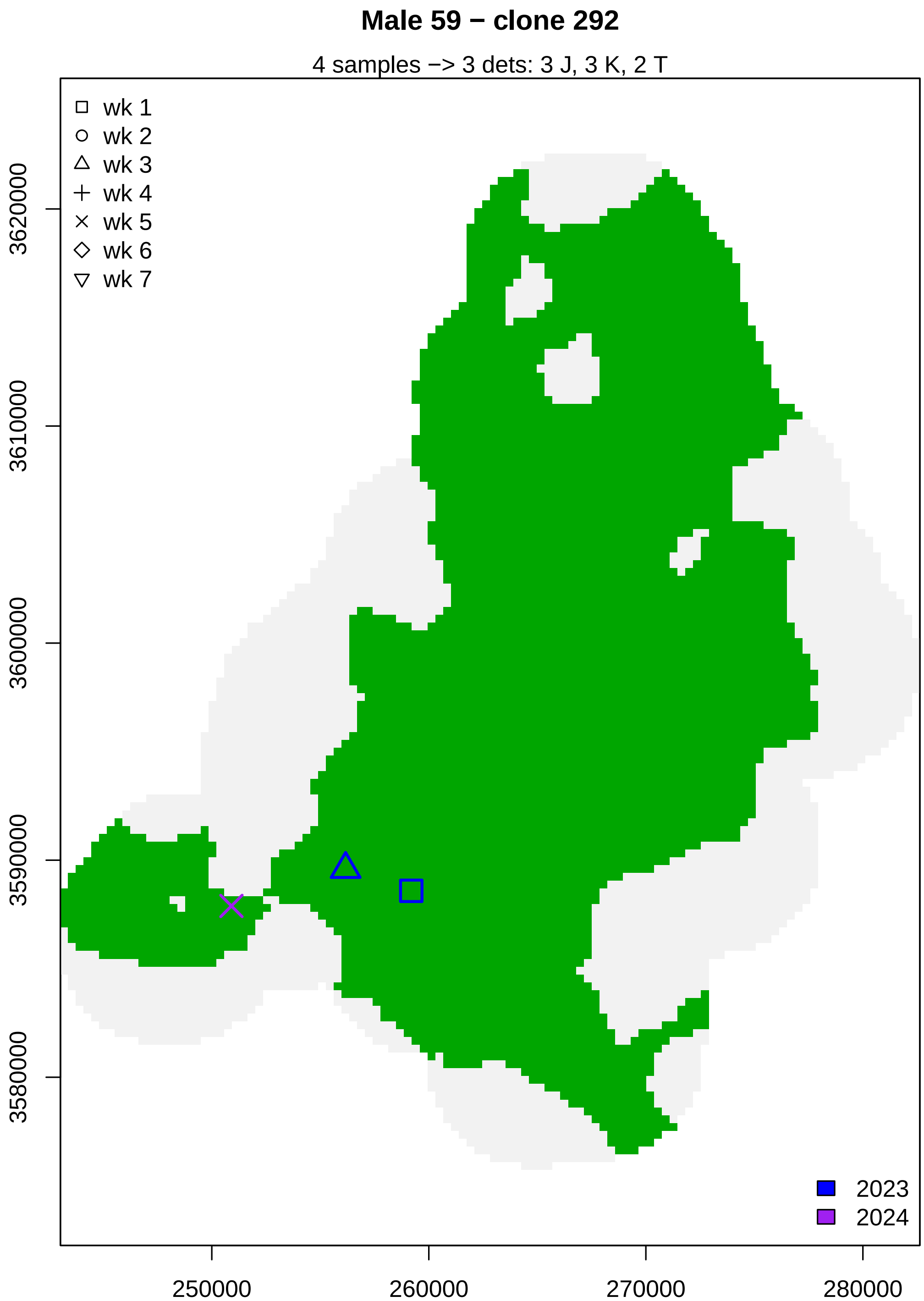


Male 52 − clone 211
2 samples −> 2 dets: 2 J, 1 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
2023
2024
250000
260000
270000
280000

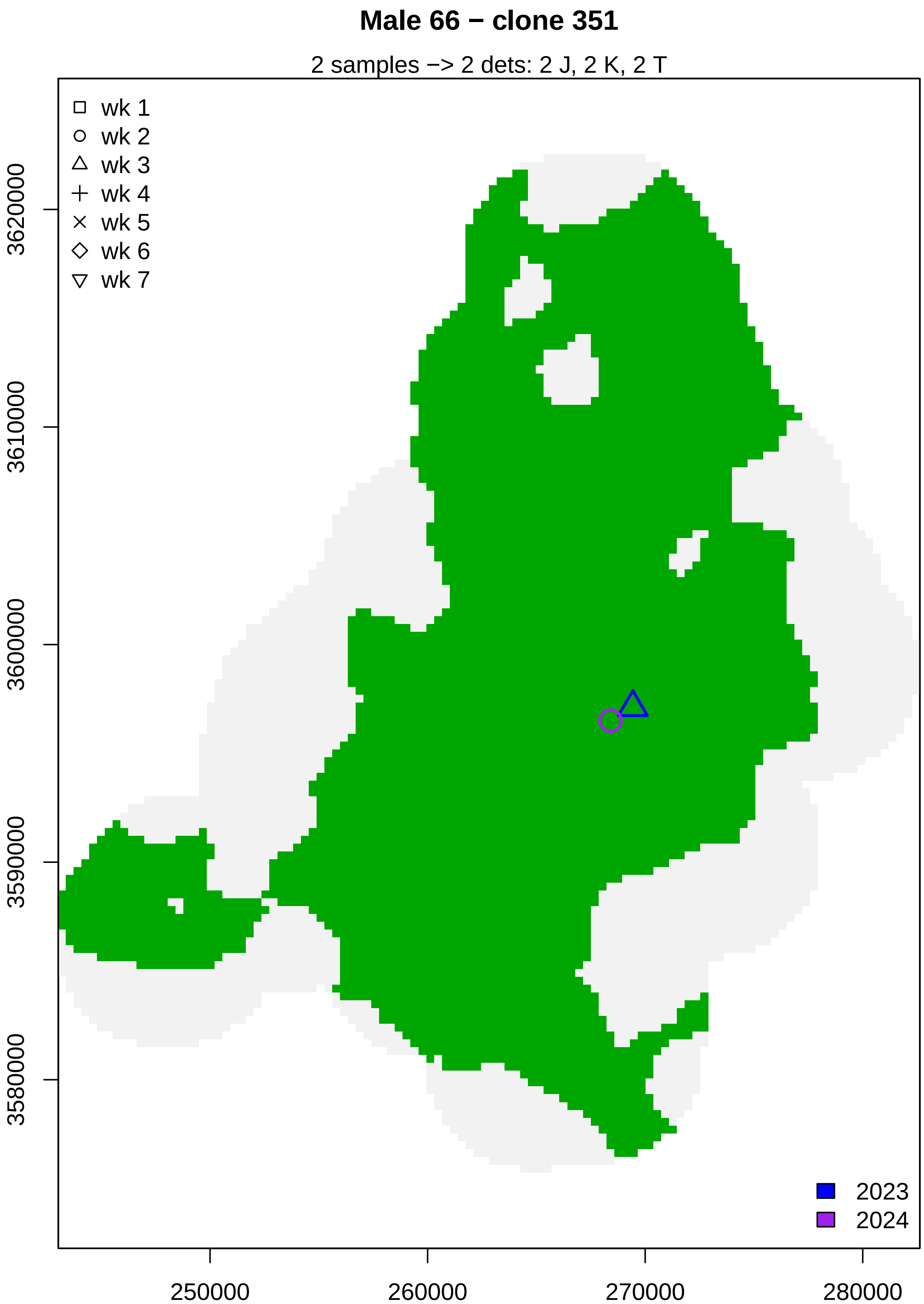
Male 55 − clone 246

3 samples −> 2 dets: 2 J, 1 K, 1 T





Male 59 − clone 292
4 samples −> 3 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



Overall, it seems we have little evidence for movement of activity centers between years. By far, the furthest movements recorded in our capture histories come in a single week!

Distances within and between years

Nonetheless, let's look more closely at the question about movement of activity centers. We'll do this in a couple of ways. First, we'll calculate the centroid of trap locations where each individual was detected in each year, then record a distance between these centroids. Second, we'll calculate the maximum distance between detections for each individual across the study period (both years) and compare that to the maximum distance between two detections within a year. From the maps above, we expect to see maxium distances within, rather than among, years.

Females

First, for females...

```
#Load capture history objects
load("../scr/CGA_CH.Rda")

#Load hair snare location data
load("../sampleinfo/snareLocations.Rda")
traps <- locs

#Females first
centroid.yr1 <- centroid.yr2 <- matrix(data = NA, nrow = dim(Fch$y)[1], ncol = 2)
is.spatrecap <- max.within <- max.all <- dist.centroid <- max.yr1 <- max.yr2 <- c()

for(i in 1:dim(Fch$y)[1]) {
  is.spatrecap[i] <- (sum(rowSums(Fch$y[i,,,]) > 0) > 1)
  yr1snares <- which(rowSums(Fch$y[i,,,1]) > 0)
  yr2snares <- which(rowSums(Fch$y[i,,,2]) > 0)

  if(length(yr1snares) > 0) {
    centroid.yr1[i,] <- colMeans(traps[yr1snares,])
    if(length(yr1snares) > 1) {
      max.yr1[i] <- max(dist(traps[yr1snares,], method = "euclidean"))
    } else {
      max.yr1[i] <- 0
    }
  } else {
    max.yr1[i] <- 0
  }

  if(length(yr2snares) > 0) {
    centroid.yr2[i,] <- colMeans(traps[yr2snares,])
    if(length(yr2snares) > 1) {
      max.yr2[i] <- max(dist(traps[yr2snares,], method = "euclidean"))
    } else {
      max.yr2[i] <- 0
    }
  } else {
    max.yr2[i] <- 0
  }

  if(length(yr1snares) > 0 & length(yr2snares) > 0) {
    dist.centroid[i] <- dist(rbind(centroid.yr1[i,], centroid.yr2[i,]),
```

```
                              method = "euclidean")
  }

  max.within[i] <- max(c(max.yr1[i], max.yr2[i]))
  if(length(yr1snares) > 0 & length(yr2snares > 0)) {
    max.all[i] <- max(dist(traps[c(yr1snares,yr2snares),],
                             method = "euclidean"))
  } else if(length(yr1snares) > 0 & length(yr2snares) == 0) {
    max.all[i] <- max.yr1[i]
  } else if(length(yr1snares) == 0 & length(yr2snares) > 0) {
    max.all[i] <- max.yr2[i]
  }

  rm(yr1snares, yr2snares)

}

#Focusing on the spatial recaptures
mean(dist.centroid[which(is.spatrecap == TRUE)], na.rm = TRUE)
```

```
## [1] 1238.211
```

```
sr.max.within <- max.within[which(is.spatrecap == TRUE)]
sr.max.all <- max.all[which(is.spatrecap == TRUE)]
sum(is.spatrecap)
```

```
## [1] 31
```

```
sum(sr.max.within >= sr.max.all)
```

```
## [1] 23
```

The average distance between yearly centroids for females with spatial recaptures was 1238.2 m. Among the 31 females with spatial recaptures, 23 had their greatest distance between two detections *within* a year.

Males

Then for males…

```
#Now for males
centroid.yr1 <- centroid.yr2 <- matrix(data = NA, nrow = dim(Mch$y)[1], ncol = 2)
is.spatrecap <- max.within <- max.all <- dist.centroid <- max.yr1 <- max.yr2 <- c()

for(i in 1:dim(Mch$y)[1]) {
  is.spatrecap[i] <- (sum(rowSums(Mch$y[i,,,]) > 0) > 1)
  yr1snares <- which(rowSums(Mch$y[i,,,1]) > 0)
  yr2snares <- which(rowSums(Mch$y[i,,,2]) > 0)

  if(length(yr1snares) > 0) {
    centroid.yr1[i,] <- colMeans(traps[yr1snares,])
```

```r
    if(length(yr1snares) > 1) {
      max.yr1[i] <- max(dist(traps[yr1snares,], method = "euclidean"))
    } else {
      max.yr1[i] <- 0
    }
  } else {
    max.yr1[i] <- 0
  }

  if(length(yr2snares) > 0) {
    centroid.yr2[i,] <- colMeans(traps[yr2snares,])
    if(length(yr2snares) > 1) {
      max.yr2[i] <- max(dist(traps[yr2snares,], method = "euclidean"))
    } else {
      max.yr2[i] <- 0
    }
  } else {
    max.yr2[i] <- 0
  }

  if(length(yr1snares) > 0 & length(yr2snares) > 0) {
    dist.centroid[i] <- dist(rbind(centroid.yr1[i,], centroid.yr2[i,]),
                             method = "euclidean")
  }

  max.within[i] <- max(c(max.yr1[i], max.yr2[i]))
  if(length(yr1snares) > 0 & length(yr2snares > 0)) {
    max.all[i] <- max(dist(traps[c(yr1snares,yr2snares),],
                           method = "euclidean"))
  } else if(length(yr1snares) > 0 & length(yr2snares) == 0) {
    max.all[i] <- max.yr1[i]
  } else if(length(yr1snares) == 0 & length(yr2snares) > 0) {
    max.all[i] <- max.yr2[i]
  }

  rm(yr1snares, yr2snares)

}

#Focusing on the spatial recaptures
mean(dist.centroid[which(is.spatrecap == TRUE)], na.rm = TRUE)
```

```
## [1] 2663.472
```

```r
sr.max.within <- max.within[which(is.spatrecap == TRUE)]
sr.max.all <- max.all[which(is.spatrecap == TRUE)]
sum(is.spatrecap)
```

```
## [1] 26
```

```r
sum(sr.max.within >= sr.max.all)
```

```
## [1] 16
```

The average distance between yearly centroids for males with spatial recaptures was 2663.5 m. Among the 26 males with spatial recaptures, 16 had their greatest distance between two detections *within* a year.

Digging in...

Female clone 43

Looking more closely at the 9th female plotted (clone #43), which had a large within-week movement inferred during week 4 of 2023...

```r
#Recreating the object that was used to build the female capture history...

#Load sample metadata (smd)
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)

sampsALL <- smd[which(!is.na(smd$finalClone) &
                        smd$is.SCR == TRUE &
                        smd$year %in% c(2023,2024)),]

#Switch name of finalClone column for compatibility with makeCH function
colnames(sampsALL)[colnames(sampsALL) == "finalClone"] <- "cloneID"

Fsamps <- sampsALL[which(sampsALL$finalSEX == "F"),]
length(unique(Fsamps$cloneID[!is.na(Fsamps$cloneID)]))
```

```
## [1] 81
```

```r
Fsamps[Fsamps$cloneID == 43,]
```

```
##                                       extID  ugaID is.SCR is.rep week year
## 107                   UGA_173101_P016_WA11 372-23   TRUE  FALSE    3 2023
## 168                   UGA_173101_P016_WF12 370-23   TRUE  FALSE    2 2023
## 196                   UGA_173101_P017_WA04 339-23   TRUE  FALSE    2 2023
## 222                   UGA_173101_P017_WC06  21-23   TRUE   TRUE    2 2023
## 337                   UGA_173101_P018_WE01 103-23   TRUE  FALSE    6 2023
## 359                   UGA_173101_P018_WF11 476-23   TRUE  FALSE    6 2023
## 404                   UGA_173101_P019_WD02 299-23   TRUE  FALSE    4 2023
## 753                   UGA_173101_P023_WA03  21-23   TRUE   TRUE    2 2023
## 798                   UGA_173101_P023_WD12 197-23   TRUE  FALSE    6 2023
## 996   GTseek_024_384i5_170_P098.003_996-23 996-23   TRUE  FALSE    2 2023
## 1031  GTseek_024_384i5_205_P098.001_678-23 678-23   TRUE  FALSE    2 2023
## 1165  GTseek_024_384i5_339_P098.002_895-23 895-23   TRUE  FALSE    6 2023
## 1342  GTseek_025_384i5_132_P098.007_886-23 886-23   TRUE  FALSE    6 2023
## 1404  GTseek_025_384i5_194_P098.007_987-23 987-23   TRUE  FALSE    6 2023
## 1924 GTseek_028_384i5_099_P098.UGA1_107-23 107-23   TRUE  FALSE    4 2023
## 1958 GTseek_028_384i5_133_P098.UGA1_143-23 143-23   TRUE  FALSE    3 2023
## 2014 GTseek_028_384i5_189_P098.UGA2_459-23 459-23   TRUE  FALSE    2 2023
##       snare       utmSnare
## 107  HS 068 268186x3591620
## 168  HS 068 268186x3591620
```

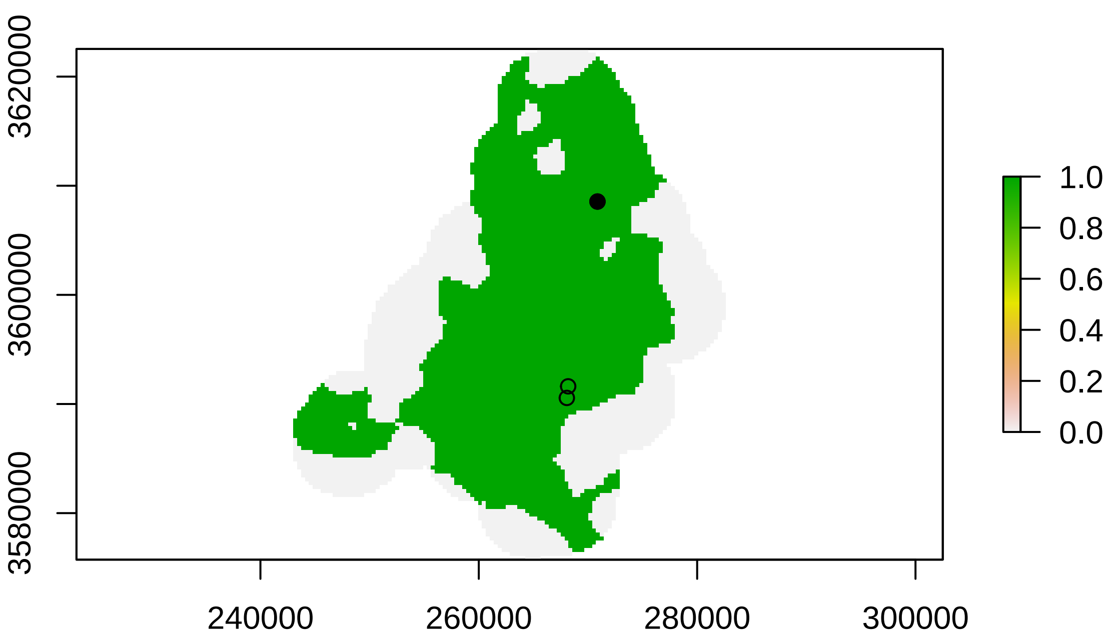
```
## 196  HS 068 268186x3591620
## 222  HS 068 268186x3591620
## 337  HS 068 268186x3591620
## 359  HS 066 268064x3590555
## 404  HS 068 268186x3591620
## 753  HS 068 268186x3591620
## 798  HS 066 268064x3590555
## 996  HS 068 268186x3591620
## 1031 HS 068 268186x3591620
## 1165 HS 066 268064x3590555
## 1342 HS 066 268064x3590555
## 1404 HS 066 268064x3590555
## 1924 HS 126 270872x3608550
## 1958 HS 068 268186x3591620
## 2014 HS 068 268186x3591620
##                                                                                   fqfile
## 107  RAPiD-Genomics_F534_UGA_173101_P016_WA11_i5-504_i7-107_S14845_L003_R1_001.fastq
## 168  RAPiD-Genomics_F534_UGA_173101_P016_WF12_i5-504_i7-168_S14905_L003_R1_001.fastq
## 196  RAPiD-Genomics_F534_UGA_173101_P017_WA04_i5-512_i7-100_S14933_L003_R1_001.fastq
## 222  RAPiD-Genomics_F534_UGA_173101_P017_WC06_i5-512_i7-126_S14959_L003_R1_001.fastq
## 337  RAPiD-Genomics_F534_UGA_173101_P018_WE01_i5-537_i7-145_S15074_L003_R1_001.fastq
## 359  RAPiD-Genomics_F534_UGA_173101_P018_WF11_i5-537_i7-167_S15098_L003_R1_001.fastq
## 404  RAPiD-Genomics_F534_UGA_173101_P019_WD02_i5-1102_i7-72_S15143_L003_R1_001.fastq
## 753   RAPiD-Genomics_F566_UGA_173101_P023_WA03_i5-1099_i7-82_S7681_L003_R1_001.fastq
## 798    RAPiD-Genomics_F566_UGA_173101_P023_WD12_i5-1099_i7-9_S7726_L003_R1_001.fastq
## 996                                                                                 <NA>
## 1031                                                                                <NA>
## 1165                                                                                <NA>
## 1342                                                                                <NA>
## 1404                                                                                <NA>
## 1924                                                                                <NA>
## 1958                                                                                <NA>
## 2014                                                                                <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 107             Hair Snare    CGA    919 UGA_173101_P016_WA11 FALSE       LGC
## 168             Hair Snare    CGA    898 UGA_173101_P016_WF12 FALSE       LGC
## 196             Hair Snare    CGA    918 UGA_173101_P017_WA04 FALSE       LGC
## 222             Hair Snare    CGA    747 UGA_173101_P017_WC06 FALSE       LGC
## 337             Hair Snare    CGA    919 UGA_173101_P018_WE01 FALSE       LGC
## 359             Hair Snare    CGA    917 UGA_173101_P018_WF11 FALSE       LGC
## 404             Hair Snare    CGA    685 UGA_173101_P019_WD02 FALSE       LGC
## 753             Hair Snare    CGA    776 UGA_173101_P023_WA03 FALSE       LGC
## 798             Hair Snare    CGA    586 UGA_173101_P023_WD12 FALSE       LGC
## 996            Hair Sample    CGA    121          GTS_996-23 FALSE    GTSeek
## 1031           Hair Sample    CGA    120          GTS_678-23 FALSE    GTSeek
## 1165           Hair Sample    CGA    121          GTS_895-23 FALSE    GTSeek
## 1342           Hair Sample    CGA    121          GTS_886-23 FALSE    GTSeek
## 1404           Hair Sample    CGA    122          GTS_987-23 FALSE    GTSeek
## 1924           Hair Sample    CGA    123          GTS_107-23 FALSE    GTSeek
## 1958           Hair Sample    CGA    123          GTS_143-23 FALSE    GTSeek
## 2014           Hair Sample    CGA    123          GTS_459-23 FALSE    GTSeek
##      sexGT COLclone cloneID finalSEX
## 107      F       43      43        F
## 168      F       43      43        F
```

```
## 196       F        43       43          F
## 222       F        43       43          F
## 337       F        43       43          F
## 359       F        43       43          F
## 404       F        43       43          F
## 753       F        43       43          F
## 798       F        43       43          F
## 996       F       282       43          F
## 1031      F       282       43          F
## 1165      F       282       43          F
## 1342      F       282       43          F
## 1404      F       282       43          F
## 1924      F       282       43          F
## 1958      F       282       43          F
## 2014      F       282       43          F
```

```
table(Fsamps$snare[which(Fsamps$cloneID == 43)])
```

```
##
## HS 066 HS 068 HS 126
##      5     11      1
```

```
plot(S360)
points(locs[rownames(locs) %in% Fsamps$utmSnare[Fsamps$cloneID == 43],])
points(locs[rownames(locs) %in% Fsamps$utmSnare[which(Fsamps$cloneID == 43 & Fsamps$snare == "HS 126")]
```



```r
Fsamps[Fsamps$cloneID == 43 & Fsamps$snare == "HS 126",]
```

```
##                                    extID  ugaID is.SCR is.rep week year
## 1924 GTseek_028_384i5_099_P098.UGA1_107-23 107-23   TRUE  FALSE    4 2023
##       snare       utmSnare fqfile knownSEX  sampSOURCE region nSNPgt      vcfID
## 1924 HS 126 270872x3608550   <NA>          Hair Sample    CGA    123 GTS_107-23
##       FILT GTservice sexGT COLclone cloneID finalSEX
## 1924 FALSE    GTSeek     F      282      43        F
```

```r
#Check kinship of this original COLONY clone
kingrel <- read.table("recapPLINK.kin0", header = FALSE)
colnames(kingrel) <- c("id1","id2","nsnp","hehet","ibs0","kinship")
kingrel$kinship[which(kingrel$id1 %in% Fsamps$vcfID[which(Fsamps$COLclone == 282)] & kingrel$id2 %in% Fs
```

```
##  [1] 0.488636 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000
##  [9] 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000
## [17] 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000
## [25] 0.500000 0.500000 0.500000 0.500000
```

```r
#Samples assigned to the same original clone by COLONY
Fsamps[which(Fsamps$COLclone == 282),]
```

```
##                                    extID  ugaID is.SCR is.rep week year
## 996   GTseek_024_384i5_170_P098.003_996-23 996-23   TRUE  FALSE    2 2023
## 1031  GTseek_024_384i5_205_P098.001_678-23 678-23   TRUE  FALSE    2 2023
## 1165  GTseek_024_384i5_339_P098.002_895-23 895-23   TRUE  FALSE    6 2023
## 1342  GTseek_025_384i5_132_P098.007_886-23 886-23   TRUE  FALSE    6 2023
## 1404  GTseek_025_384i5_194_P098.007_987-23 987-23   TRUE  FALSE    6 2023
## 1924 GTseek_028_384i5_099_P098.UGA1_107-23 107-23   TRUE  FALSE    4 2023
## 1958 GTseek_028_384i5_133_P098.UGA1_143-23 143-23   TRUE  FALSE    3 2023
## 2014 GTseek_028_384i5_189_P098.UGA2_459-23 459-23   TRUE  FALSE    2 2023
##       snare       utmSnare fqfile knownSEX  sampSOURCE region nSNPgt      vcfID
## 996  HS 068 268186x3591620   <NA>          Hair Sample    CGA    121 GTS_996-23
## 1031 HS 068 268186x3591620   <NA>          Hair Sample    CGA    120 GTS_678-23
## 1165 HS 066 268064x3590555   <NA>          Hair Sample    CGA    121 GTS_895-23
## 1342 HS 066 268064x3590555   <NA>          Hair Sample    CGA    121 GTS_886-23
## 1404 HS 066 268064x3590555   <NA>          Hair Sample    CGA    122 GTS_987-23
## 1924 HS 126 270872x3608550   <NA>          Hair Sample    CGA    123 GTS_107-23
## 1958 HS 068 268186x3591620   <NA>          Hair Sample    CGA    123 GTS_143-23
## 2014 HS 068 268186x3591620   <NA>          Hair Sample    CGA    123 GTS_459-23
##       FILT GTservice sexGT COLclone cloneID finalSEX
## 996  FALSE    GTSeek     F      282      43        F
## 1031 FALSE    GTSeek     F      282      43        F
## 1165 FALSE    GTSeek     F      282      43        F
## 1342 FALSE    GTSeek     F      282      43        F
## 1404 FALSE    GTSeek     F      282      43        F
## 1924 FALSE    GTSeek     F      282      43        F
## 1958 FALSE    GTSeek     F      282      43        F
## 2014 FALSE    GTSeek     F      282      43        F
```

```r
Fsamps[which(Fsamps$COLclone == 43),]
```

```
##                     extID  ugaID is.SCR is.rep week year  snare       utmSnare
## 107 UGA_173101_P016_WA11 372-23   TRUE  FALSE    3 2023 HS 068 268186x3591620
## 168 UGA_173101_P016_WF12 370-23   TRUE  FALSE    2 2023 HS 068 268186x3591620
## 196 UGA_173101_P017_WA04 339-23   TRUE  FALSE    2 2023 HS 068 268186x3591620
## 222 UGA_173101_P017_WC06  21-23   TRUE   TRUE    2 2023 HS 068 268186x3591620
## 337 UGA_173101_P018_WE01 103-23   TRUE  FALSE    6 2023 HS 068 268186x3591620
## 359 UGA_173101_P018_WF11 476-23   TRUE  FALSE    6 2023 HS 066 268064x3590555
## 404 UGA_173101_P019_WD02 299-23   TRUE  FALSE    4 2023 HS 068 268186x3591620
## 753 UGA_173101_P023_WA03  21-23   TRUE   TRUE    2 2023 HS 068 268186x3591620
## 798 UGA_173101_P023_WD12 197-23   TRUE  FALSE    6 2023 HS 066 268064x3590555
##                                                                        fqfile
## 107 RAPiD-Genomics_F534_UGA_173101_P016_WA11_i5-504_i7-107_S14845_L003_R1_001.fastq
## 168 RAPiD-Genomics_F534_UGA_173101_P016_WF12_i5-504_i7-168_S14905_L003_R1_001.fastq
## 196 RAPiD-Genomics_F534_UGA_173101_P017_WA04_i5-512_i7-100_S14933_L003_R1_001.fastq
## 222 RAPiD-Genomics_F534_UGA_173101_P017_WC06_i5-512_i7-126_S14959_L003_R1_001.fastq
## 337 RAPiD-Genomics_F534_UGA_173101_P018_WE01_i5-537_i7-145_S15074_L003_R1_001.fastq
## 359 RAPiD-Genomics_F534_UGA_173101_P018_WF11_i5-537_i7-167_S15098_L003_R1_001.fastq
## 404 RAPiD-Genomics_F534_UGA_173101_P019_WD02_i5-1102_i7-72_S15143_L003_R1_001.fastq
## 753  RAPiD-Genomics_F566_UGA_173101_P023_WA03_i5-1099_i7-82_S7681_L003_R1_001.fastq
## 798   RAPiD-Genomics_F566_UGA_173101_P023_WD12_i5-1099_i7-9_S7726_L003_R1_001.fastq
##     knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 107          Hair Snare    CGA    919 UGA_173101_P016_WA11 FALSE       LGC
## 168          Hair Snare    CGA    898 UGA_173101_P016_WF12 FALSE       LGC
## 196          Hair Snare    CGA    918 UGA_173101_P017_WA04 FALSE       LGC
## 222          Hair Snare    CGA    747 UGA_173101_P017_WC06 FALSE       LGC
## 337          Hair Snare    CGA    919 UGA_173101_P018_WE01 FALSE       LGC
## 359          Hair Snare    CGA    917 UGA_173101_P018_WF11 FALSE       LGC
## 404          Hair Snare    CGA    685 UGA_173101_P019_WD02 FALSE       LGC
## 753          Hair Snare    CGA    776 UGA_173101_P023_WA03 FALSE       LGC
## 798          Hair Snare    CGA    586 UGA_173101_P023_WD12 FALSE       LGC
##     sexGT COLclone cloneID finalSEX
## 107     F       43      43        F
## 168     F       43      43        F
## 196     F       43      43        F
## 222     F       43      43        F
## 337     F       43      43        F
## 359     F       43      43        F
## 404     F       43      43        F
## 753     F       43      43        F
## 798     F       43      43        F
```

```r
#And kinship of the sample in question to other members of the final clone
kingrel$kinship[which(kingrel$id1 == "GTS_107-23" & kingrel$id2 %in% Fsamps$vcfID[which(Fsamps$cloneID =
```

```
##  [1] 0.489583 0.489130 0.489583 0.468750 0.500000 0.478261 0.486111 0.480769
##  [9] 0.454545 0.500000 0.500000 0.500000 0.500000 0.500000
```

```r
kingrel$kinship[which(kingrel$id1 %in% Fsamps$vcfID[which(Fsamps$cloneID ==43)] & kingrel$id2 == "GTS_1(
```

```
## [1] 0.5 0.5
```

```
#Kinship is high to all other members of this final clone!
```

So the recapture inference in this case is from `PLINK` only, not `COLONY`, and the sample was genotyped by GTseek, so it has fewer SNP loci available to make the match. The sample is well-genotyped, though. It appears that `COLONY` split detections of this individual by dataset - LGC and GTseek. GTseek detections include the same primary hair snares (HS 066 and HS 068) as in the LGC data. Given the uniformly high kinship coefficients among members of these two `COLONY` clones, it seems unlikely that the detection at HS 126 was a false-positive recapture.
